

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
FACULTY OF LIBERAL ARTS AND BUSINESS STUDIES
DEPARTMENT OF COMPUTER SCIENCE



PRACTICAL RECORD

STUDENT NAME :

REGISTRATION NO. :

CLASS : B.SC Computer Science - A

SEMESTER & YEAR : V Semester & III Year

SUBJECT CODE : USC24D01J

SUBJECT TITLE : SOFTWARE ENGINEERING

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SRM
INSTITUTE OF SCIENCE & TECHNOLOGY
(Deemed to be University u/s 3 of UGC Act, 1956)

FACULTY OF LIBERAL ARTS AND BUSSINESS STUDIES

DEPARTMENT OF COMPUTER SCIENCE

BONAFIDE CERTIFICATE

This is to certify that this is a bonafide record of practical work done by

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During the Academic Year 2026 – 2027. Submitted for B.Sc Degree practical examination
held on _____.

FACULTY INCHARGE

H.O.D

INTERNAL EXAMINER

EXTERNAL EXAMINER

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1.1 SYSTEM REQUIREMENT SPECIFICATION FOR COLLEGE AUTOMATION SYSTEM

1.INTRODUCTION

The title of the project is COLLEGE MANAGEMENT SYSTEM (CMS). CMS is an Internet based application that aims at providing information to all the levels of management within an organization. This system can be used as a information management system for the college.

For a given user, the administrator will create a login id & password, using this user can access the system to either upload or download some information from the database.

The front-end will be HTML pages with Java Script for client side validation where as all business logics will be in

PHP reside at middle layer. And these layers will interact with third layer of database, which will be My Sql database.

The web server will be Apache. To start working on this project environment required is a server having Apache as web server, My Sql as database and XAMPP as development environment

1.1 PURPOSE

The purpose of this document is to present a detailed description of the College Management System. It will explain the purpose and features of the system, the interfaces of the system, what the system will do, the constraints under which it must operate and how the system will react to external stimuli. This document is intended for both the client and the developers of the system and will be proposed to the Administrative head for its approval.

1.2 PROJECT SCOPE AND PRODUCT FEATURES

This software system will be a College management system for a the members of an organization. This system will be

designed to maximize the administrative, academic and overall productivity by providing tools to assist in automating the technical procedures and processes, which would otherwise have to be performed manually. By maximizing the users work efficiency and production the system will meet the users needs while remaining easy to understand and use.

It is a user-friendly portal to interact, manage, access the information.

1.3 INTENDED AUDIENCE AND DOCUMENT OVERVIEW

There are different types of intended audience for this document, such as developers, testers, documentation writers and most importantly the users. We have divided the rest of the document into different subsections. We suggest that you begin with understanding the definitions, acronyms and abbreviations, then sequentially go through contents, overview section and proceeding through the detailed description sections that is most pertinent.

1.4 DEFINATIONS, ACRONYMS, AND ABBREVIATIONS

CMS- College Management System, SRS- Software Requirement Specification, IDLE Python, students data entry Ex- Name, Father name, Contact No. etc.

2 OVERALL DESCRIPTION

2.1 PRODUCT PERSPECTIVE

The product will be a standalone application and may be run on multiple systems within an Internet network. The product will require a keyboard, mouse and monitor to interface with the users. The minimum hardware requirements for the product are specified in this document

2.2 USER CLASSES AND CHARACTERISTICS

The target audience for CMS product is the college Administrator/students/faculty/staff (Technical/Nontechnical)

Administrator The Super user of the system. Mainly focuses on administrative and academic related issues.

- Student A user with limited access rights.
- Staff A user of the system who has more access rights than a normal user.

2.3 CONSTRAINTS

The current constraints on the project are related to the provision of hardware resources and software resources.

- At present, we have a i3 gen4 intel core processor running on top of the Linux/windows operating system.
- The documents will be present only in pdf format.
- In the feedback forms, the replies will not be frequent and the petitioner will not be anonymous.
- There will not be any moderator to filter out the fake complains with the genuine ones. The superuser have to do it himself manually.

- The Internet connection is also a constraint for the application. Since the application fetches data from the database over the Internet, it is crucial that there is an Internet connection for the application to function.
- The web portal will be constrained by the capacity of the database. Since the database may be forced to queue incoming requests and therefore increase the time it takes to fetch data.
- Mess Rebate Will at least of 3days.
- Registration will be open for short time.
- All Document should be in .Zip file.
- College will provide funds for SMS service if SMS service is not free.
- After submitting the course evaluation form, the user cannot revert his or her actions.
- The user cannot change his/her all personal or academic details. He/she first have to get permission from the super user to do so.

2.4 ASSUMPTIONS AND DEPENDENCIES

A number of factors that may affect the requirements specified in the SRS include:

- It is assumed that only one person from the different department will have the access to a module. i.e. Only heads of administration, academics, HEC and Mess will have the access to their department except the faculty.
- The complaints and the feedback given by the students and other members of the organization are assumed to be reliable.
- The schedule for the exam , the registration window will be open for only few days only after that these pages will be inactive until next exams or registration period.
- Apportioning of requirements

In the case that the project is delayed, there are some requirements that could be transferred to the next version of the application. Those requirements are to be developed in the third release.

3 SPECIFIC REQUIREMENTS

This section contains all the software requirements at a level of detail sufficient to enable designers to design a system to satisfy those requirements, and testers to test that the system satisfies those requirements. Throughout this section, every stated requirement should be externally perceivable by users, administrator, or, other external systems.

3.1 EXTERNAL INTERFACES

Client:

- Hardware platform:
 - PIII or above with
 - RAM of 512 or above MB – Hard Disk 20GB or above GB
- Software Platform: Browser :
 - Mozilla Fire-Fox v12.0 or higher
 - Google Chrome v27.0.1453.116m or higher.

Server:

- User Interface:

A first-time user of the web portal should see the log-in page when he/she opens the portal. If the user has not registered, he/she should be able to do that on the log-in page. It will also have a remember me button. If the user is not a first time user, he/she should be able to see the dashboard which contains different domains like academics, Hostel, Profile, Mess, Transport. A news bulletin, some general information, list of holidays and different timetables will also be visible on this page. Every user should have a profile page where they can edit their e-mail address, phone number and password and other personal details.

- Communications interfaces

The communication between the client and the server will be done through internet.

3.2 FUNCTIONAL REQUIREMENTS

This section includes the requirements that specify all the fundamental actions of the software system

LOGIN

This section contains students login menu where students have to login by their username as well as password

MARKSHEET

This section contains student's stored data, student can find their marks by entering detail in 'student detail'

Option, and after feeling their data he/she may automatically get their marks in 'grades point option'.

MENU

This section includes menu's for students details such as student profile, library system, fee report and Marksheet.

SEARCH PAGE

Here student can search their stored data entering roll no.

STUDENT INFORMATION

Here student can store their data in database form by entering data into 'student information' section.

3.3 NON FUNCTIONAL REQUIREMENTS

3.3.1 PERFORMANCE REQUIREMENTS

Performance should not be an issue because all of our server queries involve small pieces of data. Changing screens will require very little computation and thus will occur very quickly. Server updates should only take a few seconds as long as the phone can maintain a steady signal.

3.3.2 RELIABILITY

Must maintain data integrity. Computer crashes and misuse should not affect a user's history

3.3.3 AVAILABILITY

The CMS Portal shall be available, up and running for 24*7 throughout the year except due to the routine maintenance activities.

3.3.4 SECURITY REQUIREMENTS

Administrator and Users with valid credentials will be able to log in to Portal. Administrator will have access to the database structures at backend. Administrator will have the rights for modifications as well as any updation work for the datasets and website. Access to the various subsystems will be protected by a user log in screen that requires a user name and password. To be updated in future.

1.2 SYSTEM REQUIREMENT SPECIFICATION FOR LIBRARY INFORMATION SYSTEM

1. INTRODUCTION

1.1 PURPOSE

The main objective of this document is to illustrate the requirements of the project Library Management system. The document gives the detailed description of the both functional and non-functional requirements proposed by the client. The purpose of this project is to provide a friendly environment to maintain the details of books and library members. The main purpose of this project is to maintain easy circulation system using computers and to provide different reports. This project describes the hardware and software interface requirements using ER diagrams and UML diagrams.

1.2 Document Conventions

- Entire document should be justified.
- Convention for Main title
 - Font face: Times New Roman
 - Font style: Bold
 - Font Size: 14
- Convention for Sub title
 - Font face: Times New Roman
 - Font style: Bold
 - Font Size: 12
- Convention for body
 - Font face: Times New Roman
 - Font Size: 12

1.3 SCOPE OF DEVELOPMENT PROJECT

Library Management System is basically updating the manual library system into an internet based application so that the users can know the details of their accounts, availability of books and maximum limit for borrowing.

The project is specifically designed for the use of librarians and library users. The product will work as a complete user interface for library management process and library usage from ordinary users. Library Management System can be used by any existing or new library to manage its books and book borrowing, insertion and monitoring. It is especially useful for any educational institute where modifications in the content can be done easily according to requirements.

The project can be easily implemented under various situations. We can add new features as and when we require, making reusability possible as there is flexibility in all the modules. The language used for developing the project is Java as it is quite advantageous than other languages in terms of performance, tools available, cross platform compatibility, libraries, cost (freely available), and development process.

1.4 DEFINITIONS, ACRONYMS AND ABBREVIATIONS

JAVA -> platform independence

SQL-> Structured query Language

ER-> Entity Relationship

UML -> Unified Modelling Language

IDE-> Integrated Development Environment

SRS-> Software Requirement Specification

ISBN -> International Standard Book Number

IEEE ->Institute of Electrical and Electronics Engineers

1.5 REFERENCES

➤ Books

- Software Requirements and Specifications: A Lexicon of Practice, Principles and Prejudices (ACM Press) by Michael Jackson
- Software Requirements (Microsoft) Second Edition By Karl E. Wiegers
- Software Engineering: A Practitioner's Approach Fifth Edition By Roger S. Pressman

➤ Websites

- <http://www.slideshare.net/> • <http://ebookily.net/doc/srs-library-management-system>

2. OVERALL DESCRIPTIONS

2.1 PRODUCT PERSPECTIVE

This is a broad level diagram of the project showing a basic overview. The users can be either staff or student.. This System will provide a search functionality to facilitate the search of resources. This search will be based on various categories viz. book name or the ISBN. Further the library staff personnel can add/update the resources and the resource users from the system. The users of the system can request issue/renew/return of books for which they would have to follow certain criteria.

2.2 Product Function

The Online Library System provides online real time information about the books available in the Library and the user information. The main purpose of this project is to reduce the manual work. This software is capable of managing Book Issues, Returns, Calculating/Managing Fine, Generating various Reports for Record-Keeping according to end user requirements. The Librarian will act as the administrator to control members and manage books. The member's status of issue/return is maintained in the library database. The member's details can be fetched by the librarian from the database as and when required. The valid members are also allowed to view their account information.

2.3 User Classes and Characteristics

The system provides different types of services based on the type of users [Member/Librarian]. The Librarian will be acting as the controller and he will have all the privileges of an administrator. The member can be either a student or staff of the university who will be accessing the Library online.

- The features that are available to the Librarian are:- ➤ A librarian can issue a book to the member.
 - Can view the different categories of books available in the Library
 - Can view the List of books available in each category
 - Can take the book returned from students
 - Add books and their information to the database
 - Edit the information of existing books
 - Can check the report of the existing books
 - Can check the report of the issued books
 - Can access all the accounts of the students
- The features that are available to the Members are:-
 - Can view the different categories of books available in the Library • Can view the List of books available in each category
 - Can own an account in the library.
 - Can view the books issued to him
 - Can put a request for a new book
 - Can view the history of books issued to him previously
 - Can search for a particular book

2.4 OPERATING ENVIRONMENT

The product will be operating in windows environment. The Library Management System is a website and shall operate in all famous browsers, for a model we are taking Microsoft Internet Explorer, Google, Chrome and Mozilla, Firefox. Also it will be compatible with the IE 6.0. Most of the features will be compatible with the Mozilla Firefox & Opera 7.0 or higher version. The only requirement to use this online product would be the internet connection. The hardware configuration include Hard Disk: 40 GB, Monitor: 15” Colour monitor, Keyboard: 122 keys. The basic input devices required are keyboard, mouse and output devices are monitor, printer etc.

2.5 ASSUMPTIONS AND DEPENDENCIES

- The assumptions are:-
 - The coding should be error free
 - The system should be user-friendly so that it is easy to use for the users
 - The information of all users, books and libraries must be stored in a database that is accessible by the website
 - The system should have more storage capacity and provide fast access to the database
 - The system should provide search facility and support quick transactions
 - The Library System is running 24 hours a day
 - Users may access from any computer that has Internet browsing capabilities and an Internet connection
 - Users must have their correct usernames and passwords to enter into their online accounts and do actions
- The dependencies are:-
 - The specific hardware and software due to which the product will be run
 - On the basis of listing requirements and specification the project will be developed and run
 - The end users (admin) should have proper understanding of the product
 - The system should have the general report stored
 - The information of all the users must be stored in a database that is accessible by the Library System

- Any update regarding the book from the library is to be recorded to the database and the data entered should be correct

2.6 REQUIREMENT

Software Configuration:-

This software package is developed using java as front end which is supported by sun micro system. Microsoft SQL Server as the back end to store the database. Operating System: Windows NT, windows 98, Windows XP Language: Java Runtime Environment, Net beans 7.0.1 (front end) Database: MYSQL Server (back end)

Hardware Configuration:-

Processor: Pentium(R)Dual-core CPU

Hard Disk: 40GB

RAM: 256 MB or more

2.7 DATA REQUIREMENT

The inputs consist of the query to the database and the output consists of the solutions for the query. The output also includes the user receiving the details of their accounts. In this project the inputs will be the queries as fired by the users like create an account, selecting books and putting into account. Now the output will be visible when the user requests the server to get details of their account in the form of time, date and which books are currently in the account.

3. EXTERNAL INTERFACE REQUIREMENT

3.1 GUI

The software provides good graphical interface for the user and the administrator can operate on the system, performing the required task such as create, update, viewing the details of the book.

- It allows user to view quick reports like Book Issued/Returned in between particular time.
- It provides stock verification and search facility based on different criteria.
- The user interface must be customizable by the administrator
- All the modules provided with the software must fit into this graphical user interface and accomplish to the standard defined
- The design should be simple and all the different interfaces should follow a standard template
- The user interface should be able to interact with the user management module and a part of the interface must be dedicated to the login/logout module

Login Interface:-

In case the user is not yet registered, he can enter the details and register to create his account. Once his account is created he can 'Login' which asks the user to type his username and password. If the user entered either his username or password incorrectly then an error message appears.

Search:-

The member or librarian can enter the type of book he is looking for and the title he is interested in, then he can search for the required book by entering the book name.

Categories View:-

Categories view shows the categories of books available and provides ability to the librarian to add/edit or delete category from the list.

Librarian's Control Panel:-

This control panel will allow librarian to add/remove users; add, edit, or remove a resource. And manage lending options.

4. SYSTEM FEATURES

The users of the system should be provided the surety that their account is secure.

This is possible by providing:-

- User authentication and validation of members using their unique member ID
- Proper monitoring by the administrator which includes updating account status, showing a popup if the member attempts to issue number of books that exceed the limit provided by the library policy, assigning fine to members who skip the date of return
- Proper accountability which includes not allowing a member to see other member's account. Only administrator will see and manage all member accounts

5. OTHER NON-FUNCTIONAL REQUIREMENTS

5.1 PERFORMANCE REQUIREMENT

The proposed system that we are going to develop will be used as the Chief performance system within the different campuses of the university which interacts with the university staff and students. Therefore, it is expected that the database would perform functionally all the requirements that are specified by the university.

- The performance of the system should be fast and accurate
- Library Management System shall handle expected and non-expected errors in ways that prevent loss in information and long downtime period. Thus it should have inbuilt error testing to identify invalid username/password
- The system should be able to handle large amount of data. Thus it should accommodate high number of books and users without any fault

5.2 SAFETY REQUIREMENT

The database may get crashed at any certain time due to virus or operating system failure. Therefore, it is required to take the database backup so that the database is not lost. Proper UPS/inverter facility should be there in case of power supply failure.

5.3 SECURITY REQUIREMENT

- System will use secured database
- Normal users can just read information but they cannot edit or modify anything except their personal and some other information.
- System will have different types of users and every user has access constraints
- Proper user authentication should be provided

- No one should be able to hack users' password
- There should be separate accounts for admin and members such that no member can access the database and only admin has the rights to update the database.

5.4 REQUIREMENT ATTRIBUTES

- There may be multiple admins creating the project, all of them will have the right to create changes to the system. But the members or other users cannot do changes
 - The project should be open source
 - The Quality of the database is maintained in such a way so that it can be very user friendly to all the users of the database
 - The user be able to easily download and install the system

5.5 BUSINESS RULES

A business rule is anything that captures and implements business policies and practices. A rule can enforce business policy, make a decision, or infer new data from existing data. This includes the rules and regulations that the System users should abide by. This includes the cost of the project and the discount offers provided. The users should avoid illegal rules and protocols. Neither admin nor member should cross the rules and regulations.

5.6 USER REQUIREMENT

The users of the system are members and Librarian of the university who act as administrator to maintain the system. The members are assumed to have basic knowledge of the computers and internet browsing. The administrators of the system should have more knowledge of the internals of the system and is able to rectify the small problems that may arise due to disk crashes, power failures and other catastrophes to maintain the system. The proper user interface, user manual, online help and the guide to install and maintain the system must be sufficient to educate the users on how to use the system without any problems.

- The admin provides certain facilities to the users in the form of:-
 - Backup and Recovery
 - Forgot Password
 - Data migration i.e. whenever user registers for the first time then the data is stored in the server
 - Data replication i.e. if the data is lost in one branch, it is still stored with the server
 - Auto Recovery i.e. frequently auto saving the information
 - Maintaining files i.e. File Organization
 - The server must be maintained regularly and it has to be updated from time to time.

1.3 SYSTEM REQUIREMENT SPECIFICATION FOR BANKING MANAGEMENT SYSTEM

1. INTRODUCTION

1.1 Purpose

This document gives detailed functional and non functional requirements for the bank management system. This product will support online banking transaction using ATM. The purpose of this document is that the requirements mentioned in it should be utilized by software developer to implement the system.

1.2 Scope

This Product will automate of banking transaction process.

1.3 Overview

The system provides easy solution to banks.

1.4 Additional Information

This system will work together with bank's computer. It will not be operated independently. Various banks might be networked together.

2. GENERAL DESCRIPTION

Using Automatic Teller Machine (ATM) customer can withdraw the desired amount of money. ATM is basically computerized telecommunication device that helps the customer to perform banking transactions. The customer is identified by inserting a plastic ATM card with a magnetic stripe or a plastic smartcard with a chip. This magnetic media contains a unique card number and some security information. Security is provided by the customer entering a Personal Identification Number (PIN). When a customer wants to perform some transaction, he/she will enter the ATM card in the machine and then enter the PIN. Only authentic customer will be allowed to perform this transaction.

3. FUNCTIONAL REQUIREMENTS

This section provides the requirement overview of the product. The project will require the PHP as a front end and at the back end the database MYSQL will be running. Various functional modules that can be implemented by the product will be –

1. Login
2. Validation
3. Get balance information
4. Withdrawal of money
5. Transfer Money
6. Report Generation

3.1 DESCRIPTION

Login:

Customer logs in by entering the card and typing the PIN.

Validation:

When a customer enters the ATM card, its validity must be ensured. Then customer is allowed to enter the valid PIN. The validation can be for following conditions –

☐ Validation for lost or stolen card

When card is already reported as lost or stolen then the message “Lost/Stolen card!!!”.

☐ Validation for card’s expiry date

If the card inserted by the customer has crossed the expiry date then the system will prompt “Expired Card”.

☐ Validation for PIN

After validating the card, the validity of PIN must be ensured. If he/she fails to enter valid code for three times then the card will not be returned to him. That means the account can be locked.

The counter for number of logins must be maintained

Get balance information:

This system must be networked to the bank’s computer. The updated database of every customer is maintained with bank. Hence the balance information of every account is available in the database and can be displayed to the customer.

Withdrawal of Money:

A customer is allowed to enter the amount which he/she wishes to withdraw. If the entered amount is less than the available balance and if after withdraw if the minimum required balance is maintained then allow the transaction.

Transfer of Money:

The customer can deposit or transfer the desired amount of money.

Report:

The bank statement showing credit and debit information of corresponding account must be printed by the machine.

3.2 TECHNICAL ISSUES

This product will work on client-server architecture. It will require an internet server and which will be able to run PHP applications. The product should support some commonly used browsers such as Internet Explorer, Mozilla Firefox.

4. INTERFACE REQUIREMENTS

4.1 GUI

This interface must be highly intuitive or interactive because there will not be an assistance for the user who is operating the ATM. The sample GUI for withdrawal of money can be If user wants to perform another transaction he will press – Enter key otherwise press Quit. After pressing the Quit button the card must be ejected.

4.2 Hardware Interface

Various interfaces for the product could be

1. The Hardware interface that can read the ATM card.
2. Touch screen/Monitor
3. Keypad
4. Continuous battery backup
5. Printer which can produce the hard copy.
6. Interface that connects the device to bank's computer.
7. An interface that can count currency notes.

4.3 Software Interface

1. Any windows operating system.
2. The PHP must be installed. For the database handling MYSQL must be installed. These products are open source³ products.
3. The final application must be packaged in a set up program, so that the products can be easily installed on ATM machines. This application must be networked to corresponding banks.

5. Performance Requirements

The high and low temperature should not affect the performance of the device. An uninterrupted transaction must be performed.

6. Design Constraints None.

7. Non Functional Requirements

7.1 Security

The ATM system must be in separate cabin.

Its door must have ATM card swipe slot so that it can be opened to only authentic user.

There must be emergency phone located outside the cabin.

Some security watchman should be at ATM center

7.2 Reliability

The application should be highly reliable and it should generate all the updated information in correct order.

7.3 Availability

Any information about the account should be quickly available from any computer to the authorized user. The previously visited customer's data must not be cleared.

7.4 Maintainability

The application should be maintainable in such a manner that if any new requirement occurs then it should be easily incorporated in an individual module.

7.5 Portability

The application should be portable on any windows based system.

1.4 SYSTEM REQUIREMENT SPECIFICATION FOR RAILWAY TICKET RESERVATION SYSTEM

1.Introduction

The introduction of the Software Requirements Specification (SRS) provides an overview of the entire SRS purpose, scope, definitions, acronyms, abbreviations, references and overview of SRS. A Software Requirements Specification (SRS) - a requirements specification for a software system - is a complete description of the behaviour of a system to be developed. It includes a set of use cases that describe all the interactions the users will have with the software. Use cases are also known as functional requirements. In addition to use cases, the SRS also contains non-functional (or supplementary) requirements. Non-functional requirements are requirements which impose constraints on the design or implementation (such as performance engineering requirements, quality standards, or design constraints). The aim of this document is to gather and analyse and give an in-depth insight of the complete Marvel Electronics and Home Entertainment software system by defining the problem statement in detail. This is a documentation of the project Railways Reservation System done sincerely and satisfactorily by my group members. A Software has to be developed for automating the manual Railway Reservation System.

- RESERVE SEATS – Reservation form has to be filled by passenger. If seats are available entries like train name, number, destination are made.
- CANCEL RESERVATION- The clerk deletes the entry in the System and changes in the Reservation Status.
- VIEW RESERVATION STATUS-The user need to enter the PIN number printed on ticket.

1.1 Objective:

The purpose of this source is to describe the railway reservation system which provides the train timing details, reservation, billing and cancellation on various types of reservation namely,

- Confirm Reservation for confirm Seat.
- Reservation against Cancellation.
- Waiting list Reservation.
- Online Reservation.
- Tatkal Reservation.

The origin of most software systems is in the need of a client, who either wants to automate the existing manual system or desires a new software system. The software system is itself created by the developer. Finally, the end user will use the completed system. Thus, there are three major parties interested in a new system: the client, the user, and the developer. Somehow the requirements for the system that will satisfy the needs of the clients and the concerns of the users have to be communicated to the developer. The problem is that the client doesn't usually design the software or the software development process and the developer does not understand the client's problem and the application area. This causes a communication gap between the parties involved in the development of the project.

The basic purpose of Software Requirement Specification (SRS) is to bridge this communication gap. SRS is the medium through which the client's and the user's needs are accurately specified; indeed SRS forms the basis of software development.

Another important purpose of developing an SRS is helping the clients understanding their own needs. An SRS establishes the basis for agreement between the client and the supplier on what the software product will do.

An SRS provides a reference for validation of the final product. A high quality SRS is a prerequisite to high quality software and it also reduces the development cost.

A few factors that direct us to develop a new system are given below -:

1. Faster System
2. Accuracy
3. Reliability
4. Informative
5. Reservations and cancellations from anywhere to any place

1.2 Scope:

“Railways Reservation System” is an attempt to simulate the basic concepts of an online Reservation system. The system enables to perform the following functions:

- SEARCH FOR TRAIN
- BOOKING OF A SELECTED FLIGHT
- PAYMENT
- CANCELLATION
- Freight Revenue enhancement
- Passenger Revenue enhancement
- Improved & optimized service

1.3 Glossary:

This should define all technical terms and abbreviations used in the document

- NTES – National Train Enquiry System
- IVRS – Interactive Voice Response system
- PRS – passenger reservation system
- DFD :- Data Flow Diagram
- ERD :- Entity Relationship Diagram
- SRS :- Software Requirements Specification
- STD :- State Transition Diagram

1.4 Overview:

The remaining sections of this document provide a general description, including characteristics of the users of this project, the product's hardware, and the functional and data requirements of the product. General description of the project is discussed in section 2 of this document. Section 3 gives the functional requirements, data requirements and constraints and assumptions made while designing the E-Store. It also gives the user viewpoint of product. Section 3 also gives the specific requirements of the product. Section 3 also discusses Section 4 is for supporting information. the external interface requirements and gives detailed description of functional requirements.

2. Overall Description

This document contains the problem statement that the current system is facing which is hampering the growth opportunities of the company. It further contains a list of the stakeholders and users of the proposed solution. It also illustrates the needs and wants of the stakeholders that were identified in the brainstorming exercise as part of the requirements workshop. It further lists and briefly describes the major features and a brief description of each of the proposed system.

2.1 Product Perspective:

Before the automation, the system suffered from the following DRAWBACKS:

- Ø The existing system is highly manual involving a lot of paper work and calculation and therefore may be erroneous. This has lead to inconsistency and inaccuracy in the maintenance of data.
- Ø The data, which is stored on the paper only, may be lost, stolen or destroyed due to natural calamity like fire and water.
- Ø The existing system is sluggish and consumes a lot of time causing inconvenience to customers and the airlines staff.
- Ø Due to manual nature, it is difficult to update, delete, add or view the data.
- Ø Since the number of passengers have drastically increased therefore maintaining and retrieving detailed record of passenger is extremely difficult.
- Ø An railways has many offices around the world, an absence of a link between these offices lead to lack of coordination and communication.

Hence the railways reservation system is proposed with the following

- Ø The computerization of the reservation system will reduce a lot of paperwork and hence the load on the airline administrative staff.
- Ø The machine performs all calculations. Hence chances of error are nil.
- Ø The passenger, reservation, cancellation list can easily be retrieved and any required addition, deletion or updation can be performed.
- Ø The system provides for user-ID validation, hence unauthorized access is prevented.

2.2 Project Functions:

Booking agents with varying levels of familiarity with computers will mostly use this system. With this in mind, an important feature of this software is that it be relatively simple to use. The scope of this project encompasses: -

“ Search: This function allows the booking agent to search for train that are available between the two travel cities, namely the "Departure city" and "Arrival city" as desired by the traveller. The system initially prompts the agent for the departure and arrival city, the date of departure, preferred time slot and the number of passengers. It then displays a list of train available with different airlines between the designated cities on the specified date and time.

“ Selection: This function allows a particular train to be selected from the displayed list. All the details of the train are shown :-

1. train Number
2. Date, time and place of departure
3. Date, time and place of arrival
4. TRAIN Duration
5. Fare per head
6. Number of stoppages – 0, 1, 2...

“ Review: If the seats are available, then the software prompts for the booking of train. The train information is shown. The total fare including taxes is shown and flight details are reviewed.

“ Traveller Information: It asks for the details of all the passengers supposed to travel including name, address, telephone number and e-mail id.

“ Payment: It asks the agent to enter the various credit card details of the person making the reservation.

1. Credit card type
2. Credit card number
3. CVC number of the card
4. Expiration date of the card
5. The name on the card

“ Cancellation : The system also allows the passenger to cancel an existing reservation. This function registers the information regarding a passenger who has requested for a cancellation of his/her ticket. It includes entries pertaining to the train No., Confirmation No., Name, Date of Journey, Fare deducted.

2.3 User Characteristics:

- Ø EDUCATIONAL LEVEL:-At least user of the system should be comfortable with English language.
- Ø TECHNICAL EXPERTISE: - User should be comfortable using general purpose applications on the computer system.

2.4 Constrains:

Software constraints:

- Ø The system will run under windows98 or higher platforms of operating system.

2.5 Assumptions and Dependencies:

Ø Booking Agents will be having a valid user name an password to access the software

Ø The software needs booking agent to have complete knowledge of railways reservation system.

Ø Software is dependent on access to internet.

3.1 Function Requirements

3.1.1 performance requirements:

- User Satisfaction: - The system is such that it stands up to the user expectations.
- Response Time: -The response of all the operation is good. This has been made possible by careful programming.
- Error Handling: - Response to user errors and undesired situations has been taken care of to ensure that the system operates without halting.
- Safety and Robustness: - The system is able to avoid or tackle disastrous action. In other words, it should be foul proof. The system safeguards against undesired events, without human intervention.
- Portable: - The software should not be architecture specific. It should be easily transferable to other platforms if needed.
- User friendliness: - The system is easy to learn and understand. A native user can also use the system effectively, without any difficulties.

3.1.2 Design constrain :

There are a number of factors in the client's environment that may restrict the choices of a designer. Such factors include standards that must be followed, resource limits, operating environment, reliability and security requirements and policies that may have an impact on the design of the system. An SRS (Software Requirements Analysis and Specification) should identify and specify all such constraints.

Ø Standard Compliance: - This specifies the requirements for the standards the system must follow. The standards may include the report format and accounting properties.

Ø Hardware Limitations :- The software may have to operate on some existing or predetermined hardware, thus imposing restrictions on the design. Hardware limitations can include the types of machines to be used, operating system available on the system, languages supported and limits on primary and secondary storage.

Ø Reliability and Fault Tolerance: - Fault tolerance requirements can place a major constraint on how the system is to be designed. Fault tolerance requirements often make the system more complex and expensive. Requirements about system behaviour in the face of certain kinds of faults are specified. Recovery requirements are often an integral part here, detailing what the system should do if some failure occurs to ensure certain properties. Reliability requirements are very important for critical applications.

Ø Security: - Security requirements are particularly significant in defence systems and database systems. They place restrictions on the use of certain commands, control access to data, provide different kinds of access requirements for different people, require the use of passwords and cryptography techniques and maintain a log of activities in the system.

3.1.3 Hardware requirements:

For the hardware requirements the SRS specifies the logical characteristics of each interface b/w the software product and the hardware components. It specifies the hardware requirements like memory restrictions, cache size, the processor, RAM size etc... those are required for the software to run.

Minimum Hardware Requirements

Processor Pentium III

Hard disk drive 40 GB

RAM 128 MB

Cache 512 kb

Preferred Hardware Requirements

Processor Pentium IV

Hard disk drive 80 GB

RAM 256 MB

Cache 512 kb

3.1.4 Software requirements: • Any window based operating system with DOS support are primary requirements for software development. Windows XP, FrontPage and dumps are required. The systems must be connected via LAN and connection to internet is mandatory.

3.1.5 other requirements:

Software should satisfy following requirements as well:-

- SECURITY
- Ø PORTABILITY
- Ø CORRECTNESS
- Ø EFFICIENCY
- Ø FLEXIBILITY
- Ø TESTABILITY
- Ø REUSABILITY

3.2 Non-Function Requirements

3.2.1 Security:

- The system use SSL (secured socket layer) in all transactions that include any confidential customer information. The system must automatically log out all customers after a period of inactivity. The system should not leave any cookies on the customer's computer containing the user's password. The system's back-end servers shall only be accessible to authenticated management.

3.2.2 Reliability:

- The reliability of the overall project depends on the reliability of the separate components.

The main pillar of reliability of the system is the backup of the database which is continuously maintained and updated to reflect the most recent changes. Also the system will be functioning inside a container. Thus the overall stability of the system depends on the stability of container and its underlying operating system.

3.2.3 Availability:

- The system should be available at all times, meaning the user can access it using a web browser, only restricted by the down time of the server on which the system runs. A customer friendly system which is in access of people around the world should work 24 hours. In case of a of a hardware failure or database corruption, a replacement page will be shown. Also in case of a hardware failure or database corruption, backups of the database should be retrieved from the server and saved by the Organizer. Then the service will be restarted. It means 24 x 7 availability.

3.2.4 Maintainability:

- A commercial database is used for maintaining the database and the application server takes care of the site. In case of a failure, a re-initialization of the project will be done. Also the software design is being done with modularity in mind so that maintainability can be done efficiently.

3.2.5 Supportability:

- The code and supporting modules of the system will be well documented and easy to understand. Online User Documentation and Help System Requirements.

AIM:

Develop DFD model (level-0, level-1 DFD and Data dictionary) of the project.

DESCRIPTION:

Data drive business activities and can trigger events (e.g. new sales order data) or be processed to provide information about the activity. Data flow analysis, as the name suggests, follows the flow of data through business processes and determines how organisation objectives are accomplished. In the course of handling transactions and completing tasks, data are input, processed, stored, retrieved, used, changed and output. Data flow analysis studies the use of data in each activity and documents the findings in data flow diagrams, graphically showing the relation between processes and data.

Physical and Logical DFDs:

There are two types of data flow diagrams, namely physical data flow diagrams and logical dataflow diagrams and it is important to distinguish clearly between the two:

Physical Data Flow Diagrams:

An implementation-dependent view of the current system, showing what tasks are carried out and how they are performed. Physical characteristics can include:

Names of people, Form and document names or numbers, Master and transaction files, Equipment and devices used.

Logical Data Flow Diagrams:

An implementation-independent view of the system, focusing on the flow of data between processes without regard for the specific devices, storage locations or people in the system. The physical characteristics listed above for physical data flow diagrams will not be specified.

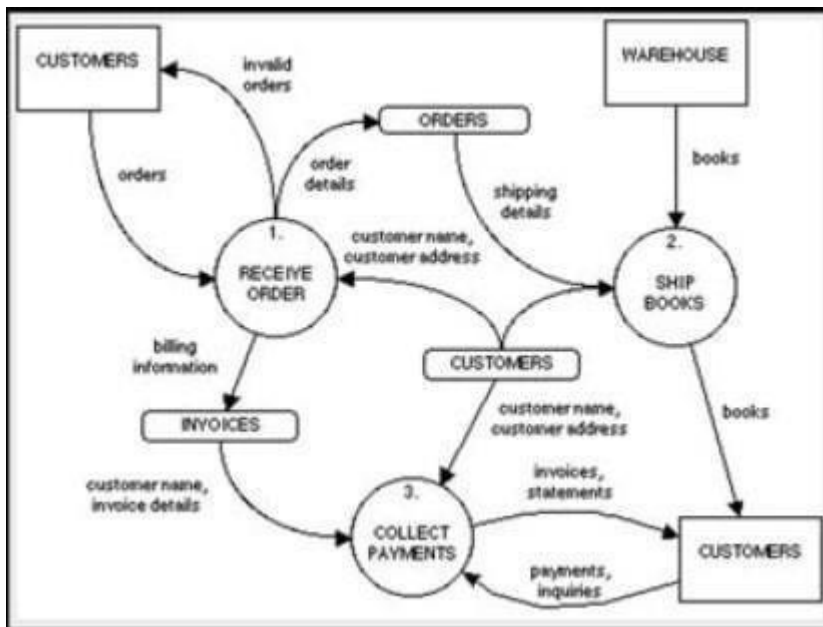


Fig. A typical DFD

Data Flow Diagram (DFD)

The DFD (also known as a bubble chart) is a hierarchical graphical model of a system that shows the different processing activities or functions that the system performs and the data interchange among these functions. Each function is considered as a processing station (or process) that consumes some input data and produces some output data. The system is represented in terms of the input data to the system, various processing carried out on these data, and the output data generated by the system. A DFD model uses a very limited number of primitive symbols to represent the functions performed by a system and the data flow among these functions. Symbols used for designing DFDs

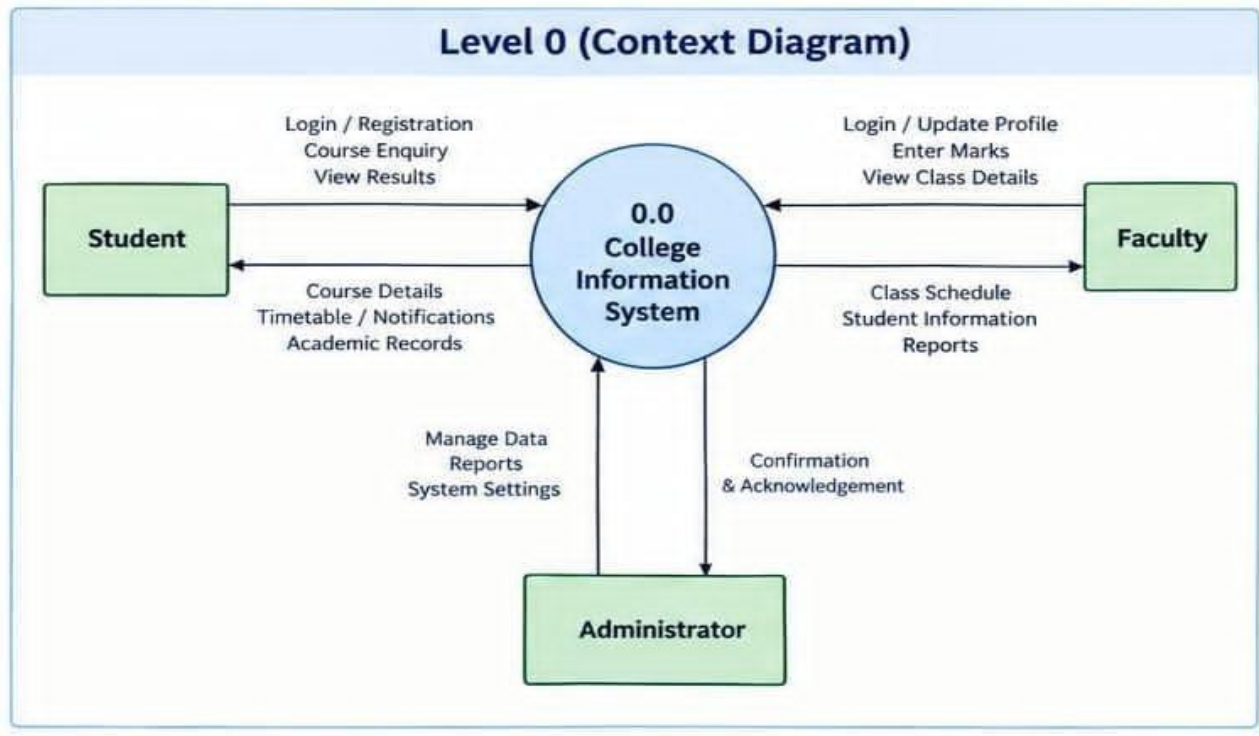
Here, two examples of data flow that describe input and validation of data are considered.

In Fig , the two processes are directly connected by a data flow. This means that the ‘validate-number’ process can start only after the ‘read-number’ process had supplied data to it. However in Fig , the two processes are connected through a data store. Hence, the operations of the two bubbles are independent. The first one is termed ‘synchronous’ and the second one ‘asynchronous’.

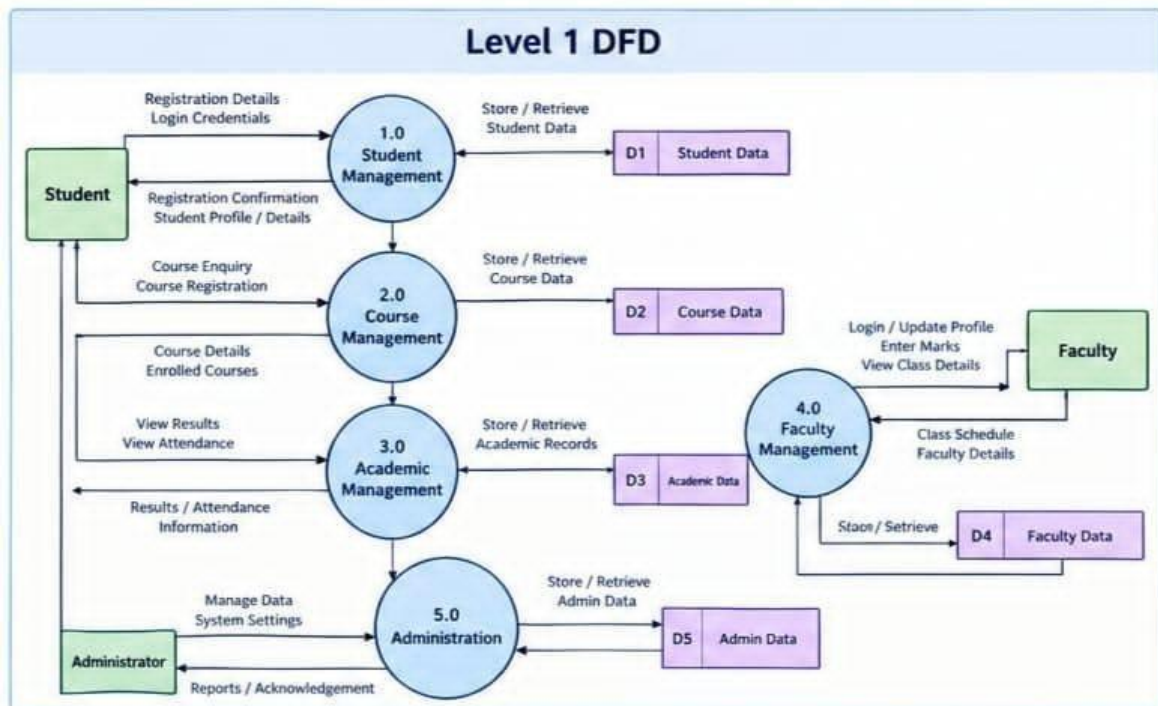
Importance of DFDs in a good software design

The main reason why the DFD technique is so popular is probably because of the fact that DFD is a very simple formalism – it is simple to understand and use. Starting with a set of high-level functions that a system performs, a DFD model hierarchically represents various sub- functions. In fact, any hierarchical model is simple to understand. Human mind is such that it can easily understand any hierarchical model of a system– because in a hierarchical model, starting with a very simple and abstract model of a system, different details of the system are slowly introduced through different hierarchies. The data flow diagramming technique also follows a very simple set of intuitive concepts and rules. DFD is an elegant modelling technique that turns out to be useful not only to represent the results of structured analysis of a software problem, but also for several other applications such as showing the flow of documents or items in an organization

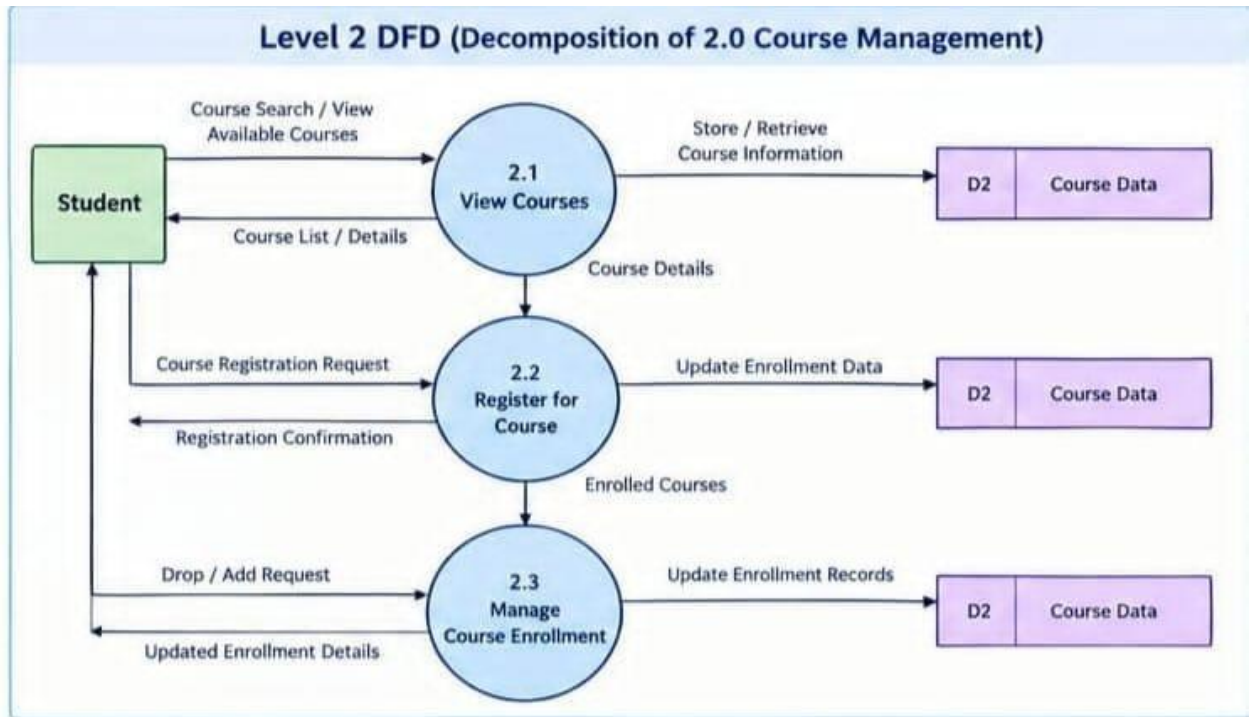
ZERO LEVEL DFD



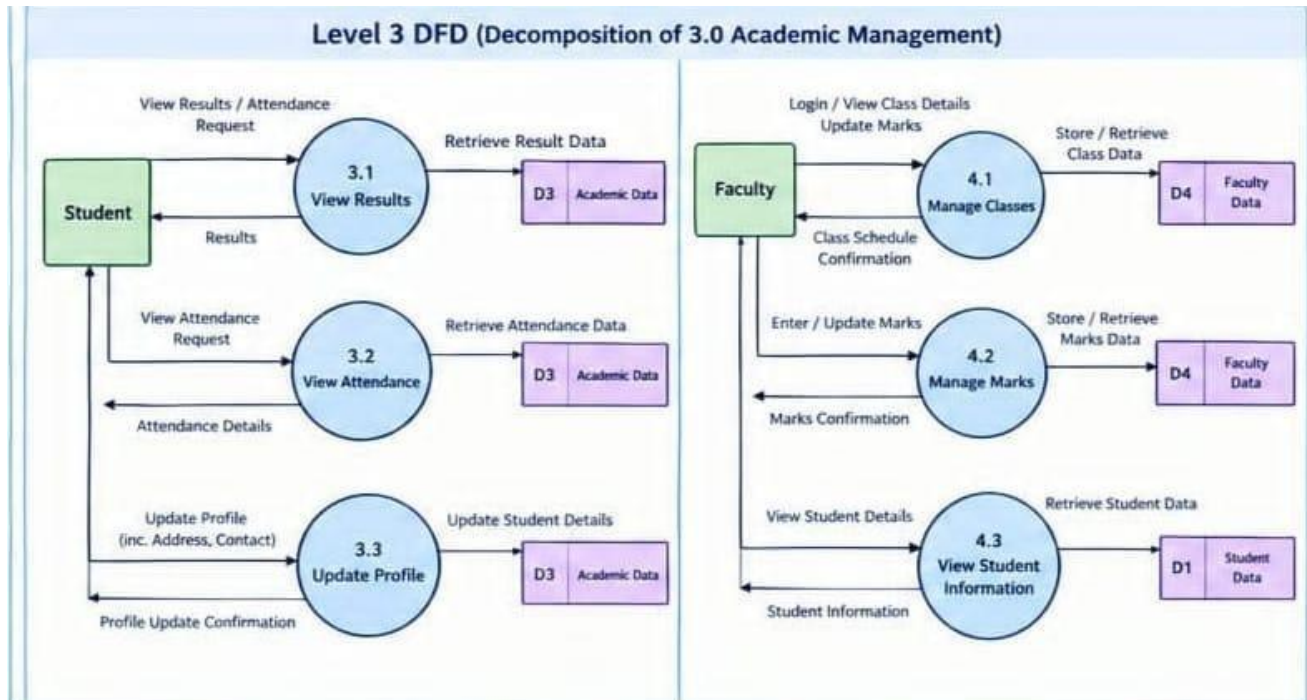
FIRST LEVEL DFD



SECOND LEVEL DFD

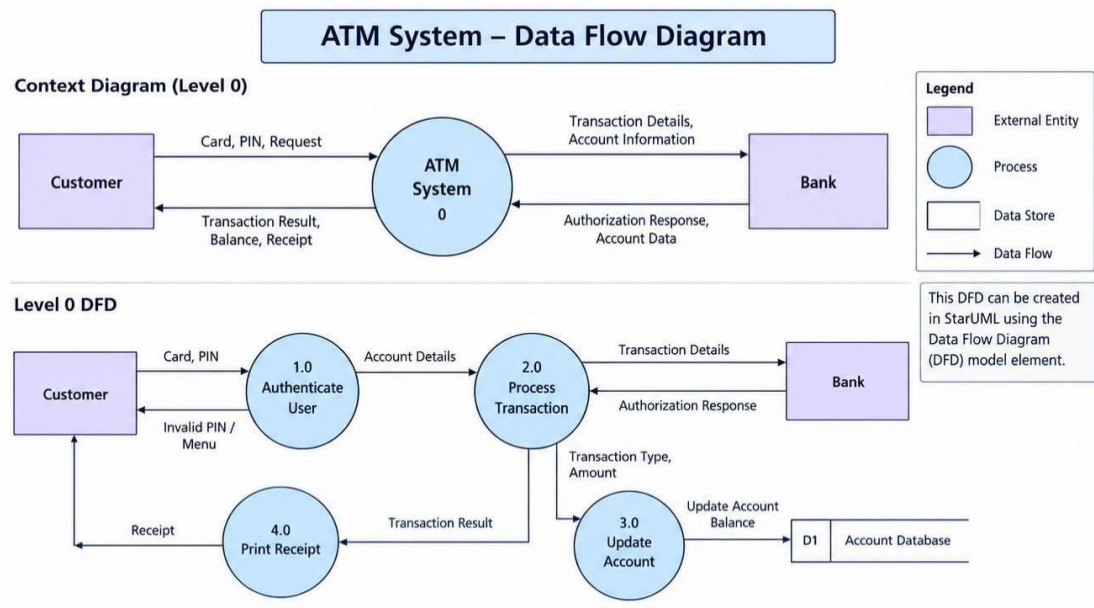


THIRD LEVEL DFD



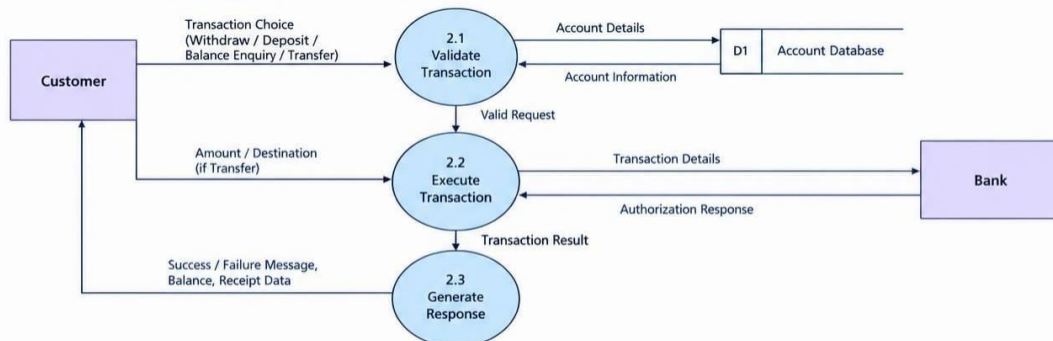
Ex.NO: 2.2 DFD OF BANKING MANAGEMENT SYSYTEM

ZERO LEVEL DFD

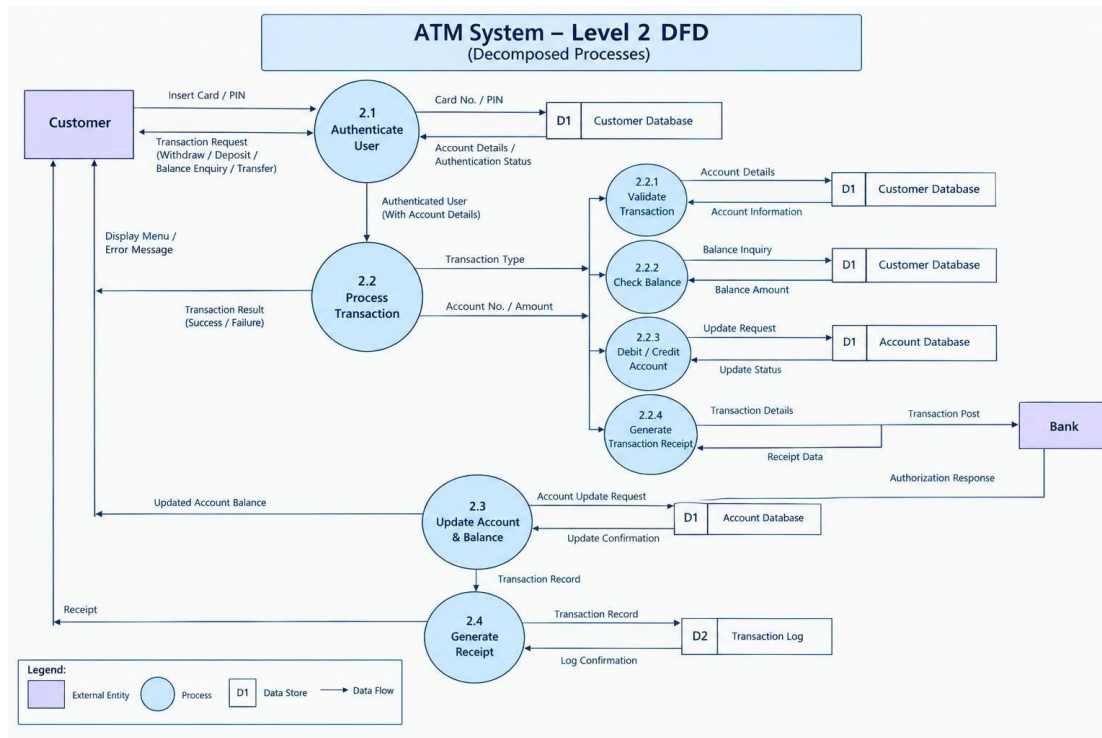


FIRST LEVEL DFD

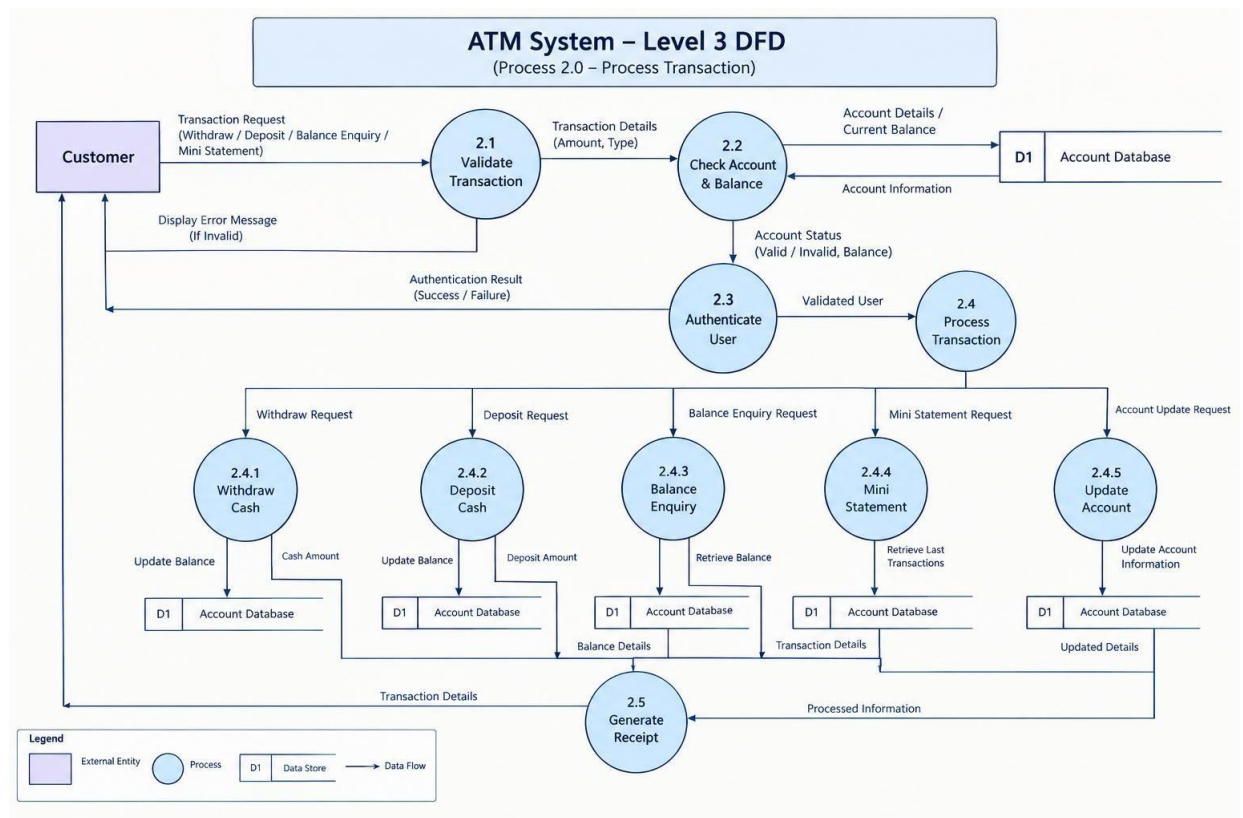
Level 1 DFD – Process 2.0 (Process Transaction)



SECOND LEVEL DFD

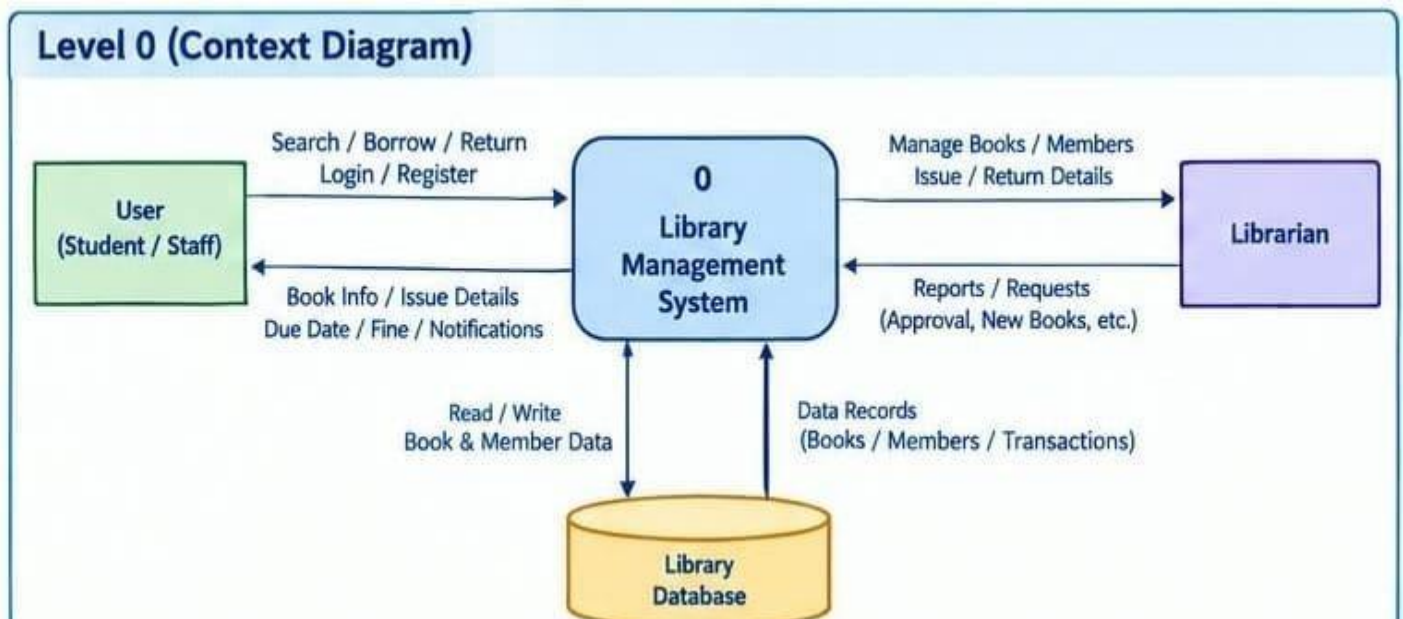


THIRD LEVEL DFD

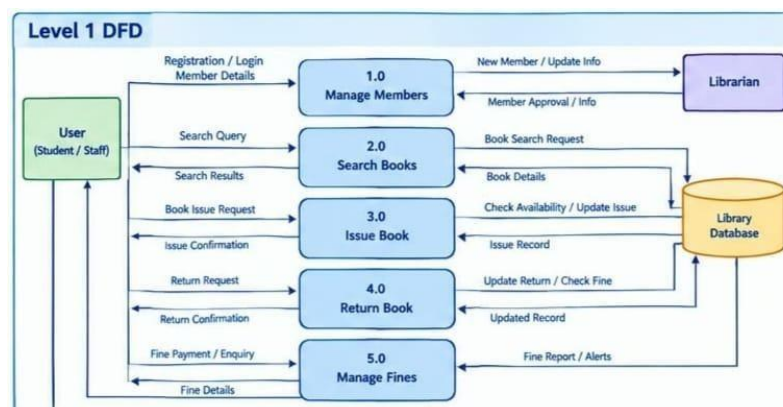


Ex.NO: 2.3 DFD OF LIBRARY MANAGEMENT SYSYTEM

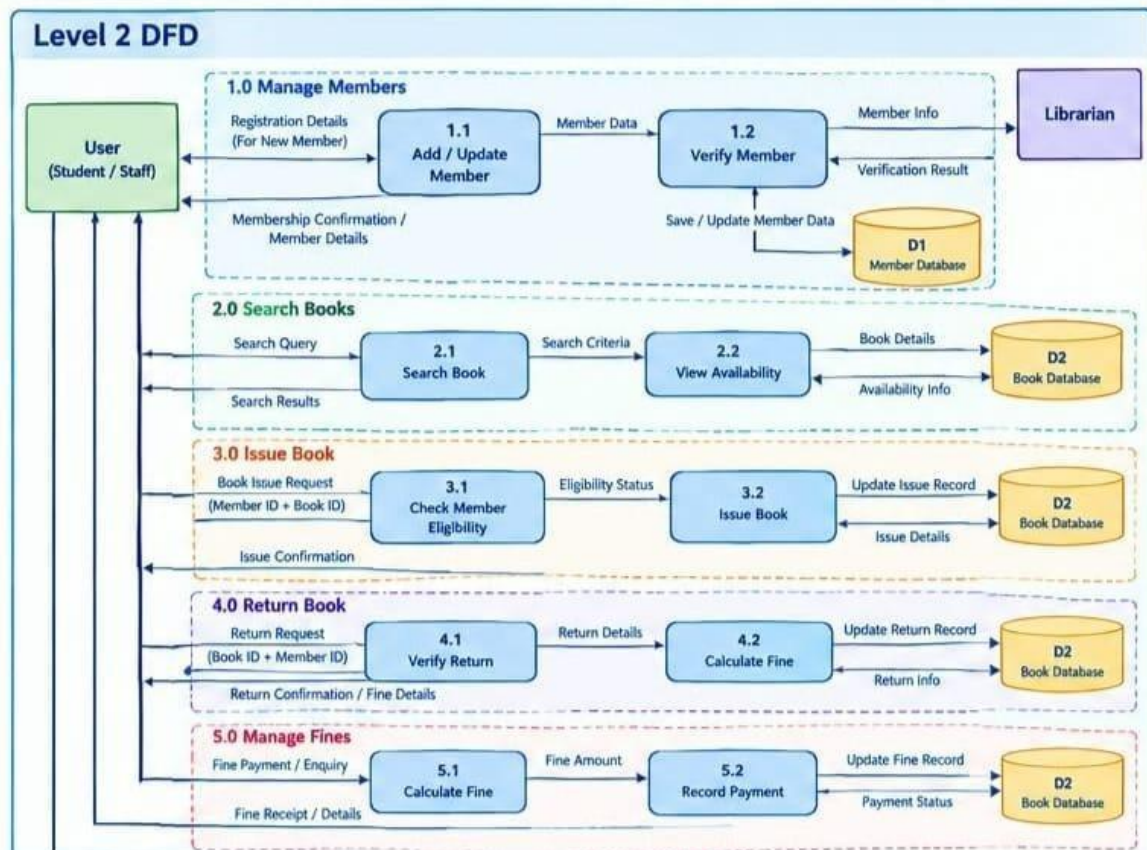
ZERO LEVEL DFD



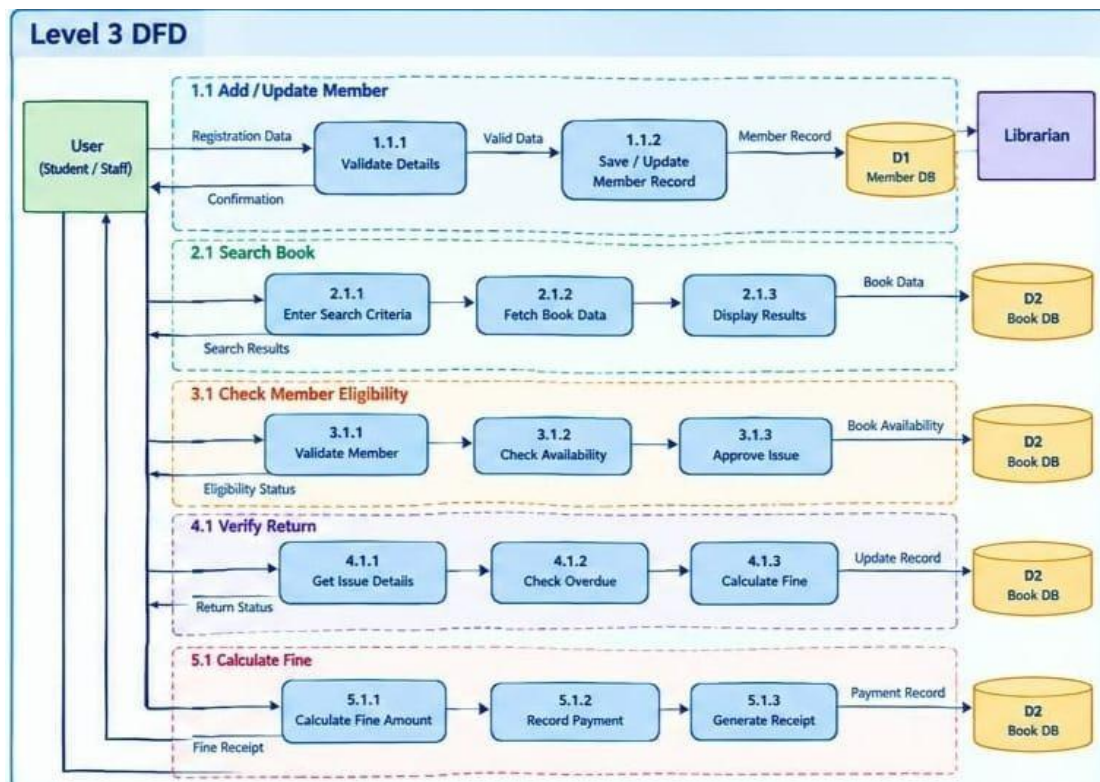
FIRST LEVEL DFD



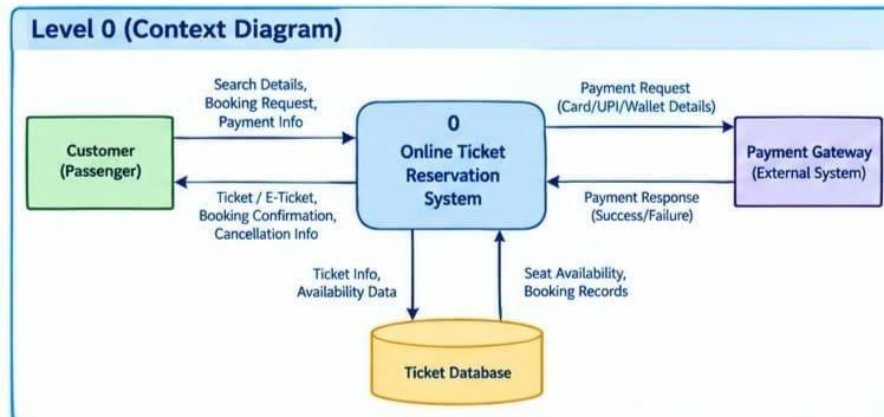
SECOND LEVEL DFD



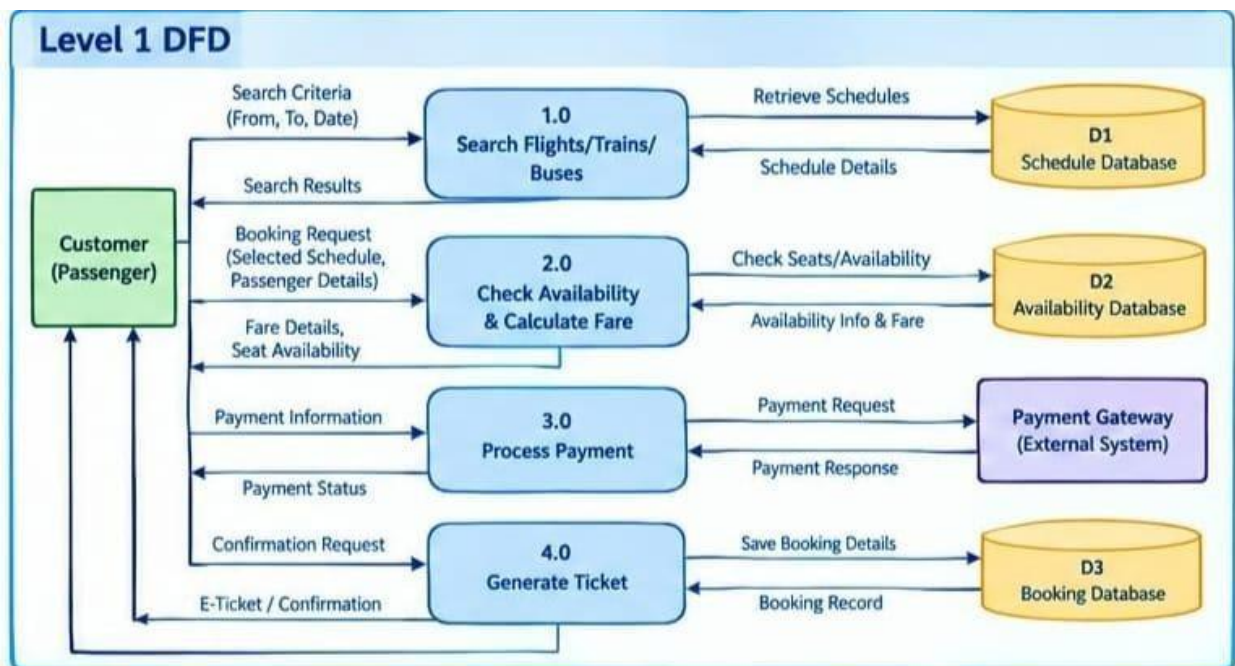
THIRD LEVEL DFD



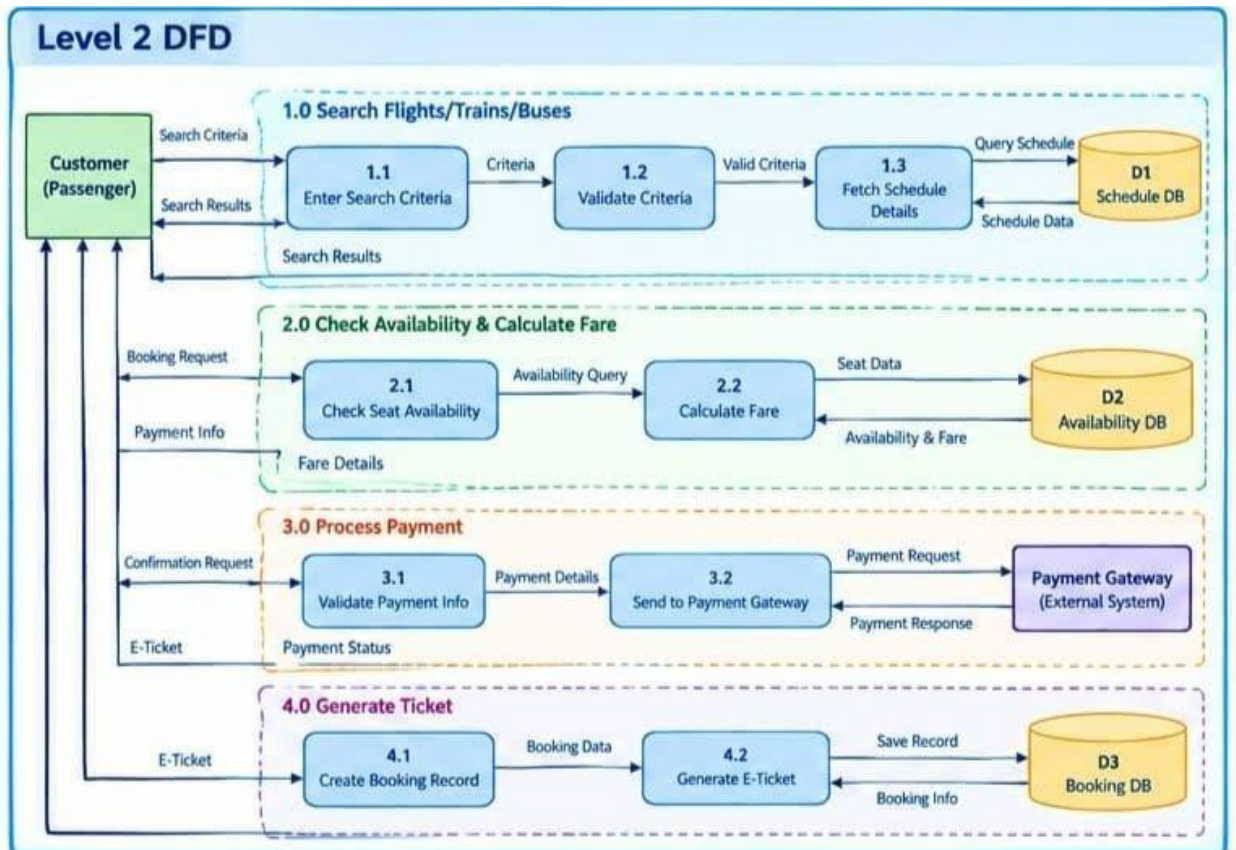
ZERO LEVEL DFD



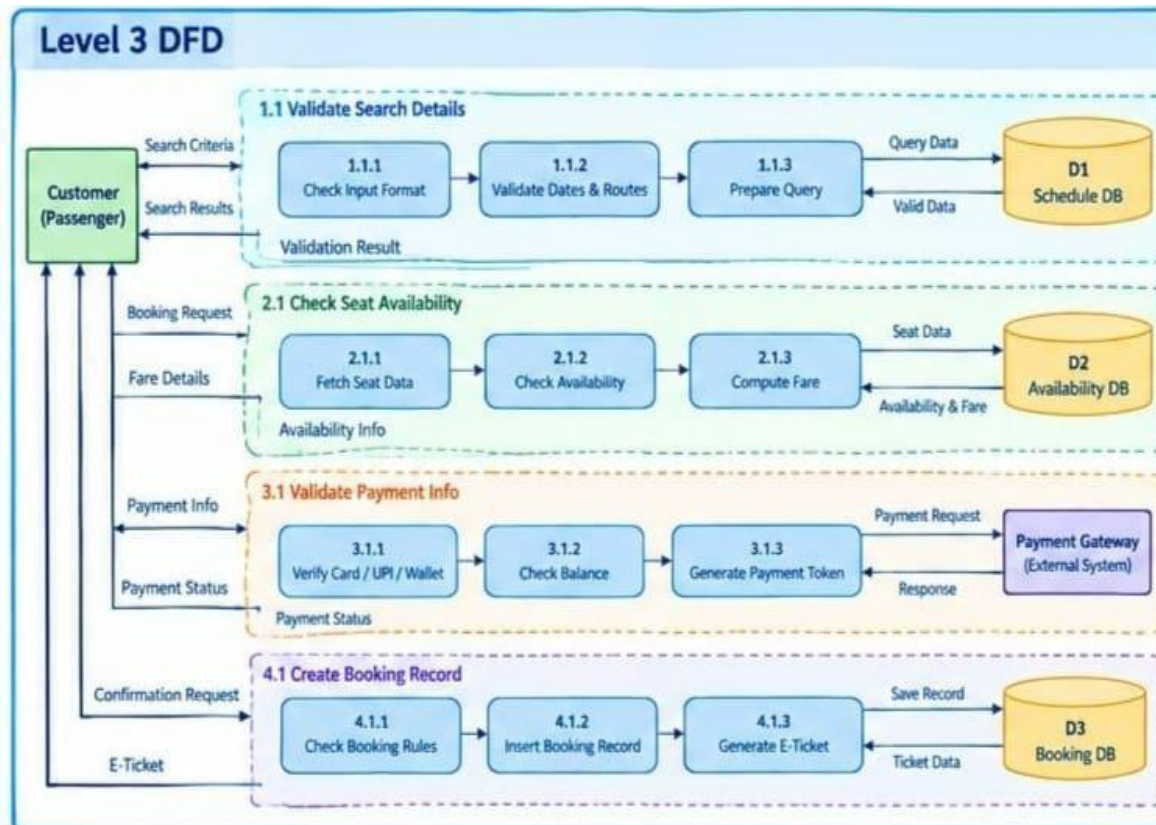
FIRST LEVEL DFD



SECOND LEVEL DFD



THIRD LEVEL DFD



To show diagrammatically the objects required and the relationships between them while developing a software product.

Software Required :-

Visual Paradigm for UML

8.2

Procedure :-

Step 1:-

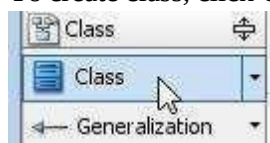
Right click Class Diagram on Diagram Navigator and select New Class Diagram from the popup menu to create a class diagram.



Step 2:-

Creating class

To create class, click Class on the diagram toolbar and then click on the diagram.

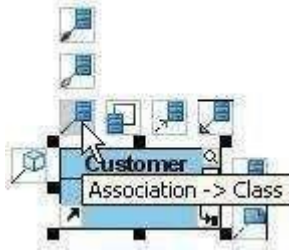


A class will be created.

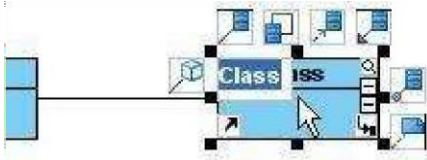


Creating association

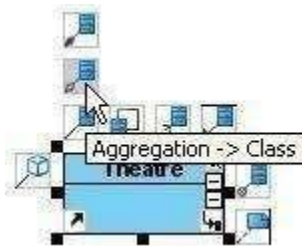
To create association from class, click the Association ->Class resource beside it and drag.



Drag to empty space of the diagram to create a new class, or drag to an existing class to connect to it. Release the mouse button to create the association.

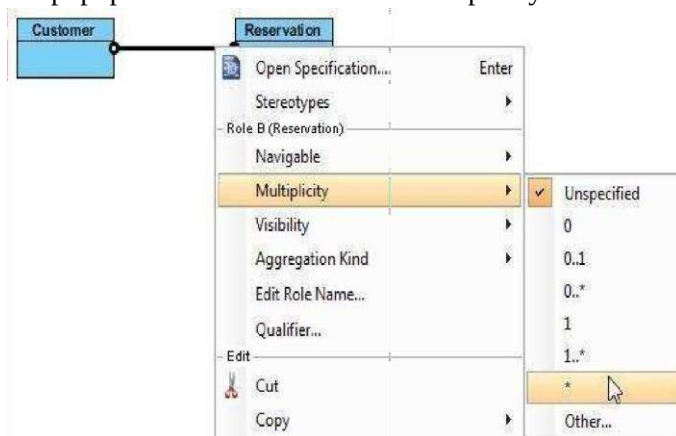


To create aggregation, use the Aggregation ->Class resource instead.

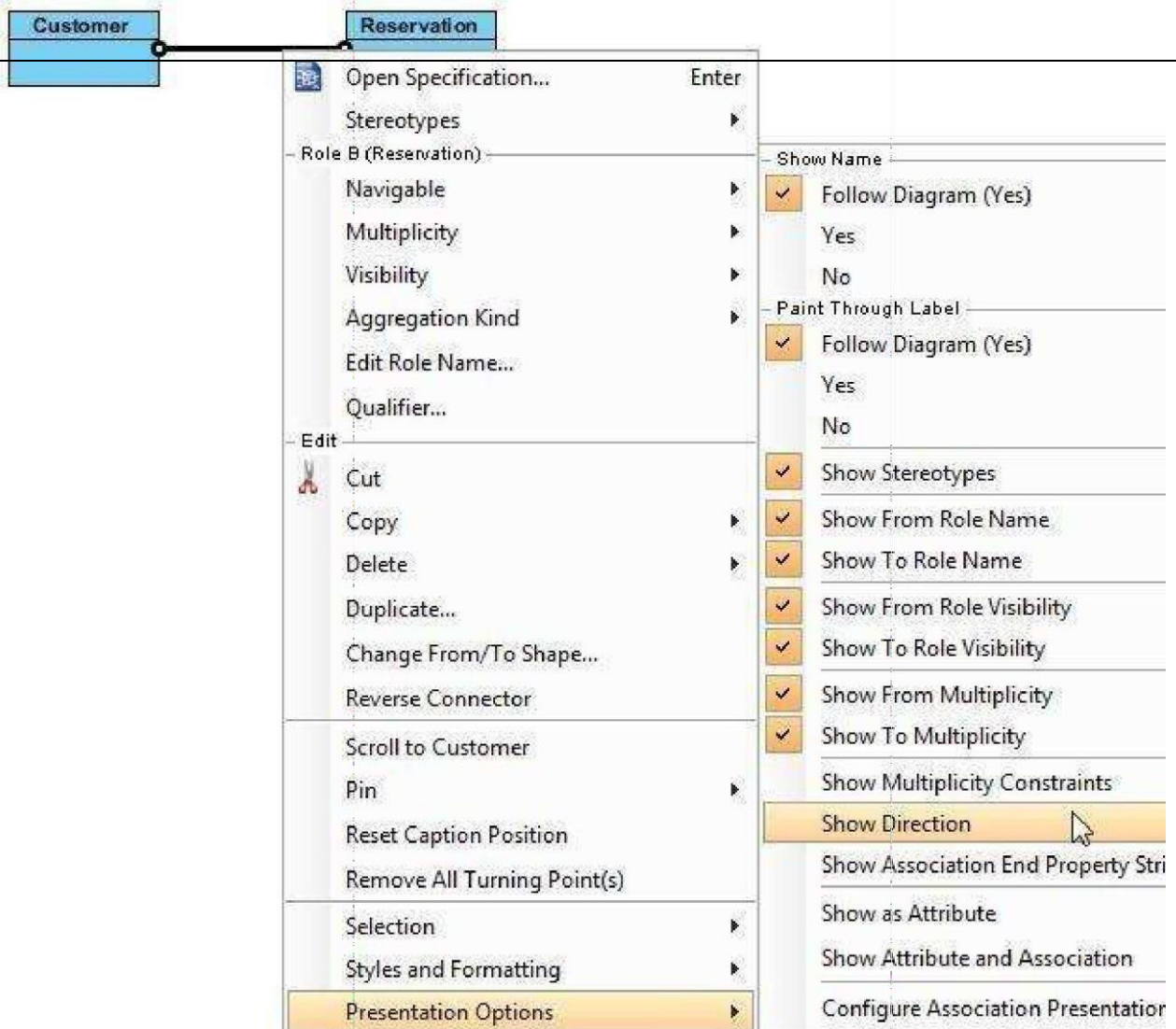


Step 3:-

To edit multiplicity of an association end, right-click near the association end, select Multiplicity from the popup menu and then select a multiplicity.

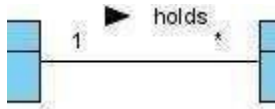


To show the direction of an association, right click on it and select Presentation Options >Show Direction from the pop-up menu.



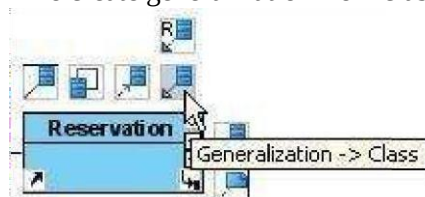
Step 4:-

The direction arrow is shown beside the association.

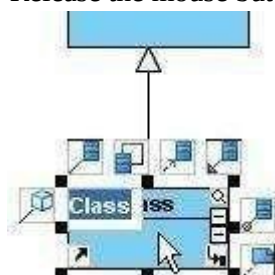


Creating generalization

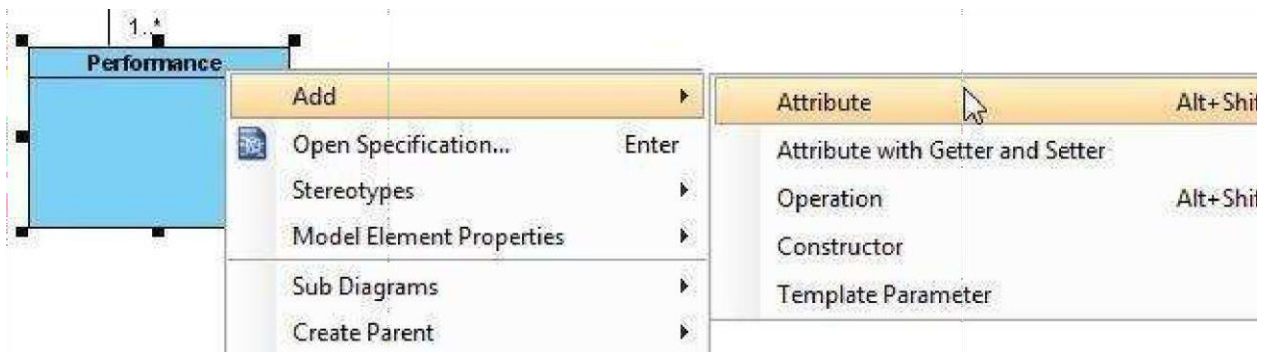
To create generalization from class, click the Generalization -> Class resource beside it and drag.



Drag to empty space of the diagram to create a new class, or drag to an existing class to connect to it. Release the mouse button to create the generalization.

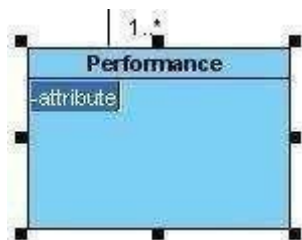


Creating attribute



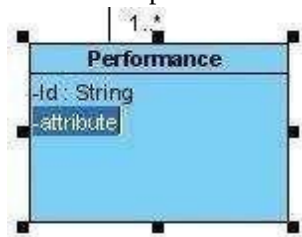
To create attribute, right click the class and select Add >Attribute from the pop-up menu.

An attribute is created.



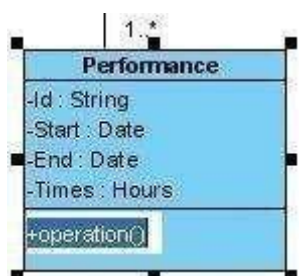
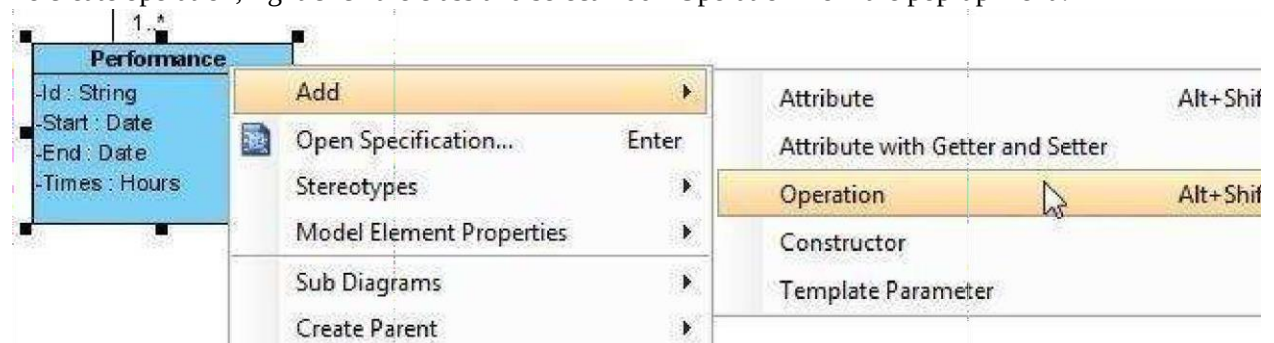
Creating attribute with enter key

After creating an attribute, press the Enter key, another attribute will be created. This method lets you create multiple attributes quickly and easily.



Creating operation

To create operation, right click the class and select Add >Operation from the pop-up menu.



An operation is created.

Similar to creating attribute, you can press the Enter key to create multiple operations continuously.
Drag-and-Drop reordering, copying and moving of class members

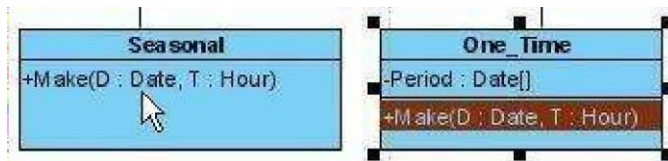


To reorder a class member, select it and drag within the compartment, you will see a thick black line appears indicating where the class member will be placed.

Release the mouse button, the class member will be reordered.

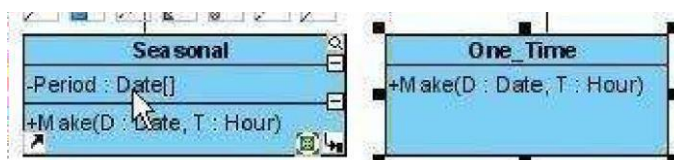


To copy a class member, select it and drag to the target class while keep pressing the Ctrl key, you will see a thick black line appears indicating where the class member will be placed. A plus sign is shown beside the mouse cursor indicating this is a copy action.



Release the mouse button, the class member will be copied.

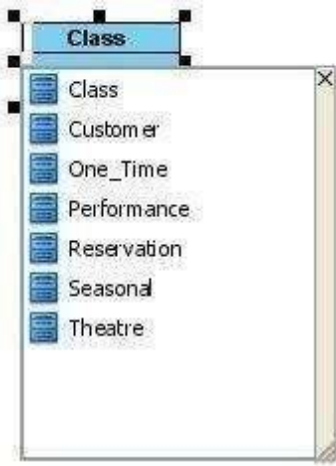
To move a class member, select it and drag to the target class, you will see a thick black line appears indicating where the class member will be placed. Unlike copy, do not press the Ctrl key when drag, the mouse cursor without the plus sign indicates this is a move action.



Release the mouse button, the class member will be moved.

Model name completion for class

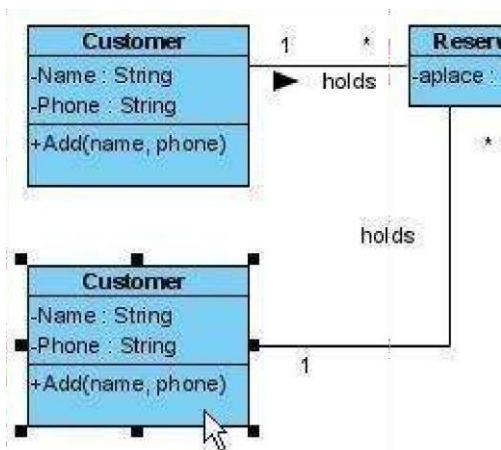
The model name completion feature enables quick creation of multiple views for the same class model. When create or rename class, the list of classes is shown.



Type text to filter classes in the list.

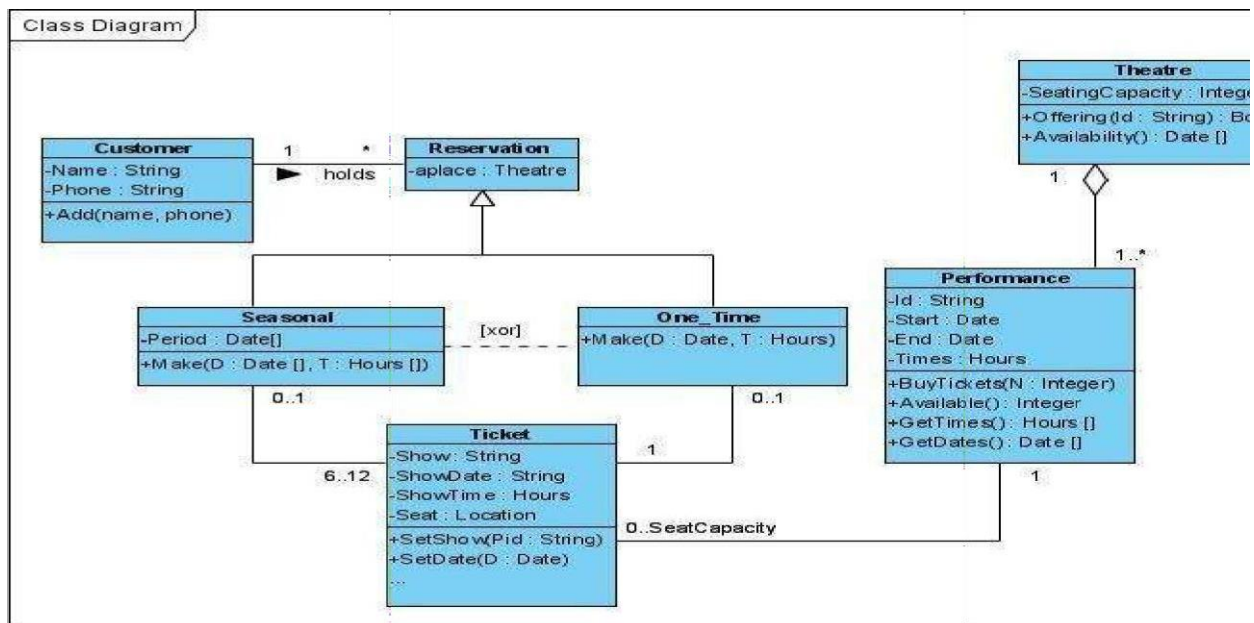


Press up or down key to select class in the list, press Enter to confirm. Upon selecting an existing class, all class members and relationships are shown immediately.



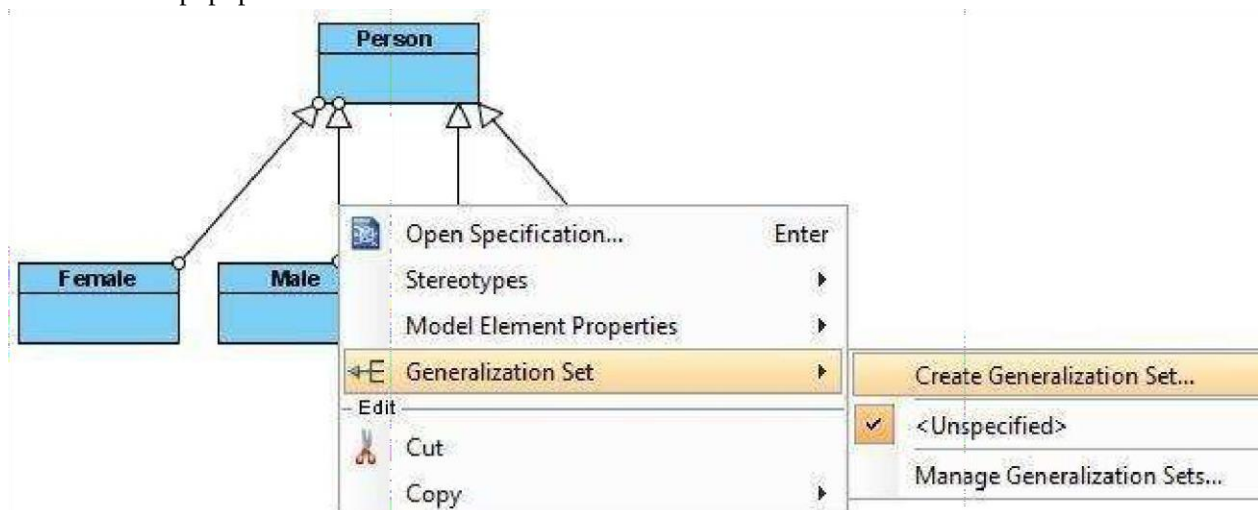
Step 5:-

Continue to complete the diagram.



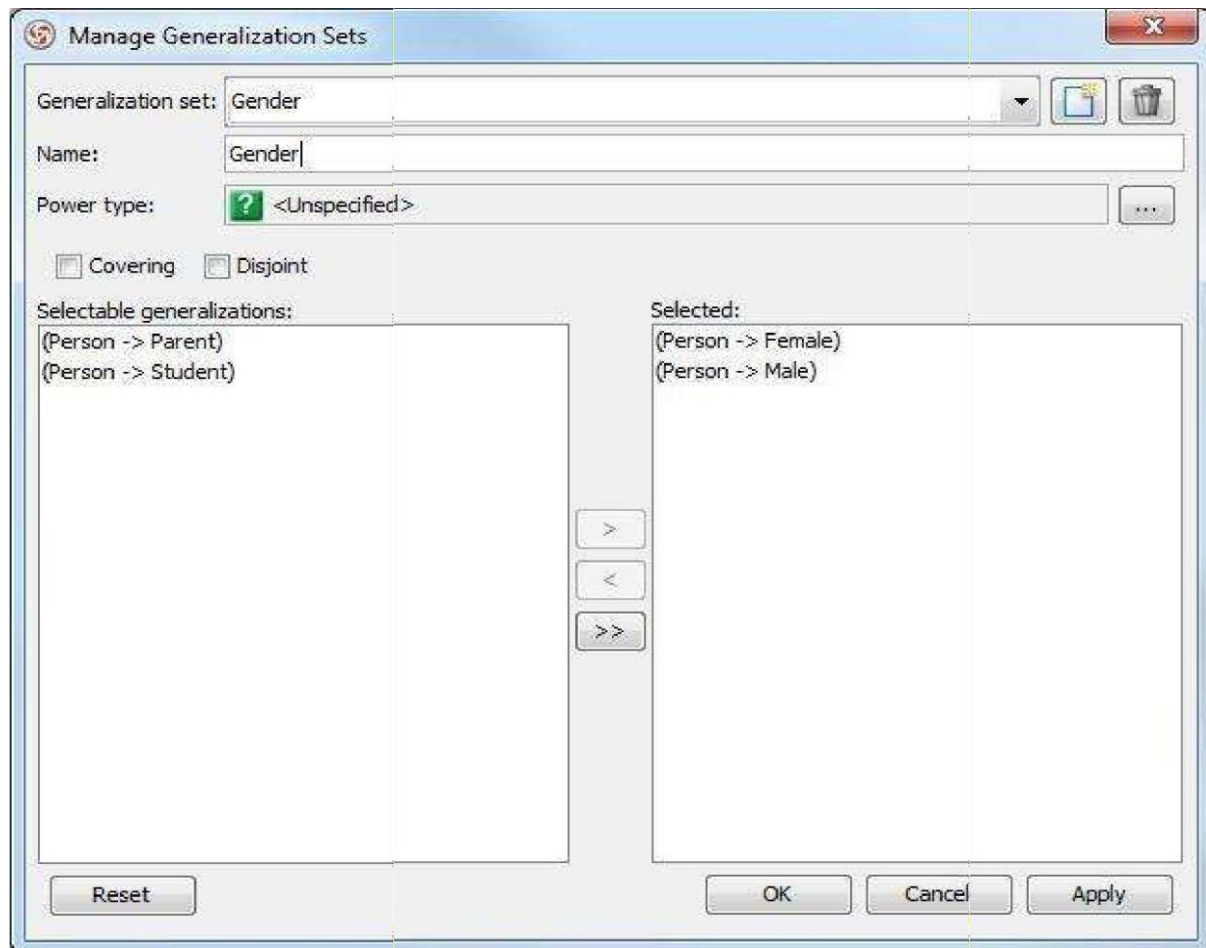
Generalization set

A generalization set defines a particular set of generalization relationships that describe the way in which a general classifier (or superclass) may be divided using specific subtypes. To define a generalization set, select the generalizations to include, right click and select Generalization set > Create Generalization Set... from the popup menu.

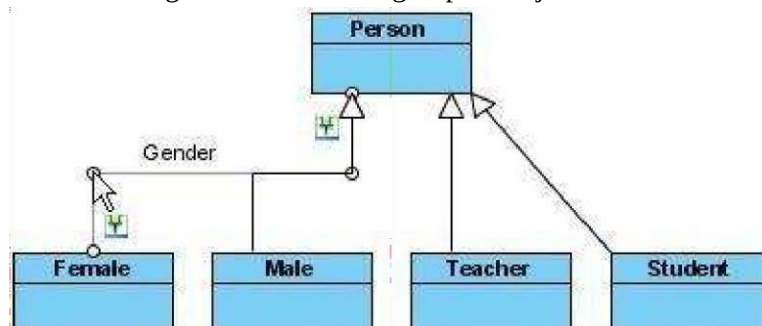


Step 6:-

Name the set in the Manage Generalization Sets dialog box, and confirm by pressing OK.

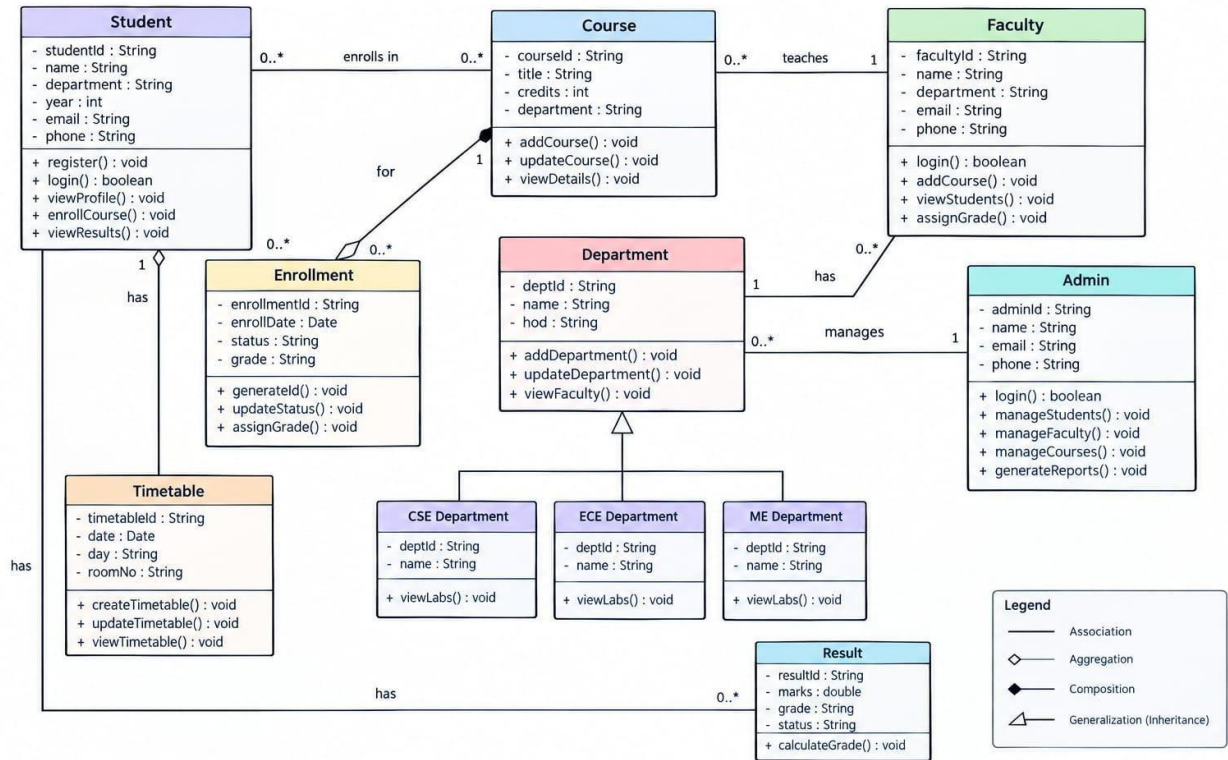


The selected generalizations are grouped. Adjust the connector to make the diagram tidy.

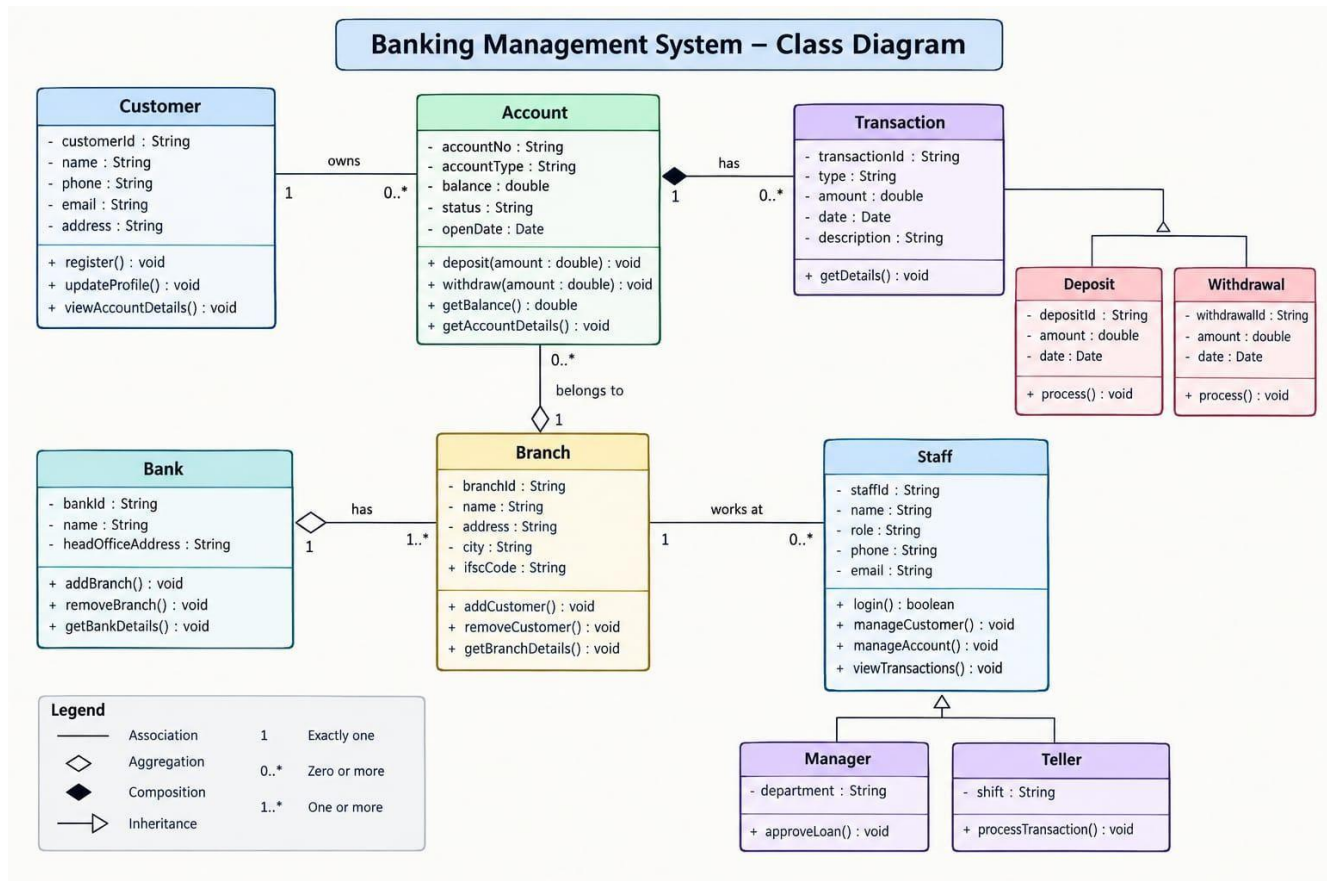


EX.NO. 3.1 CLASS DIAGRAM FOR COLLEGE AUTOMATION SYSTEM

College Automation System - Class Diagram

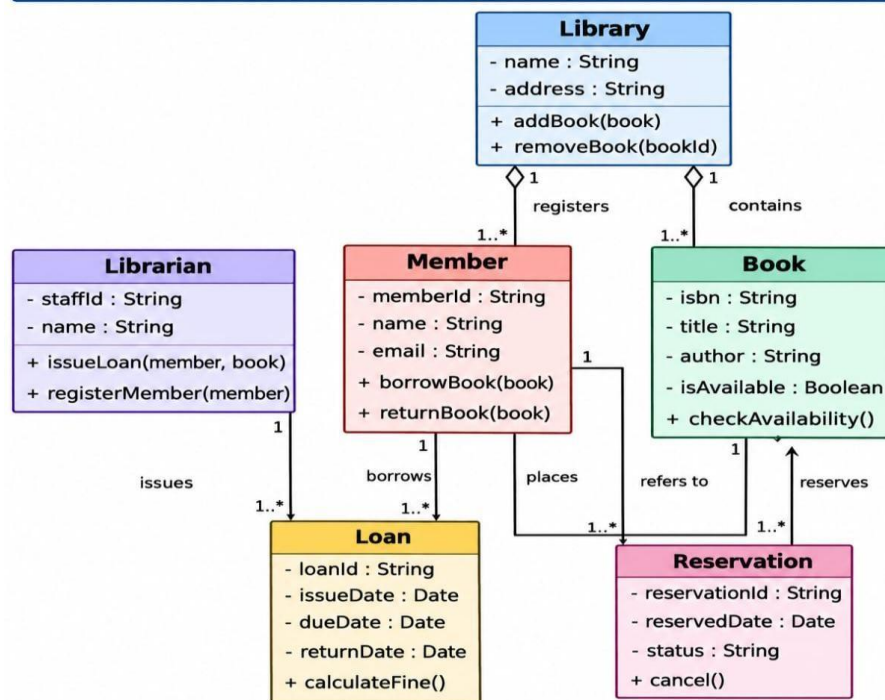


EX.NO.3.2 CLASS DIAGRAM FOR BANKING MANAGEMENT SYSTEM



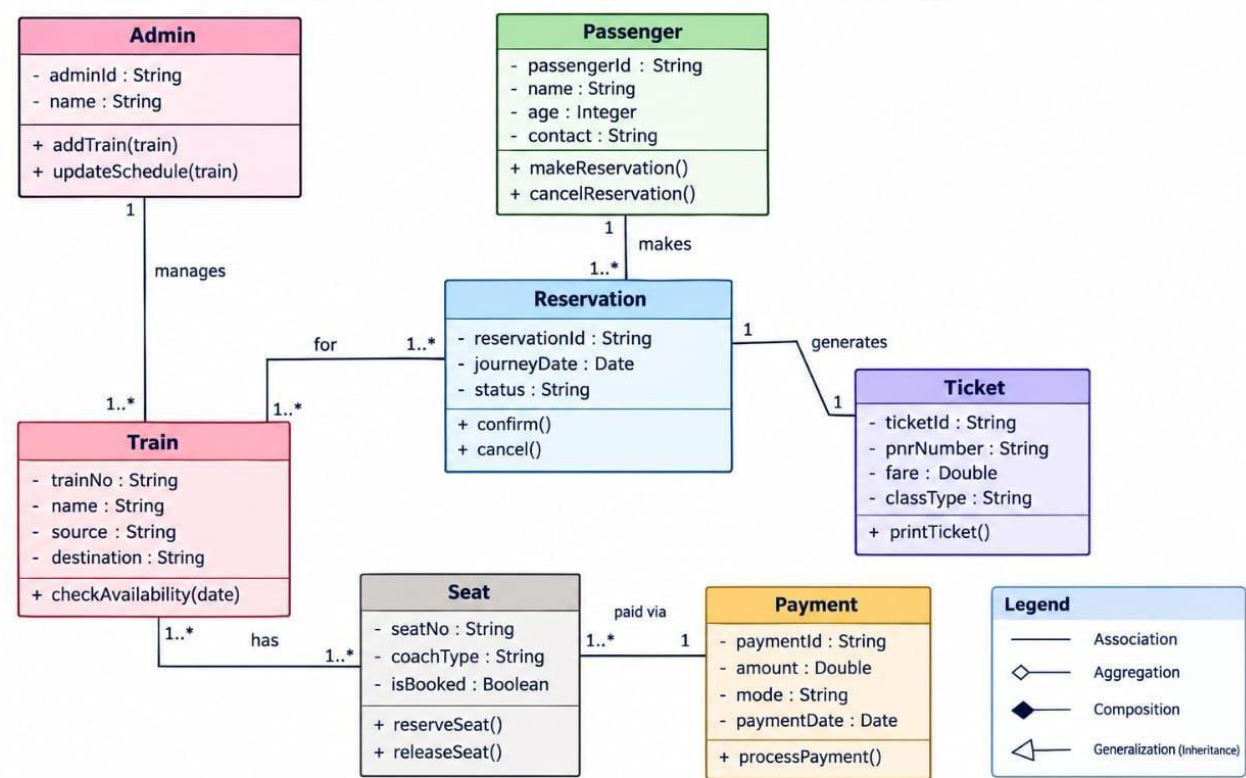
EX.NO.3.3 CLASS DIAGRAM FOR LIBRARY INFORMATION SYSTEM

Library Information System - UML Class Diagram



EX.NO.3.4 CLASS DIAGRAM FOR RAILWAY TICKET RESERVATION SYSTEM

Railway Ticket Reservation System - UML Class Diagram



To understand users view of project using use case diagram

Software Required :-

VisualUML8.2

Procedure:-

You can draw use case diagrams in VP-UML as well as to document the event flows of use cases using the flow-of-events editor of UML 8.2 .

The steps are as follows.

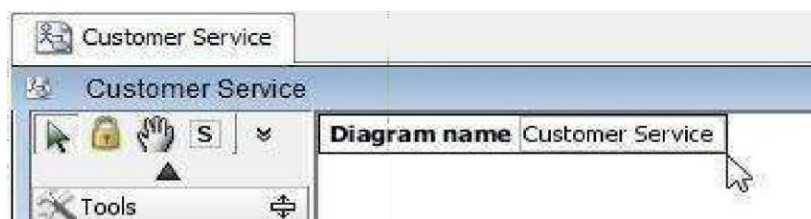
Step1:

Right click Use Case Diagram on Diagram Navigator and select New Use Case Diagram from the popup menu.



Step2:-

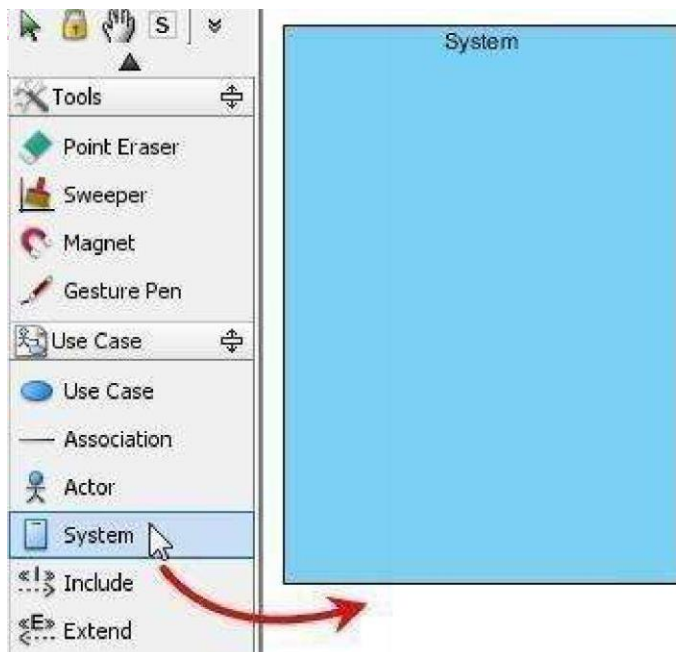
Enter name for the newly created use case diagram in the text field of pop-up box on the top left corner.



Step 3: Drawing

a system

To create a system , select System on the diagram toolbar and then click it on the diagram pane .Finally name

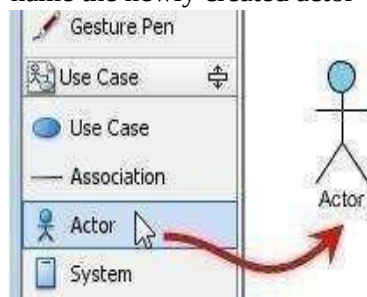


the newly created system when it is created.

Step4:

Drawing an actor

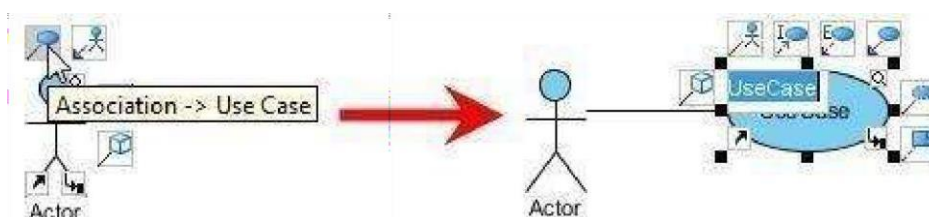
To draw an actor, select Actor on the diagram toolbar and then click it on the diagram pane. Finally, name the newly created actor when it is created.



Step 5:-

Drawing a use case

Besides creating a use case through diagram toolbar, you can also create it through resource icon. Move the mouse over a shape and press a resource icon that can create use case. Drag it and then release the mouse button until it reaches to your preferred place. The source shape and the newly created use case are connected.



Finally, name the newly created use case.

Step6:-

Create a use case through resource icon

Line wrapping use case name

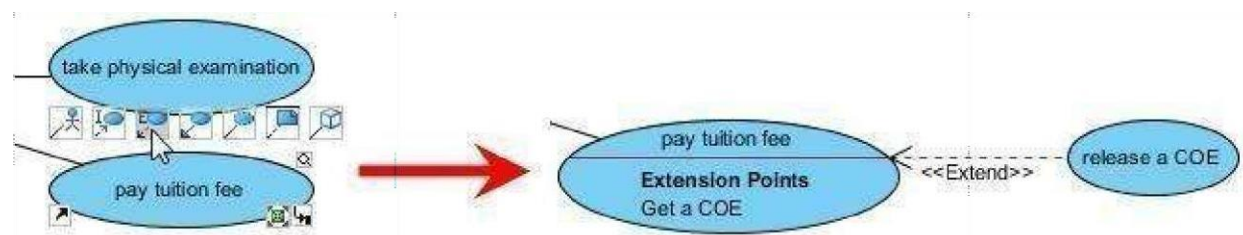
If a use case is too wide, for a better outlook, you may resize it by dragging the filled selectors. As a result, the name of use case will be line-wrapped automatically.



Step 7:

Resize a use case

To create an extend relationship, move the mouse over a use case and press its resource icon Extend>Use Case. Drag it to your preferred place and then release the mouse button. The use case with extension points and a newly created use case are connected. After you name the newly created use case, a pop-up dialog box will ask whether you want the extension point to follow the name of use case. Click Yes if you want it to do so click NO if you want to enter another name for extension point.

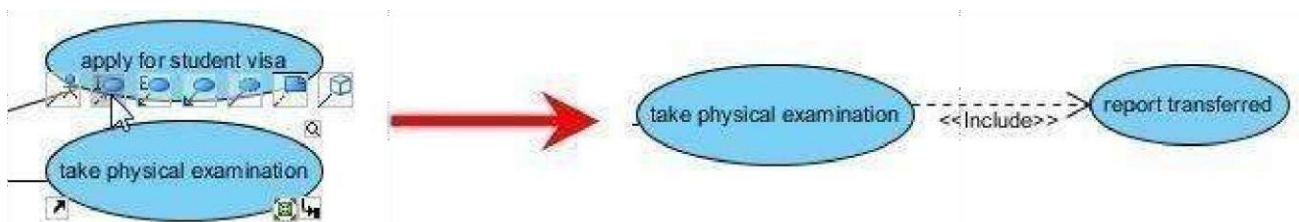


Step 8:

Create an extend relationship

Drawing<<Include>>relationship

To create an include relationship, mouse over a use case and press its resource icon Include -> Use Case. Drag it to your preferred place and then release the mouse button. A new use case together with an include relationship is created. Finally, name the newly created use case.

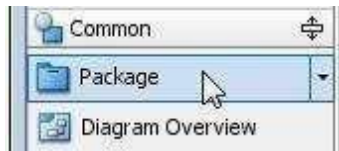


Step 9:

Include relationship is created

Structuring use cases with package

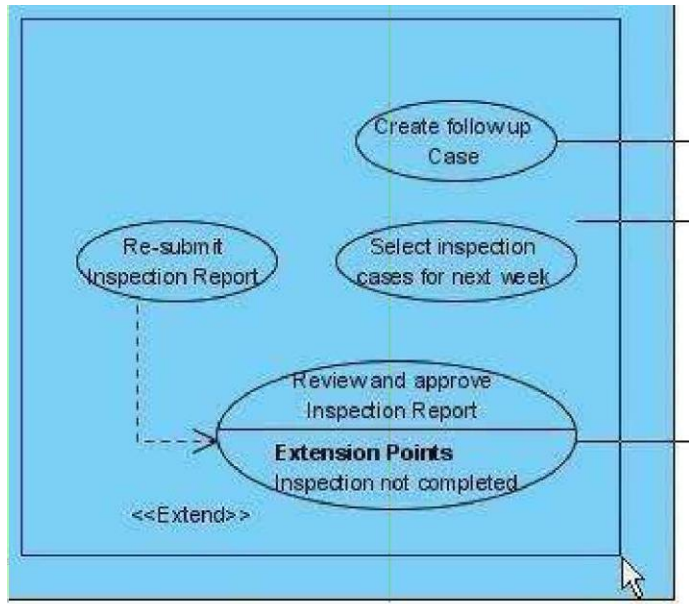
You can organize use cases with package when there are many of them on the diagram. Select Package on the diagram toolbar (under Common category).



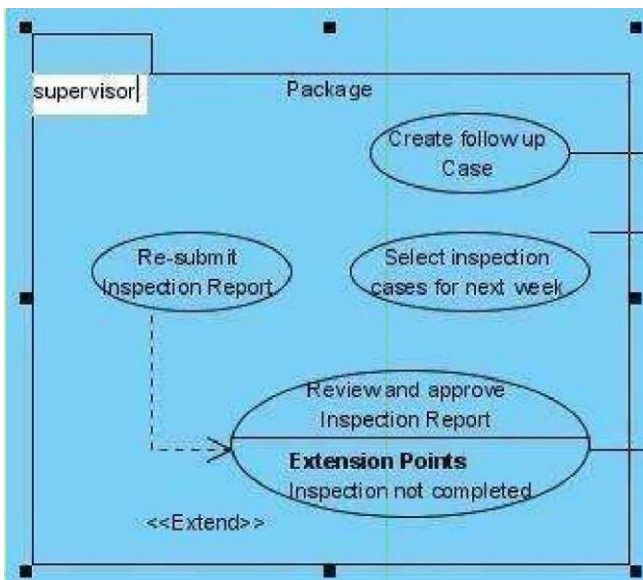
Step 10:

Create a package

Drag the mouse to create a package surrounding those use cases.



Step 11: Surround use cases with package Finally, name the package



Step 12

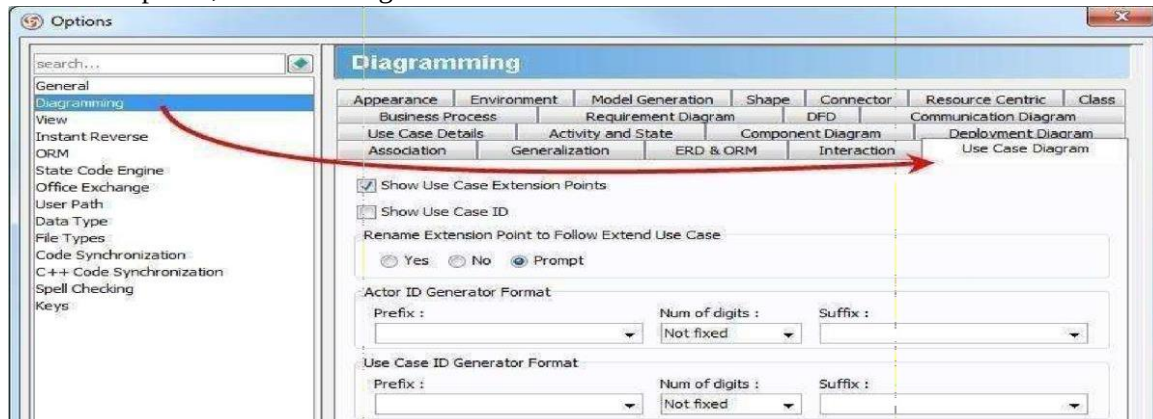
Name the package

Assigning IDs to actors/ Use cases

You may assign IDs to actors and use cases. By default, IDs are assigned with the order of object creation, starting from one onwards. However, you can define the formator even enter an ID manually.

Defining the format of ID

To define the format of ID, select Tools > Options from the main menu to unfold the Options dialog box. Select Diagramming from the list on the left hand side and select the Use Case Diagram tab on the righthand side. You can adjust the format of IDs under Use Case Diagram tab. The format of ID consists of prefix, number of digits and suffix.



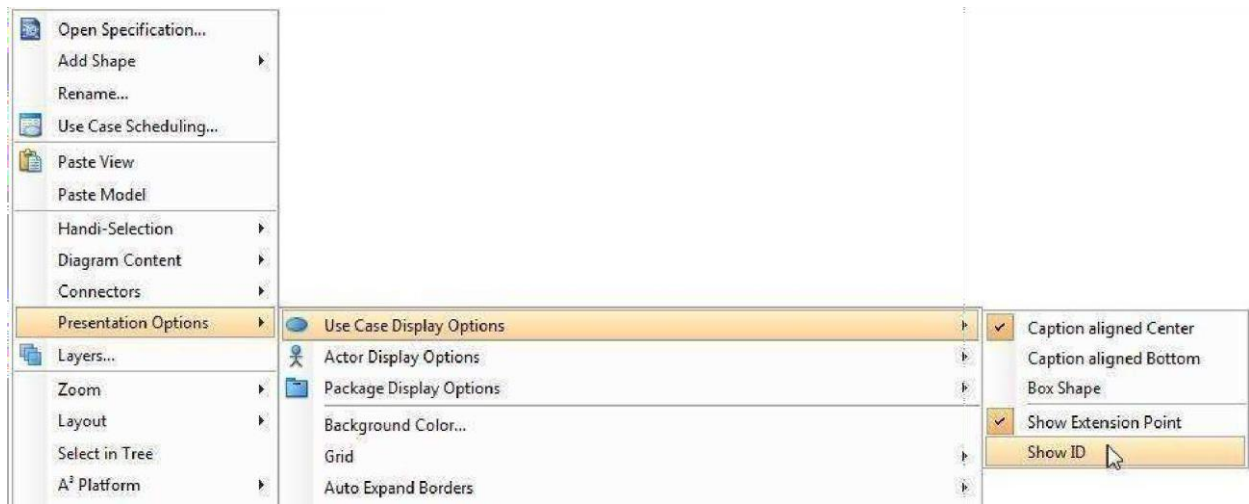
Step 13: Use Case Diagram tab

The description of options for ID generator format is shown below. Option Description

Prefix The prefix you enter in Prefix text field will be inserted before the number. Num of digits The number of digits for the number. For example, when digits is 3, ID "1" will become "001". Suffix The suffix you enter in Suffix text field will be inserted behind the number. Options for formatting ID Showing ID on diagram

By default, ID is just a text property. It usually doesn't appear on diagram. However, you can make it shown within a use case.

Right click on the diagram background, select Presentation Options and the specific model element display option from the pop-up menu.



Step 14:

Show Id on diagram

As a result, the use case is displayed with ID.



A use case with ID displayed

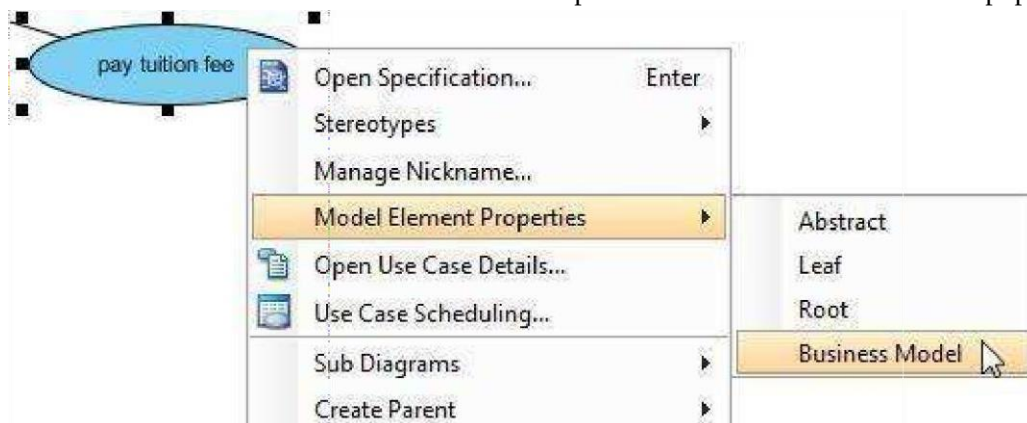
NOTE: The feature of showing ID does only support for use case, but not for actor.

ID assignment

There are several ways that you can assign an ID to a model element, including:

- Through the specification dialog box (Right click on the selected model element and select Open Specification...from the pop-up menu)
- Through the Property Pane Drawing business use case

1. Right click on a use case and select Model Element Properties > Business Model from the pop-up menu.



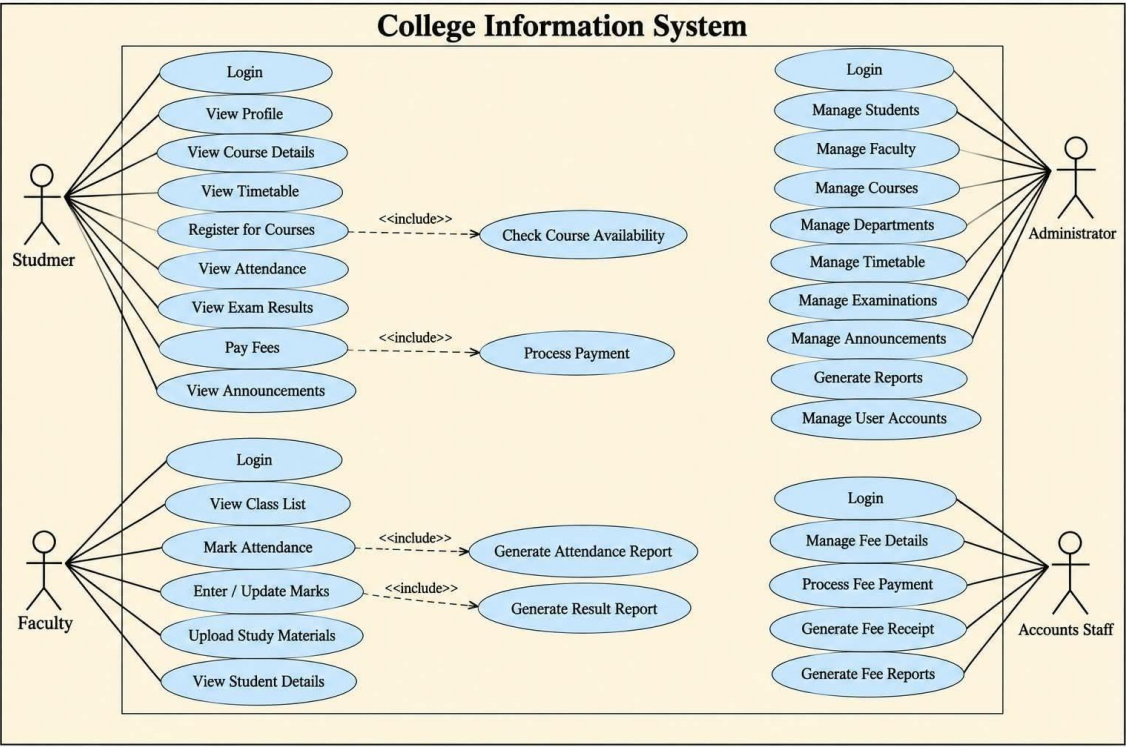
Step 15:

1. Click Business Model
2. After selected, an extra slash will be shown on the left edge of the use case.

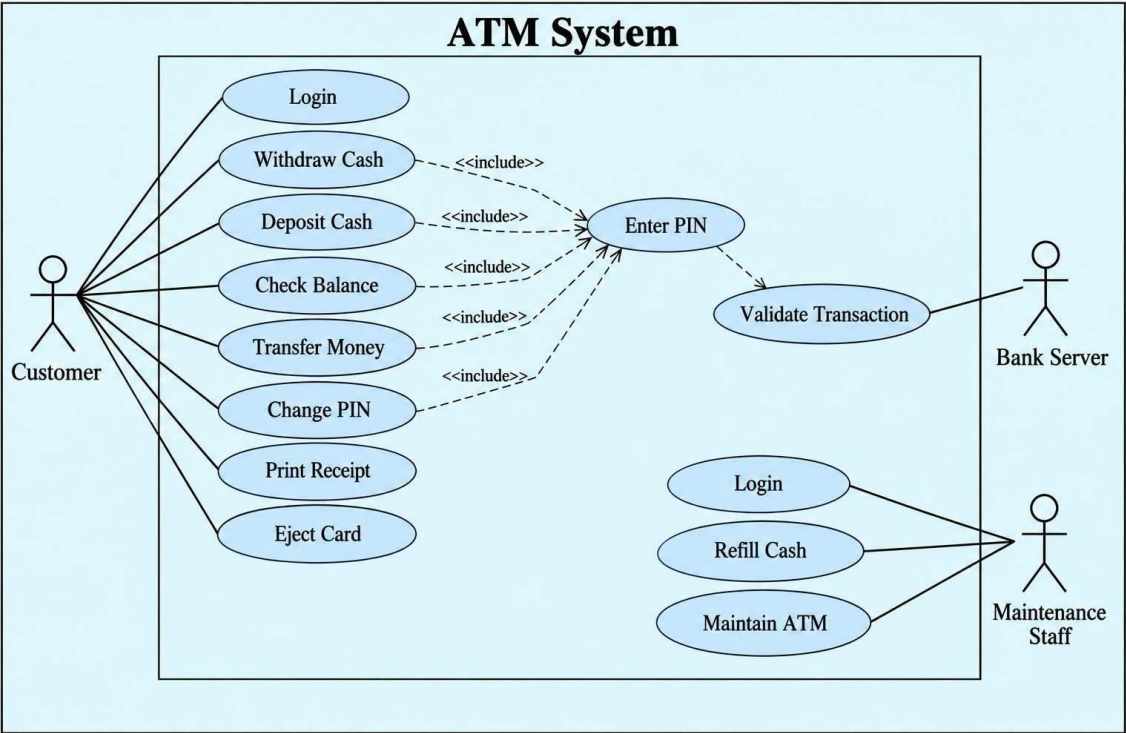


Business model And Finally The Use case Diagram is ready.

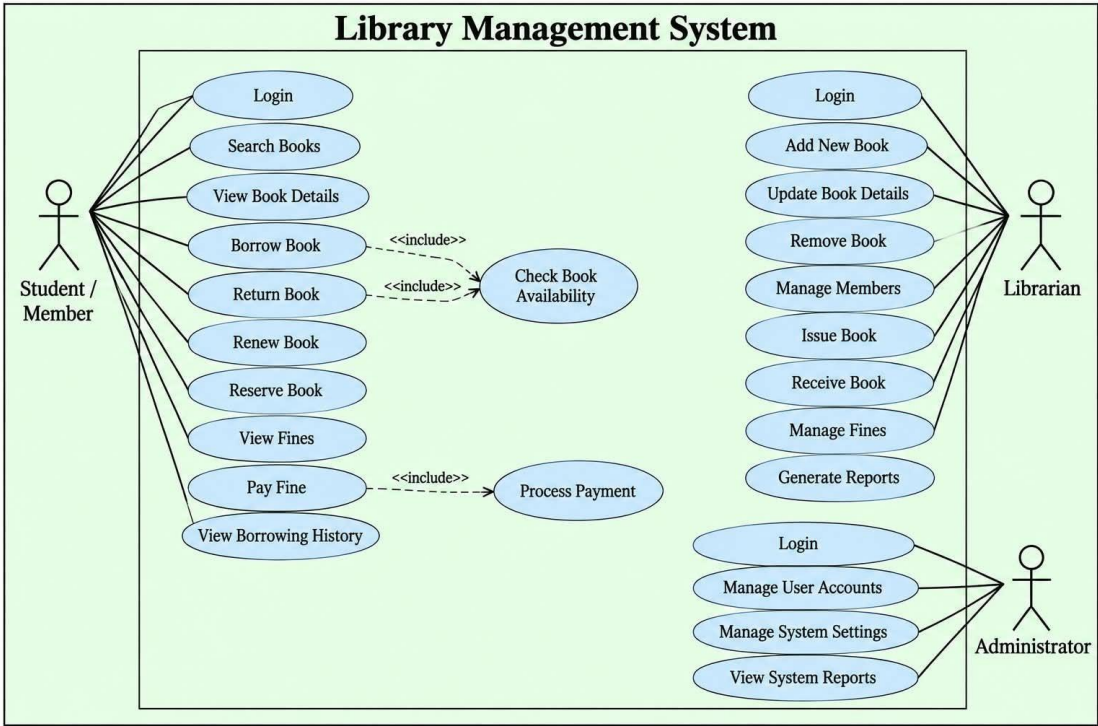
EX.NO.4.1 USECASE DIAGRAM FOR COLLEGE AUTOMATION SYSTEM



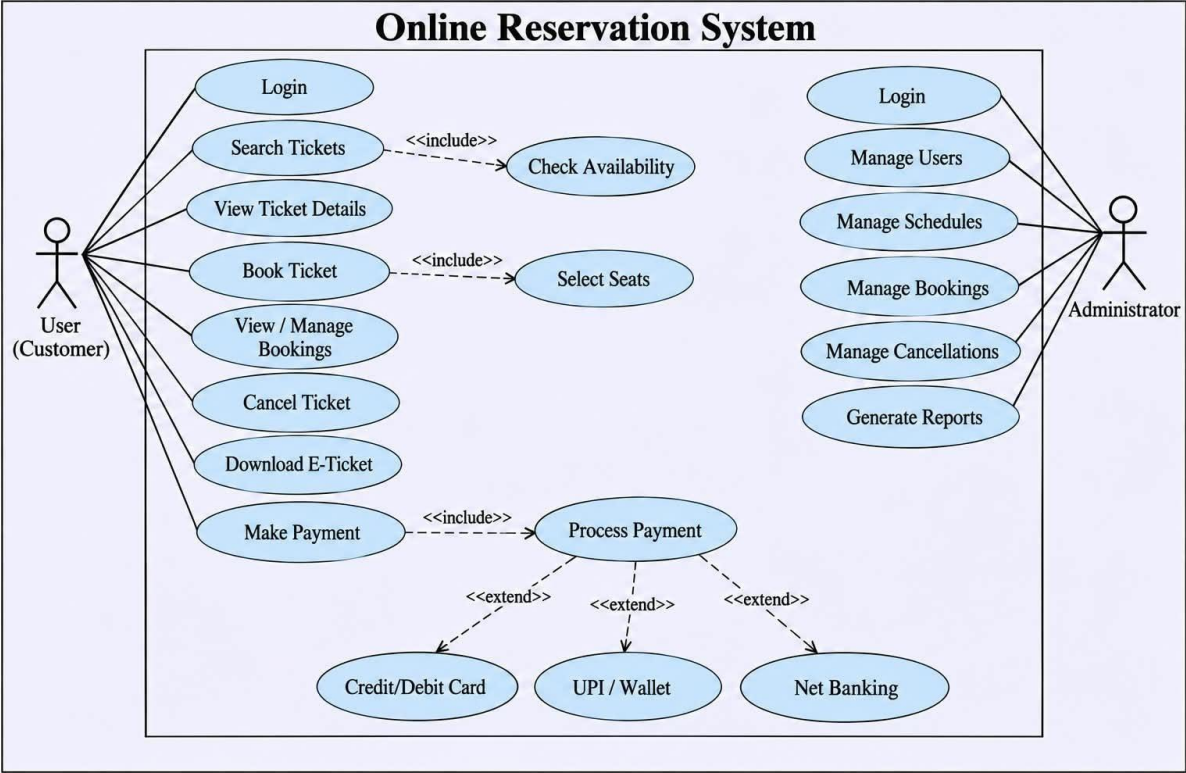
EX.NO.4.2 USECASE DIAGRAM FOR BANKING MANAGEMENT SYSTEM



EX.NO.4.3 USECASE DIAGRAM FOR LIBRARY INFORMATION SYSTEM



EX.NO.4.4 USECASE DIAGRAM FOR RAILWAY TICKET RESERVATION SYSTEM



Ex.No.5

Develop sequence diagram Objective :

To understand the interactions between objects that are represented as lifelines in a sequential order of a project using Sequence Diagram.

Software Required :-

Visual Paradigm for UML

8.2

Procedure :-

A [sequence diagram](#) is used primarily to show the interactions between objects that are represented as lifelines in a sequential order.

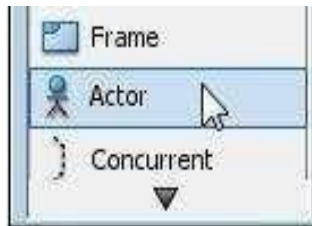
Step 1:-

Right click Sequence diagram on Diagram Navigator and select New Sequence Diagram from the pop-up menu to create a sequence diagram.

Enter name for the newly created sequence diagram in the text field of pop-up box on the top left corner.

Creating actor

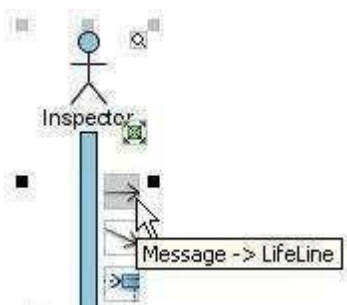
To create actor, click Actor on the diagram toolbar and then click on the diagram.



Creating lifeline

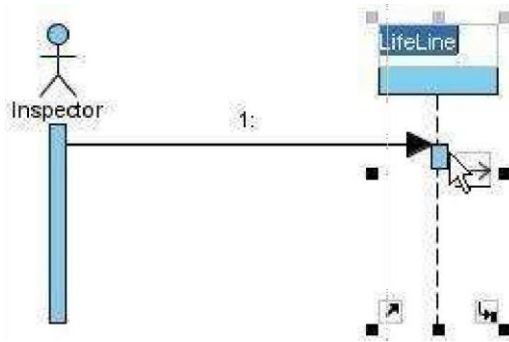
To create lifeline, you can click Life Line on the diagram toolbar and then click on the diagram.

Alternatively, a much quicker and more efficient way is to use the resource-centric interface. Click on the Message ->Life Line resource beside an actor/lifeline and drag.



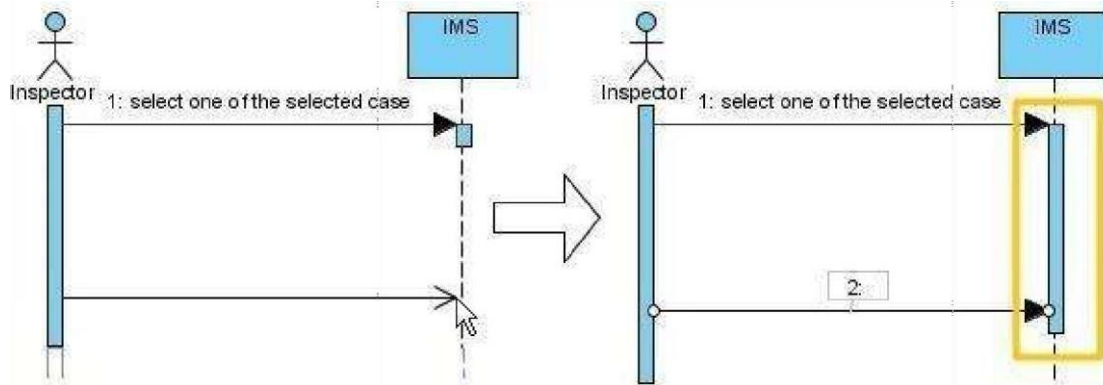
Step 3:-

Move the mouse to empty space of the diagram and then release the mouse button. A new lifeline will be created and connected to the actor/lifeline with a message.



Auto extending activation

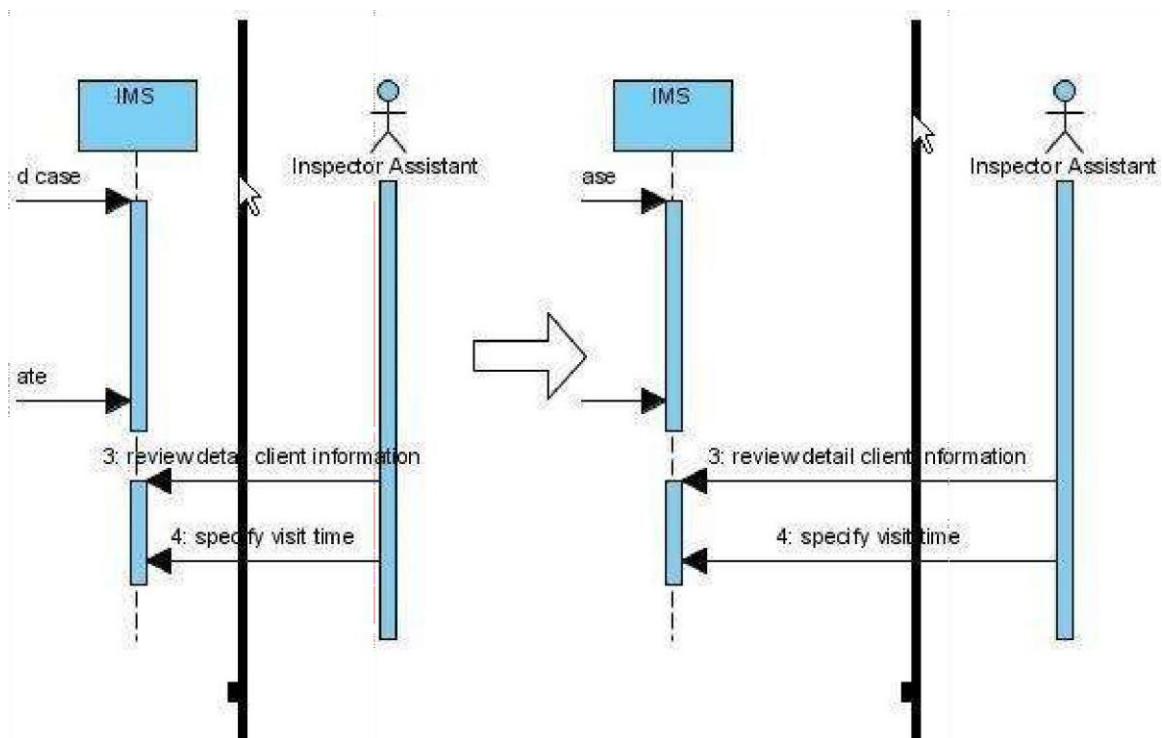
When create message between lifelines/actors, activation will be automatically extended.



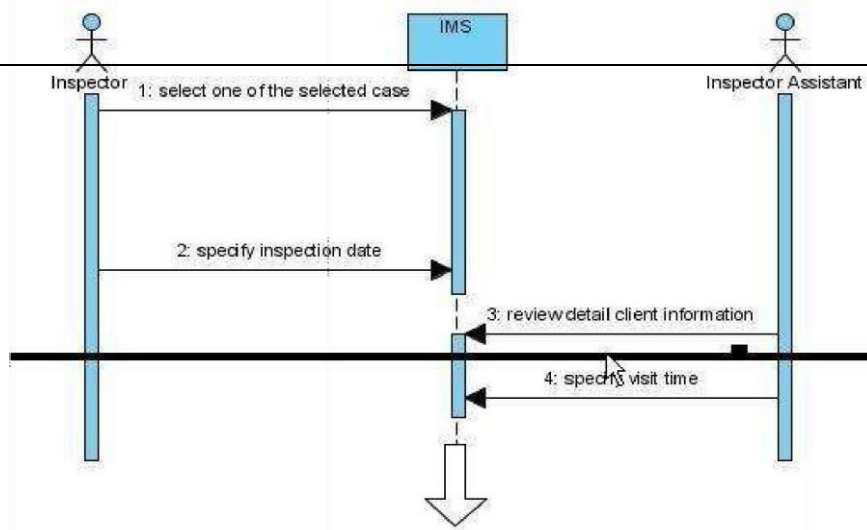
Step 4:-

Using sweeper and magnet to manage sequence diagram

Sweeper helps you to move shapes aside to make room for new shapes or connectors. To use sweeper, click Sweeper on the diagram toolbar (under the Tools category).



The picture below shows the message specify visit time is being swept downwards, thus new room is made for new messages.



Step 5:-

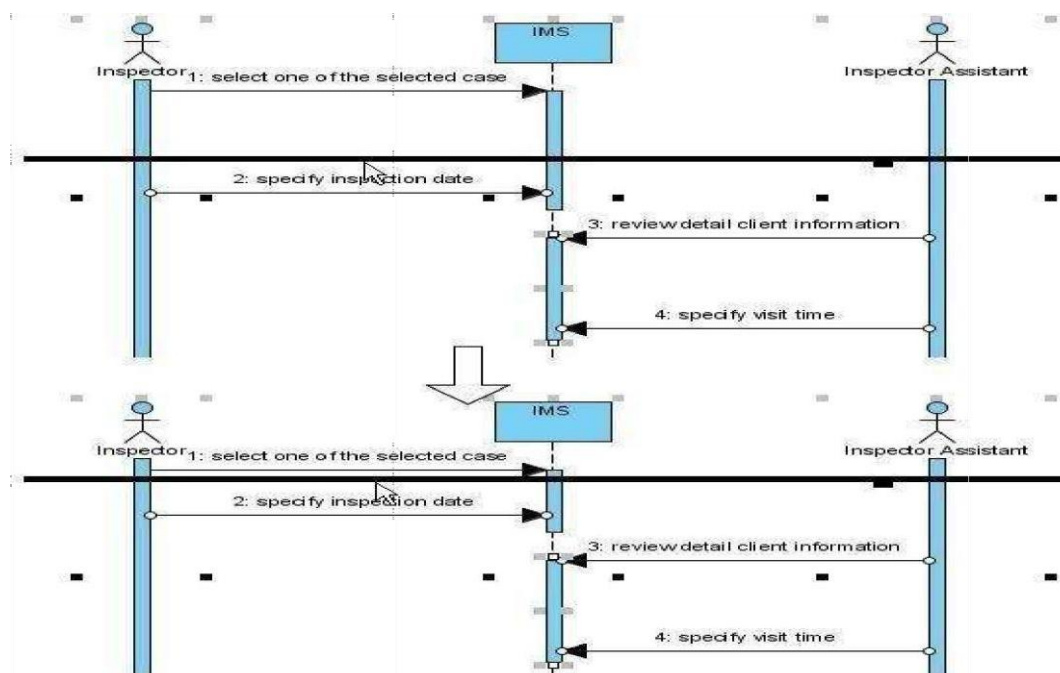
You can also use magnet to pull shapes together. To use magnet, click Magnet on the diagram toolbar (under the Tools category)



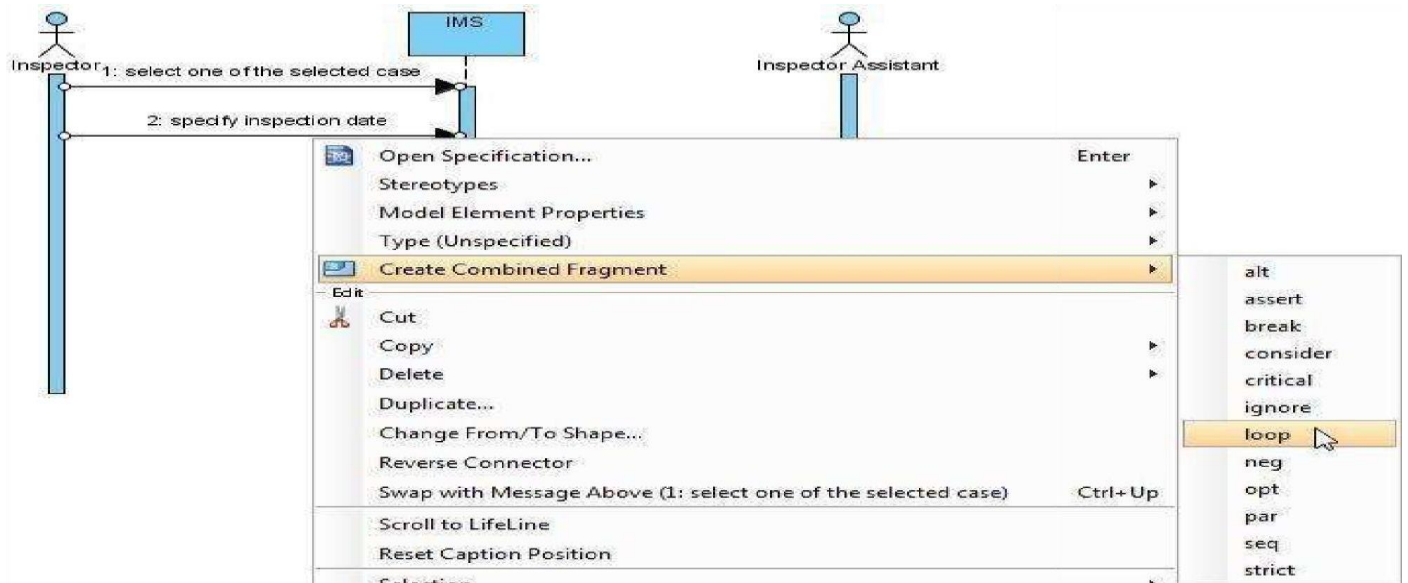
Magnet

Click on empty space of the diagram and drag towards top, right, bottom or left. Shapes affected will be pulled to the direction you dragged.

The picture below shows when drag the magnet upwards, shapes below dragged position are pulled upwards.

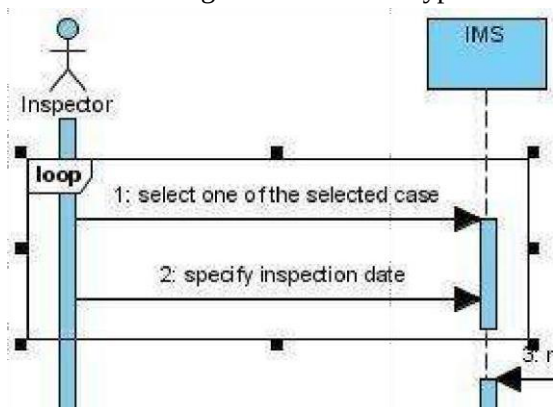


Step6:- Creating combined fragment for messages To create combined fragment to cover messages, select the messages, right-click on the selection and select Create Combined Fragment, and then select a combined fragment type (e.g. loop) from the popup menu.



Step 7:-

A combined fragment of selected type will be created to cover the messages.

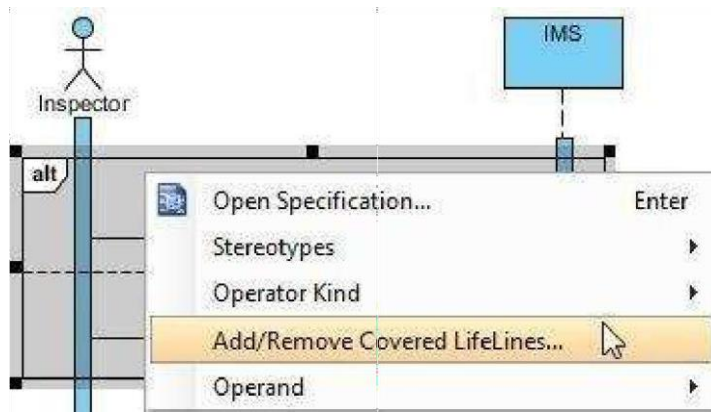


Step 8:-

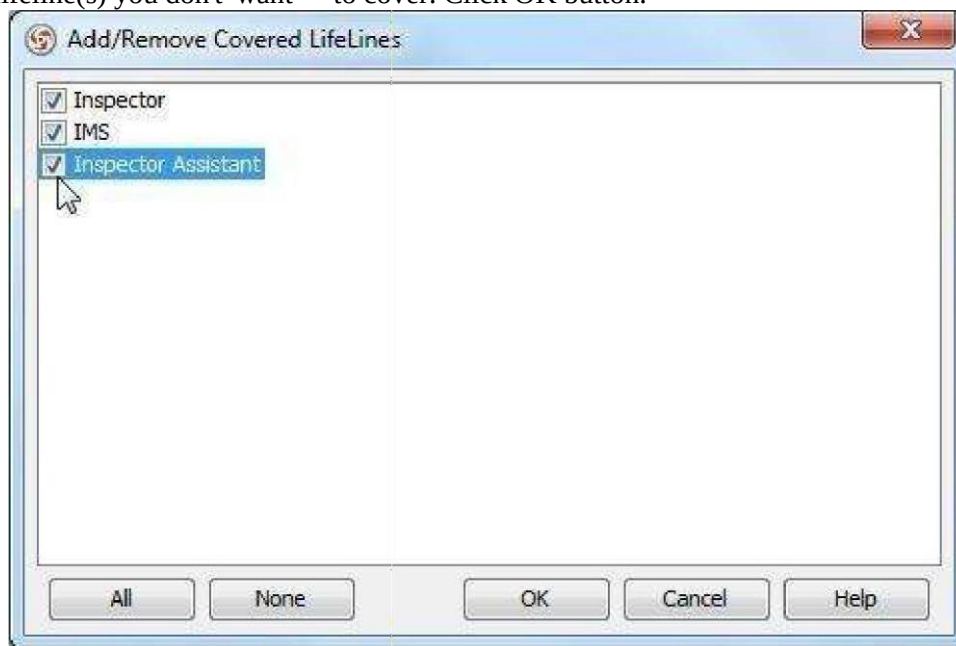
Adding/removing covered lifelines

After you've created a combined fragment on the messages, you can add or remove the covered lifelines.

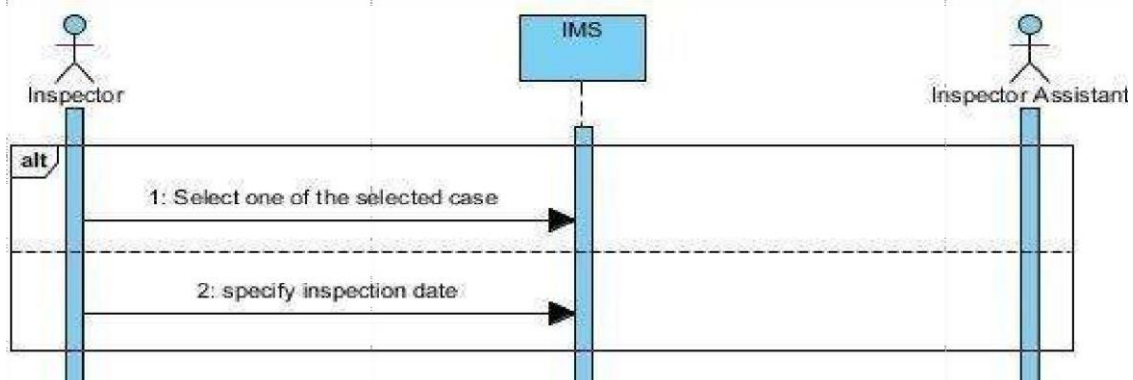
1. Move the mouse over the combined fragment and select Add/Remove Covered Lifeline... from the pop-up menu.



2. In the Add/Remove Covered Lifelines dialog box, check the lifeline(s) you want to cover or uncheck the lifeline(s) you don't want to cover. Click OK button.

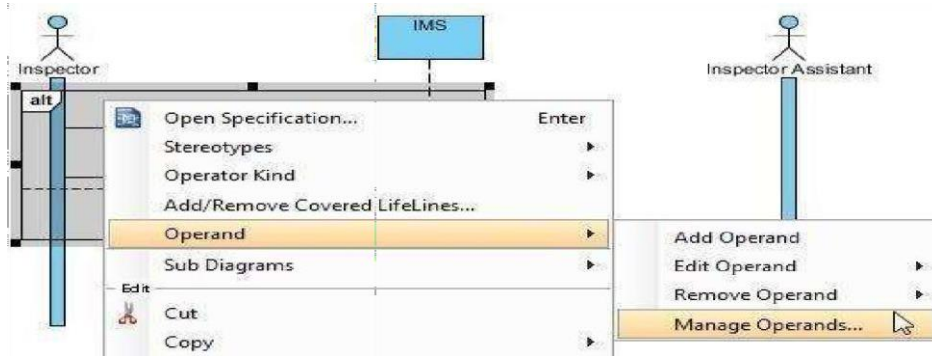


3. As a result, the area of covered lifelines is extended or narrowed down according to your selection.



Managing Operands

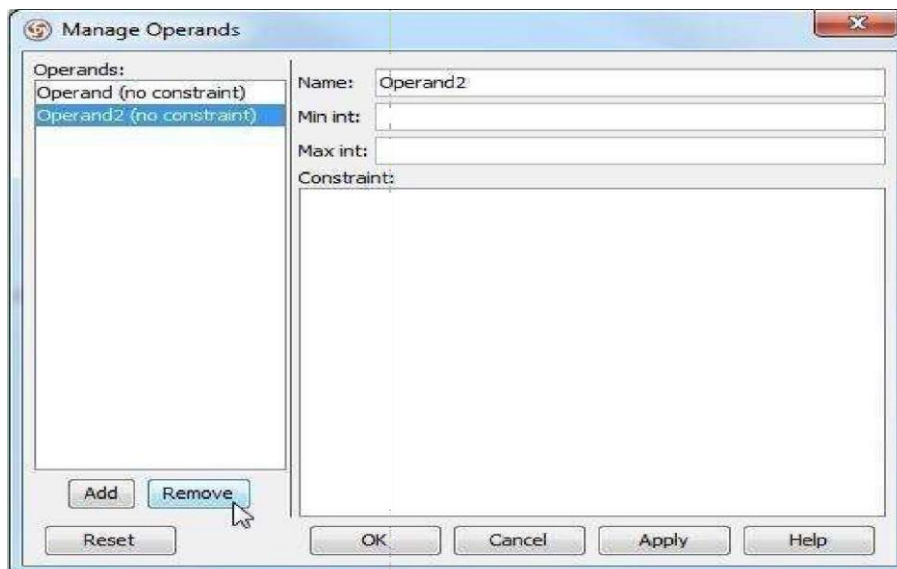
After you've created a combined fragment on the messages, you can also add or remove operand(s).



1. Move the mouse over the combined fragment and select Operand > Manage Operands... from the pop-up menu.

Step 9:-

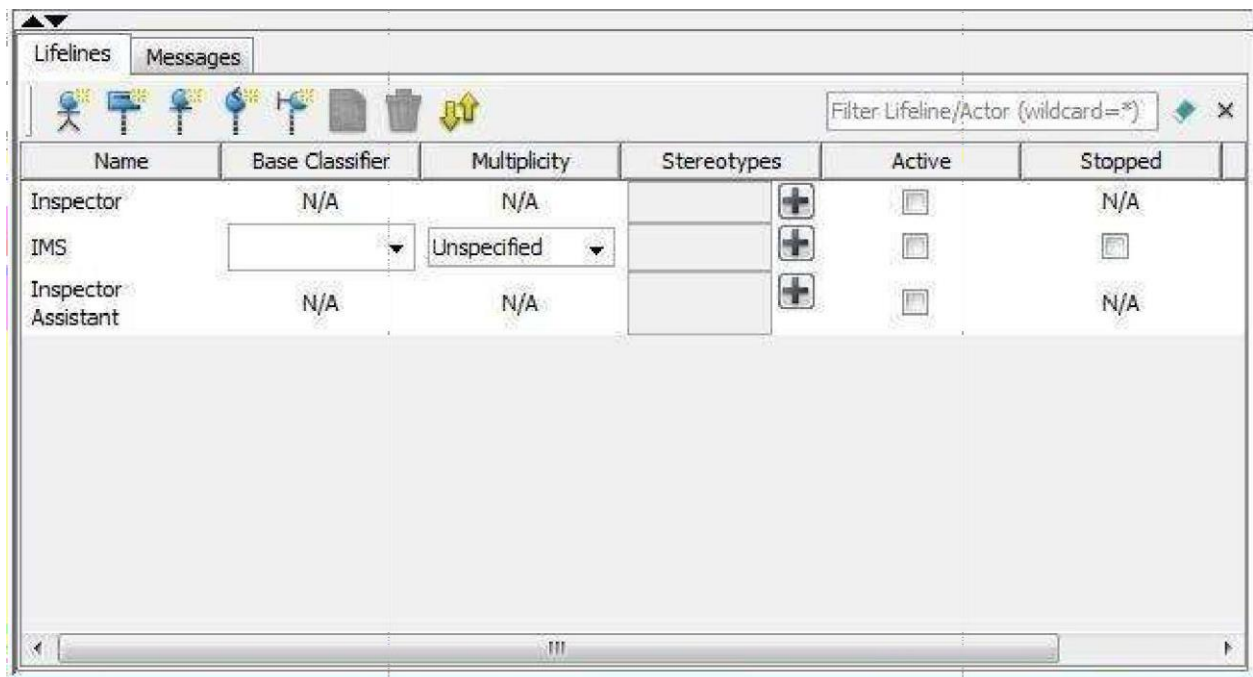
1. To remove an operand, select the target operand from Operands and click Remove button. Click OK button.



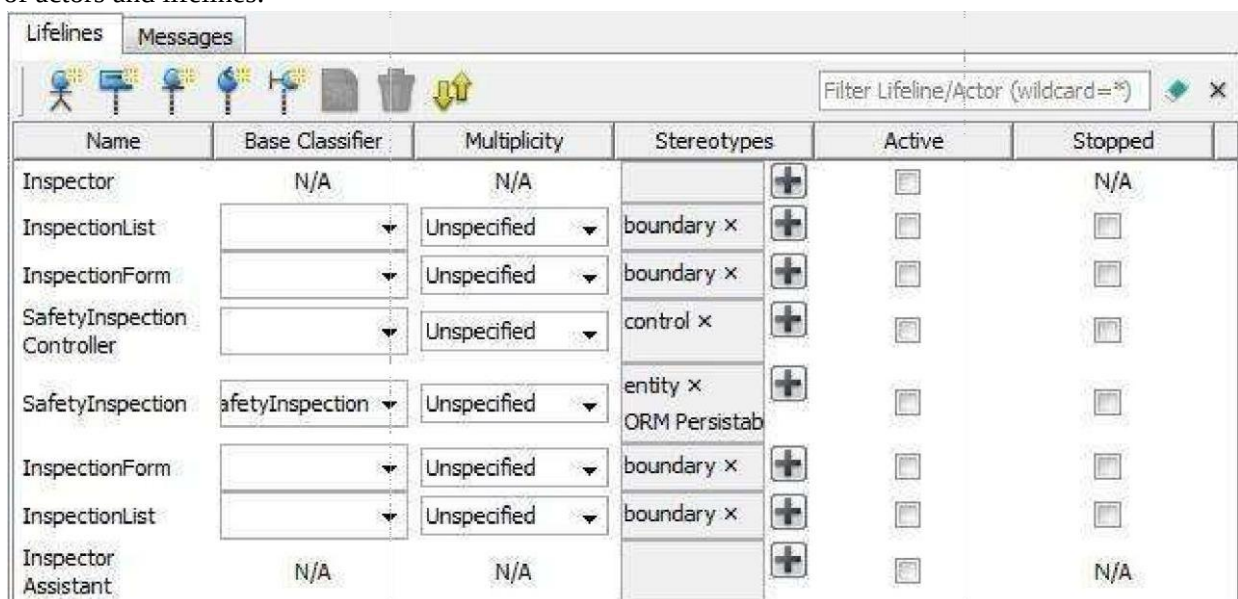
2. Otherwise, click Add button to add a new operand and then name it. Click OK button.

Developing sequence diagram with quick editor or keyboard shortcuts

In sequence diagram, an editor appears at the bottom of diagram by default, which enables you to construct sequence diagram with the buttons there. The shortcut keys assigned to the buttons provide a way to construct diagram through keyboard. Besides constructing diagram, you can also access diagram elements listing in the editor.



There are two panes, Lifelines and Messages. The Lifelines pane enables you to create different kinds of actors and lifelines.



Button Shortcut Description

Alt-Shift- A

 To create an actor

Alt-Shift- L

 To create a general lifeline

Alt-Shift- E

 To create an <<entity>> lifeline

Alt-Shift- C

 To create a <<control>> lifeline

Alt-Shift- B

 To create a <<boundary>> lifeline

Alt-Shift- O

To open the specification of the element chosen in quick editor

Ctrl-Del

To delete the element chosen in quick editor

Ctrl-L

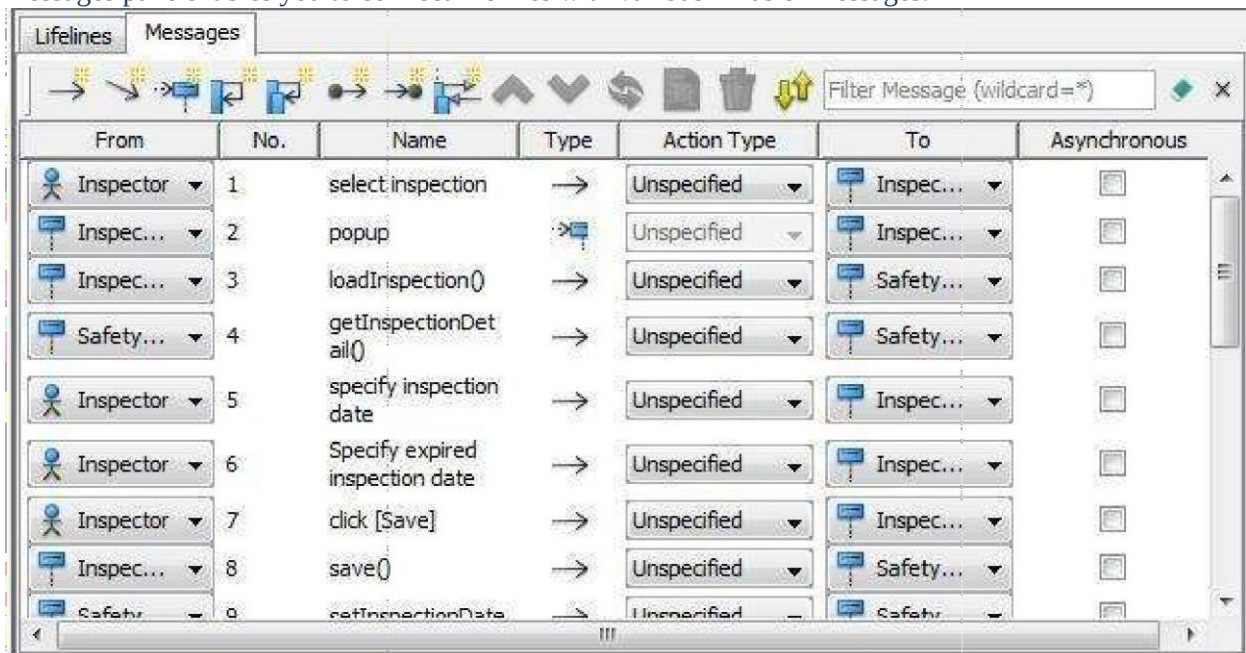
To link with the diagram, which cause the diagram element to be selected when selecting an element in editor, and vice versa

Step 10:-

Buttons in Lifelines pane

Editing messages

The Messages pane enables you to connect lifelines with various kinds of messages.



Messages pane in quick editor

Button

Shortcut Description



Alt-Shift-M

To create a message that connects actors/lifelines in diagram



Alt-Shift-D

To create a duration message that connects actors/life lines in diagram



Alt-Shift-C To create a create message that connects actors/lifelines in diagram



Alt-Shift-S

To create a self message on an actor/lifeline in diagram



- Alt-Shift-R To create a recursive message on an actor/life line in diagram
- Alt-Shift-F To create a found message that connects to an actor/lifeline
- Alt-Shift-L To create a lost message from an actor/lifeline
- Alt-Shift-E To create a re entrant message that connects actors/lifelines in diagram
- Ctrl-Shift-Up To swap the chosen message with the one above
- Ctrl-Shift-Down To swap the chosen message with the one below
- Ctrl-R To revert the direction of chosen message
- Alt-Shift-O To open the specification of the message chosen in quick editor
- Ctrl-Del To delete the message chosen in quick editor
- Ctrl-L To linkwiththediagram,whichcausethemessagetobeselectedwhenselectingamessageineditor, and vice versa

Buttons in Messages pane

Expanding and collapsing the editor

To hide the editor, click on the down arrow button that appears at the bar on top of the quick editor. To expand, click on the up arrow button.



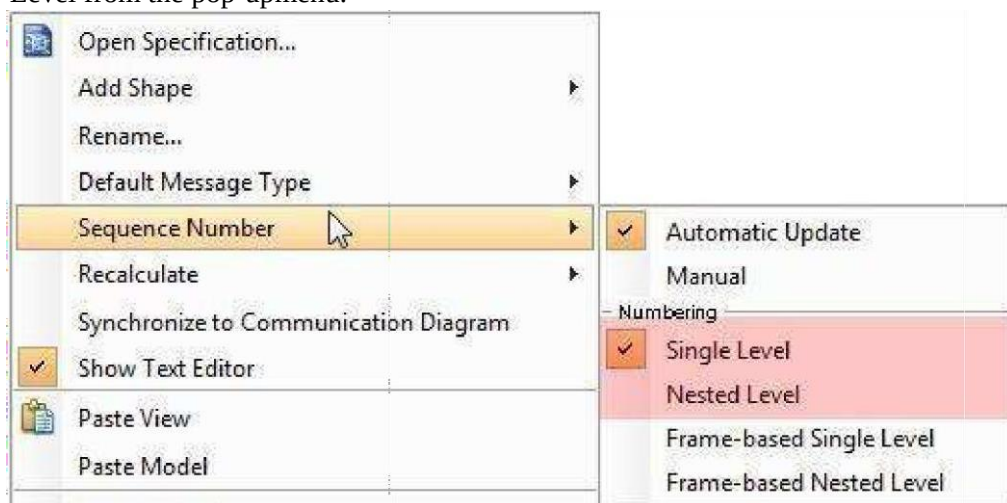
Collapse the quick editor

Setting different ways of numbering sequence messages

You are able to set the way of numbering sequence messages either on diagram base or frame base.

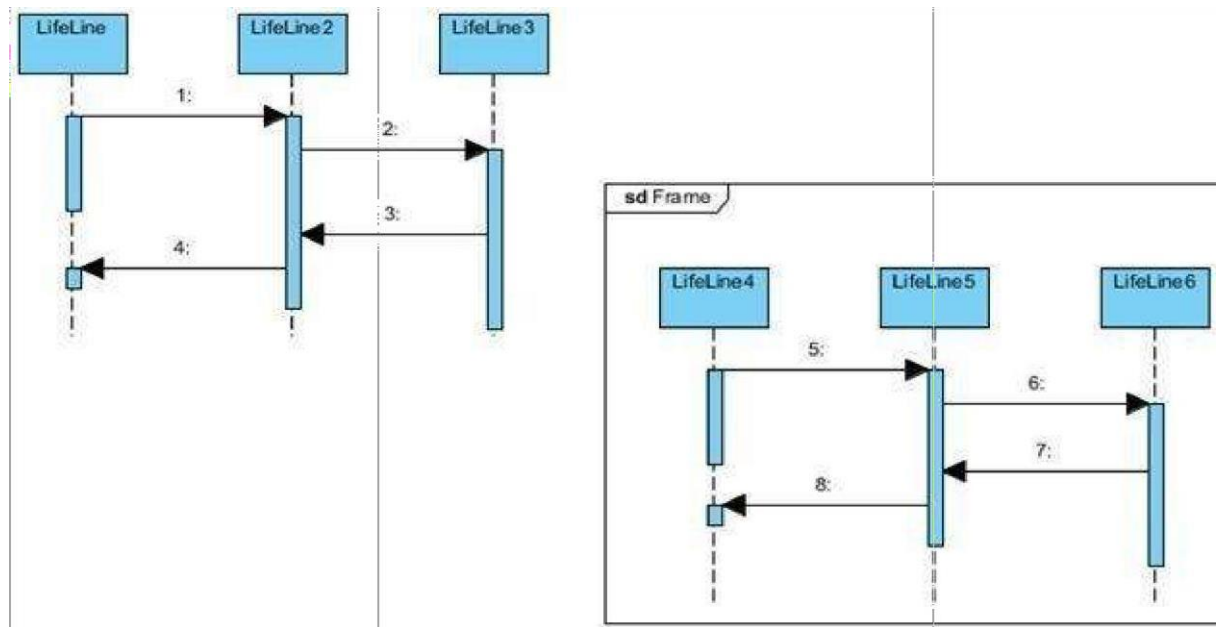
Diagram-based sequence message

Right click on the diagram's background, select Sequence Number and then either Single Level or Nested Level from the pop-up menu.

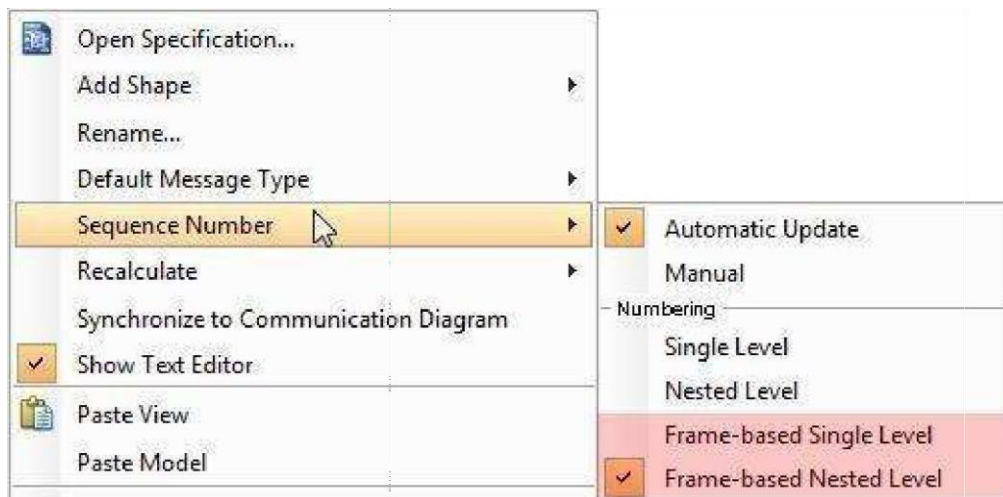


Step 11:-

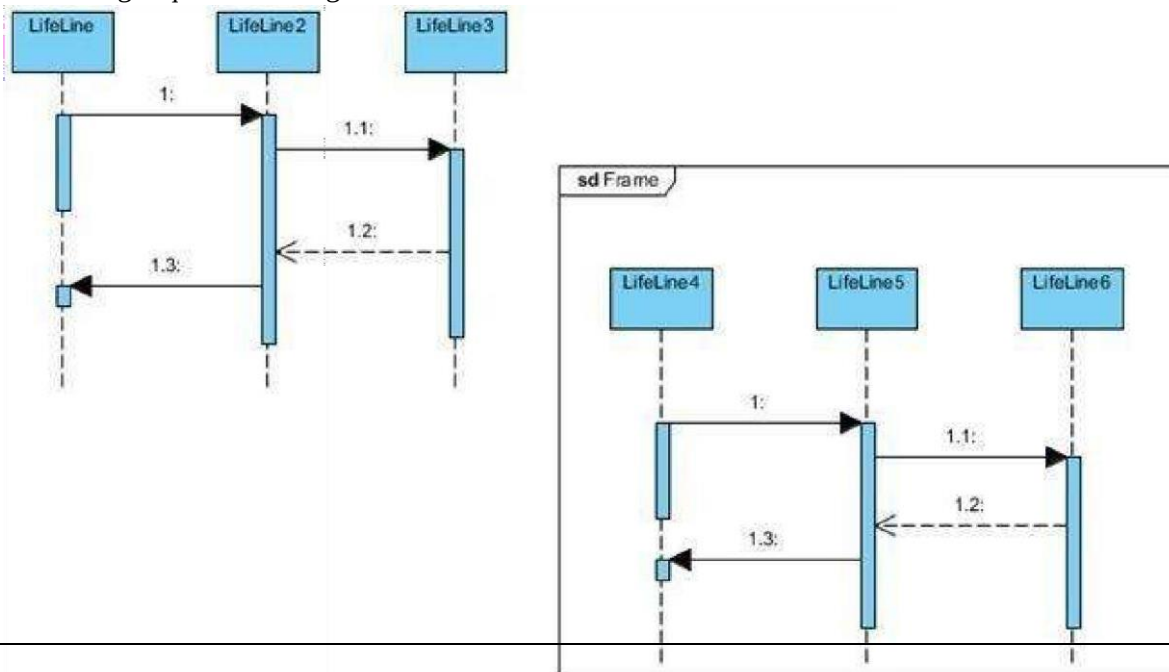
If you choose Single Level, all sequence messages will be ordered with integers on diagram base. On the other hand, if you choose Nested Level, all sequence messages will be ordered with decimal place on diagram base.



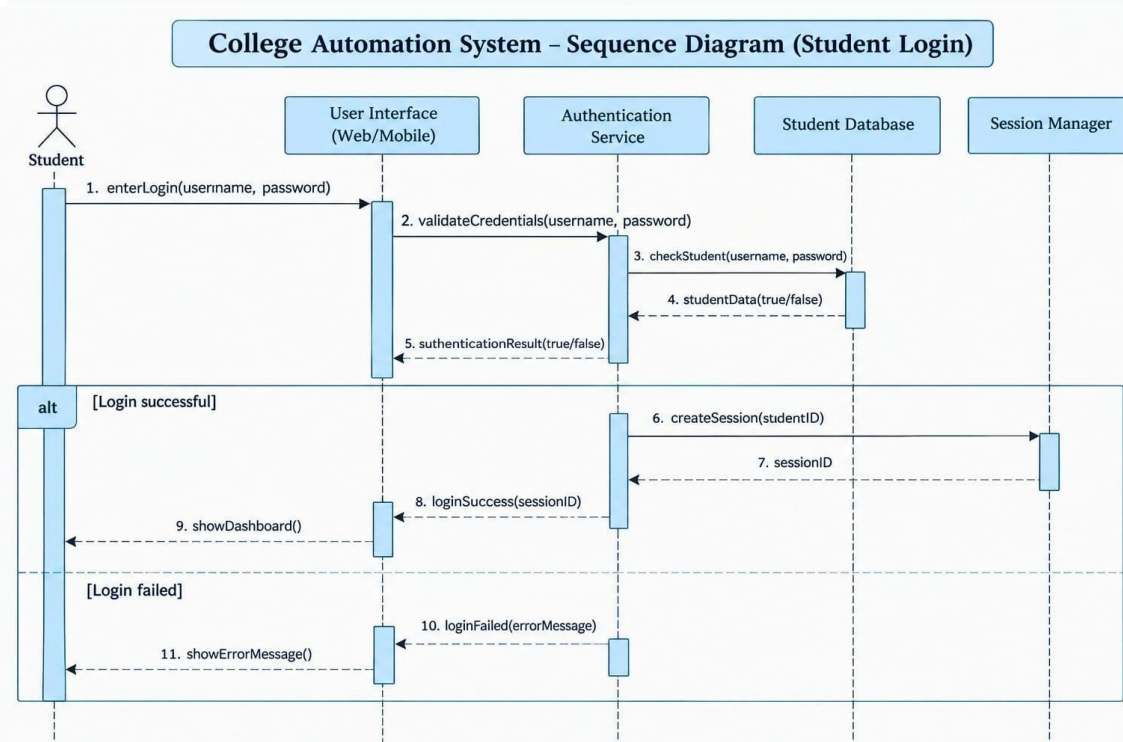
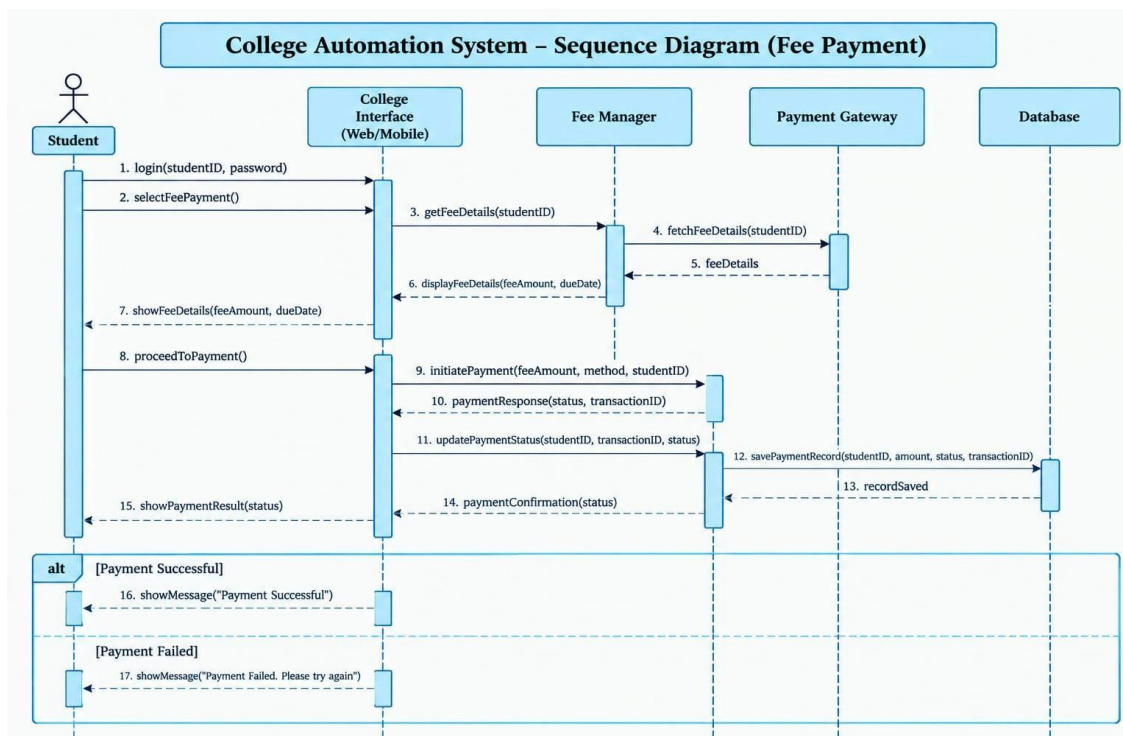
Right click on the diagram's background, select Sequence Number and then either Frame-based Single Level or Frame-based Nested Level from the pop-up menu.



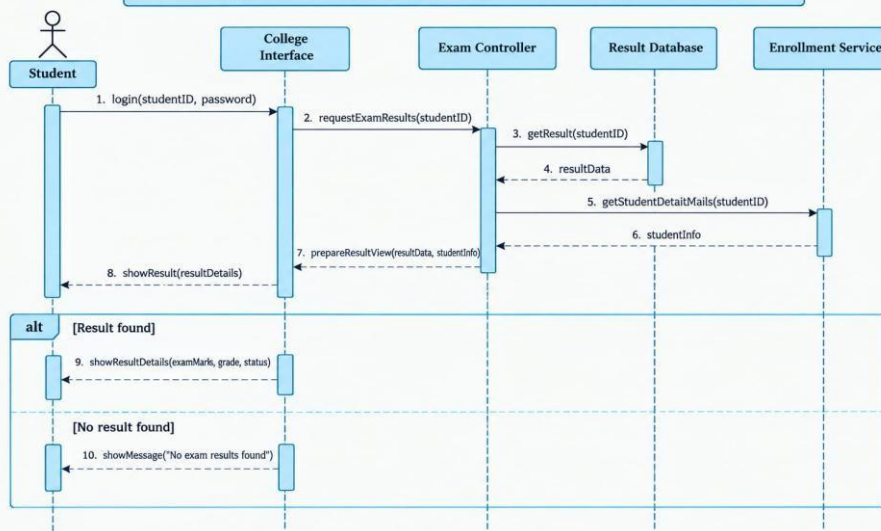
When you set the way of numbering sequence messages on frame base, the sequence messages in frame will restart numbering sequence message since they are independent and ignore the way of numbering sequence message outside the frame.



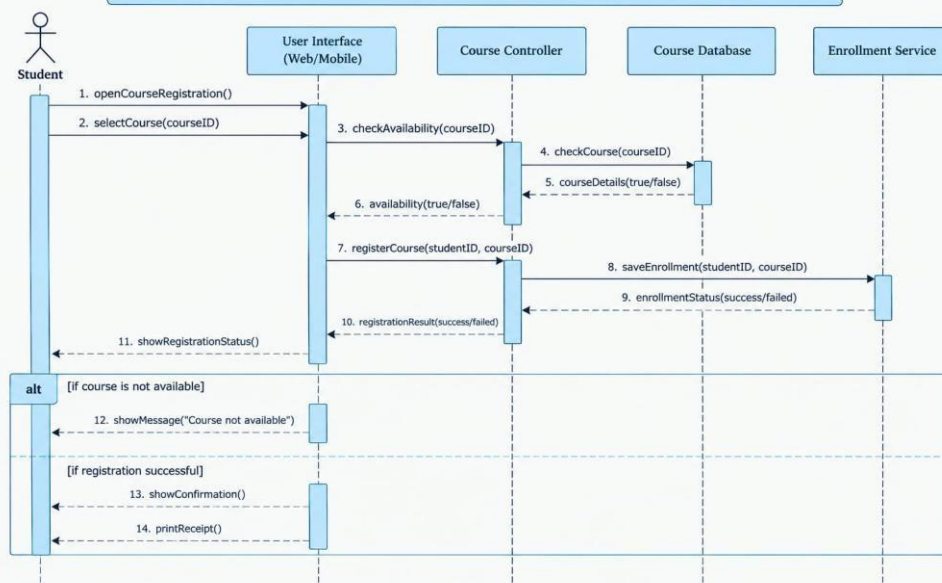
EX.NO.5.1 SEQUENCE DIAGRAM FOR COLLEGE AUTOMATION SYSTEM



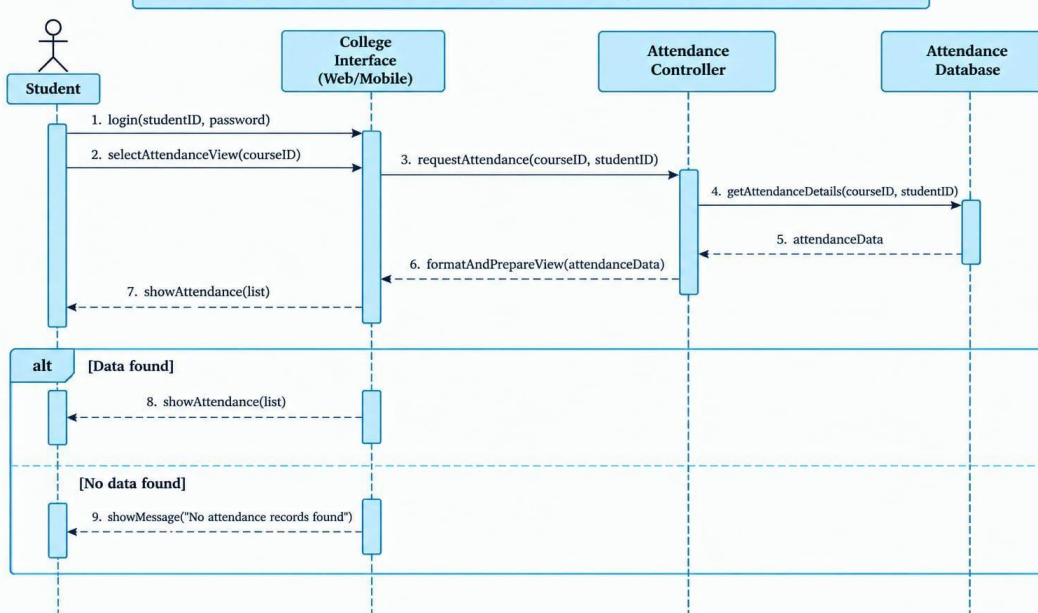
College Automation System – Sequence Diagram (Exam Results)



College Automation System – Sequence Diagram (Course Registration)

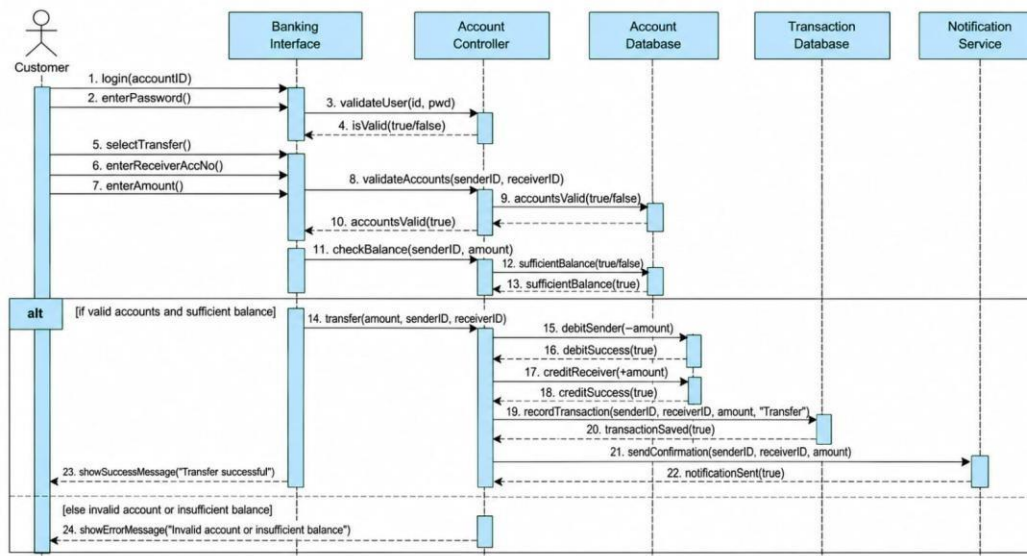


College Automation System – Sequence Diagram (Attendance View)

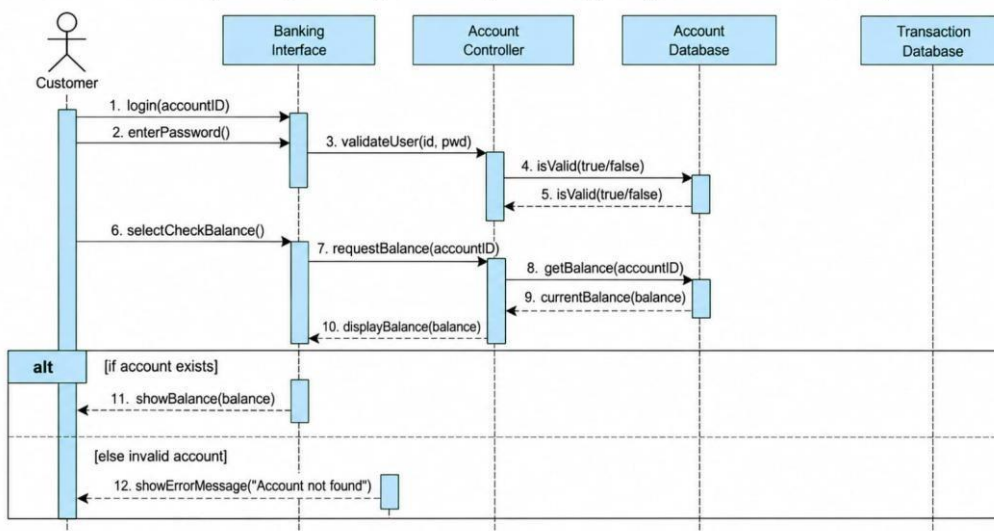


EX.NO.5.2 SEQUENCE DIAGRAM FOR BANKING MANAGEMENT SYSTEM

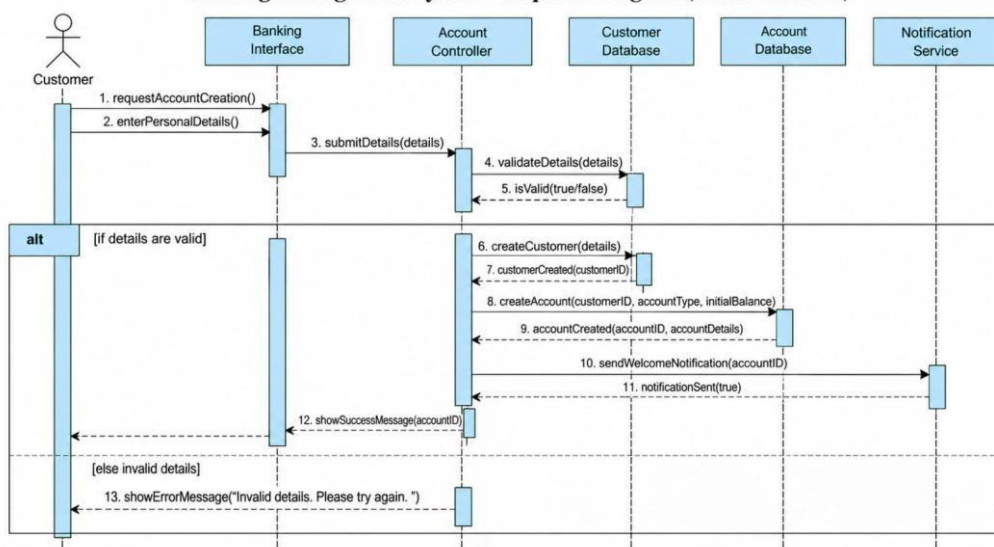
Banking Management System – Sequence Diagram (Transfer Money)



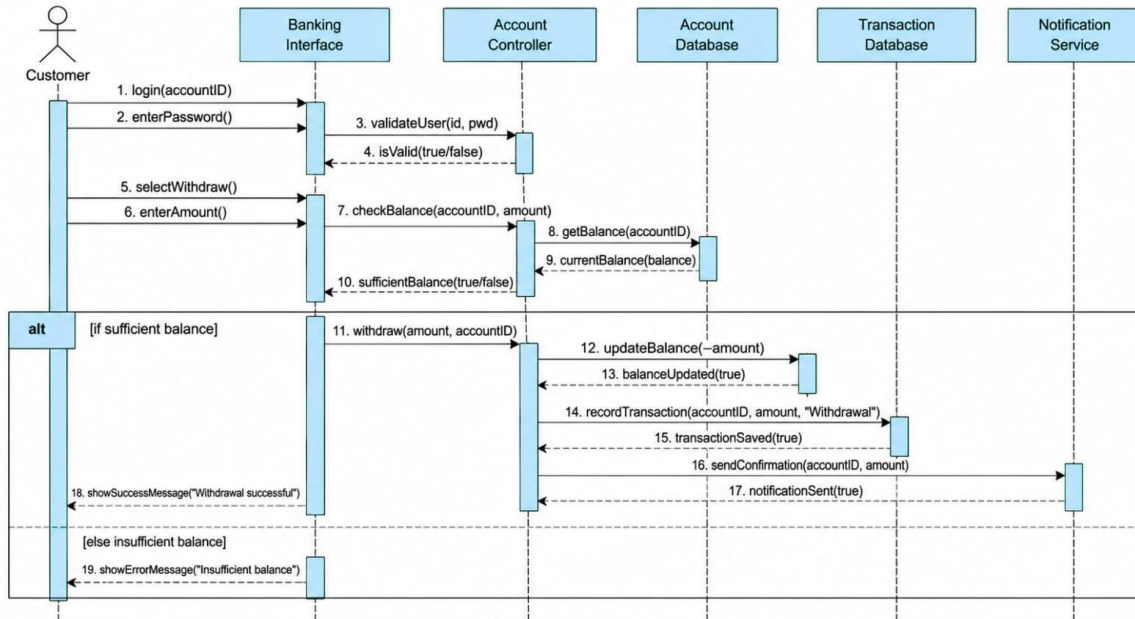
Banking Management System – Sequence Diagram (Check Account Balance)



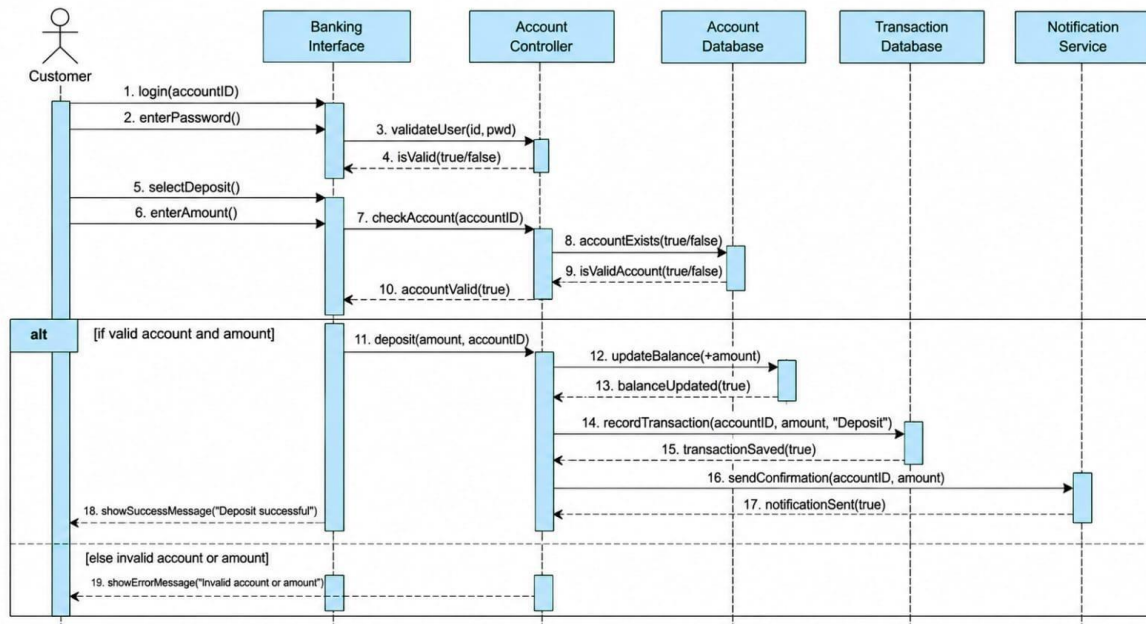
Banking Management System – Sequence Diagram (Create Account)



Banking Management System – Sequence Diagram (Withdraw Money)

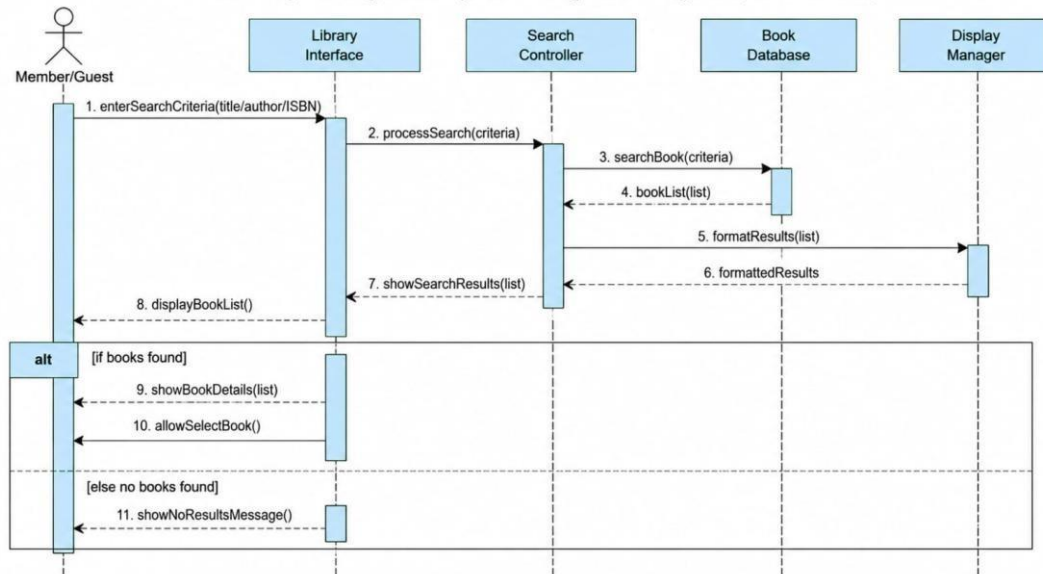


Banking Management System – Sequence Diagram (Deposit Money)

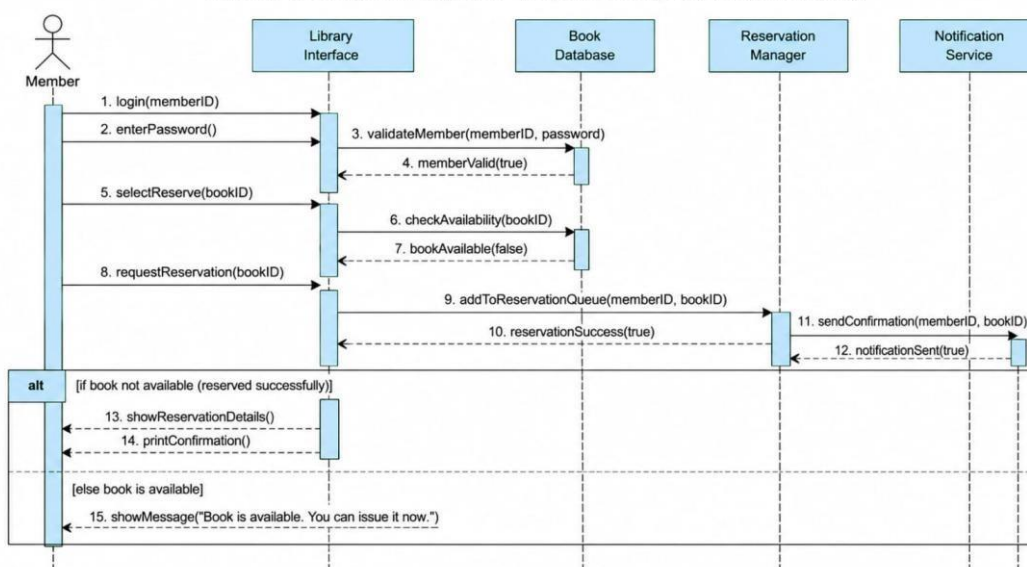


EX.NO.5.3 SEQUENCE DIAGRAM FOR LIBRARY INFORMATION SYSTEM

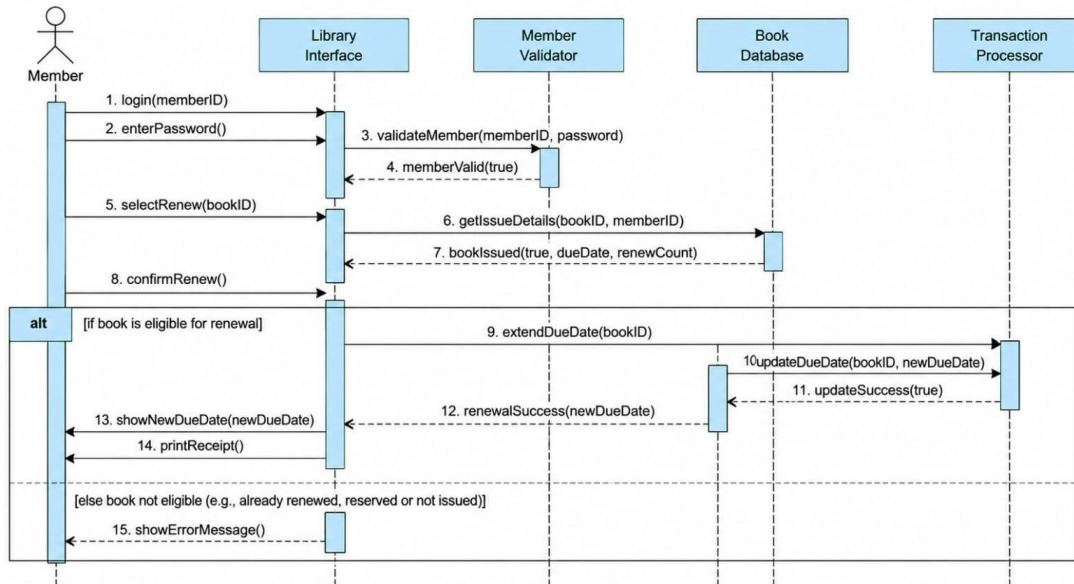
Library Management System – Sequence Diagram (Search Book)



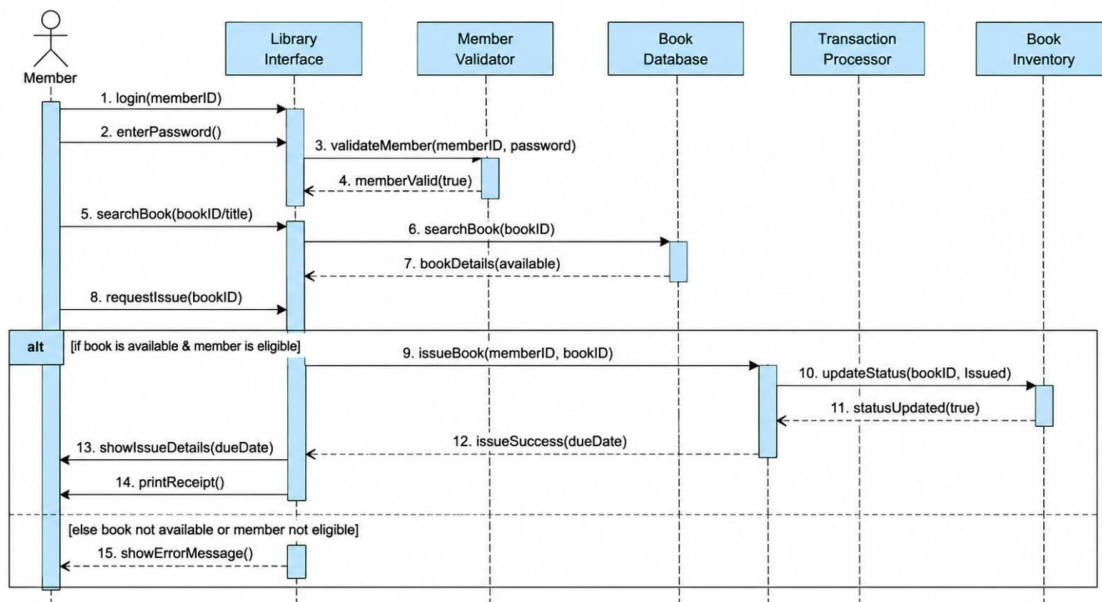
Library Management System – Sequence Diagram (Reserve Book)



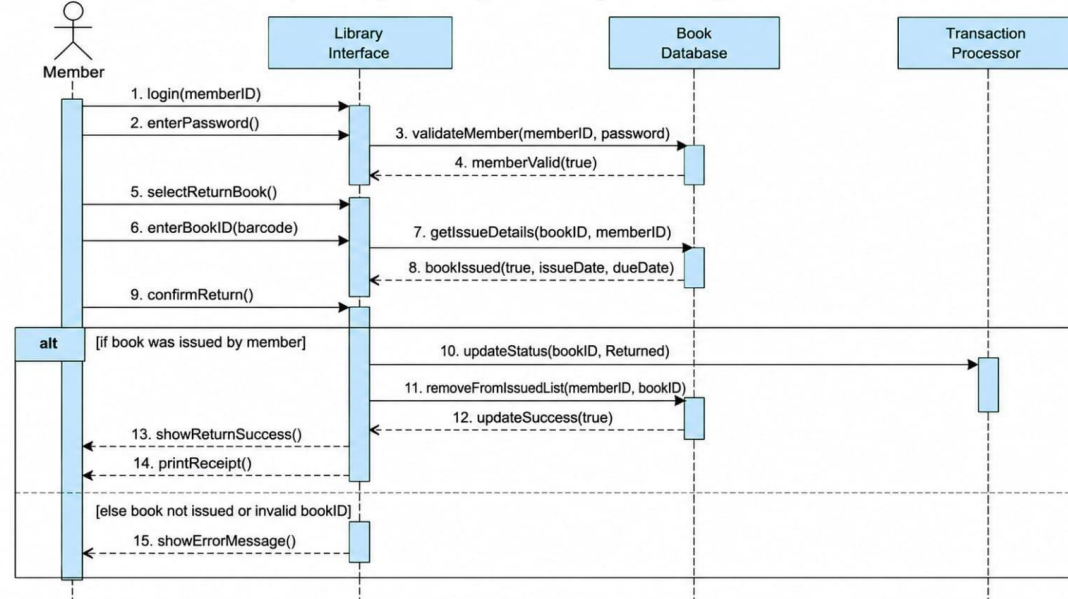
Library Management System – Sequence Diagram (Renew Book)



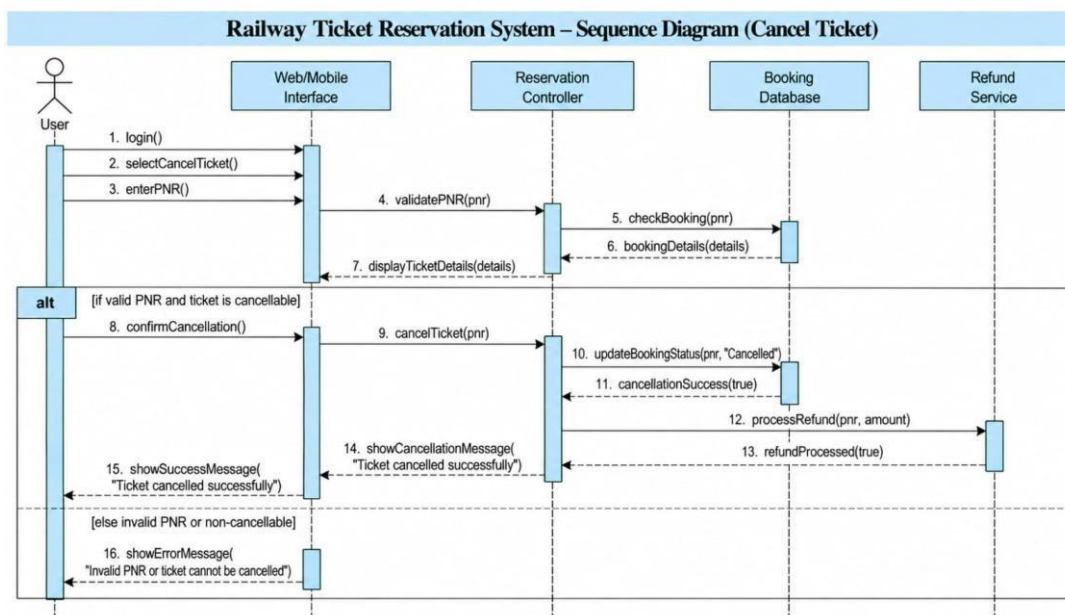
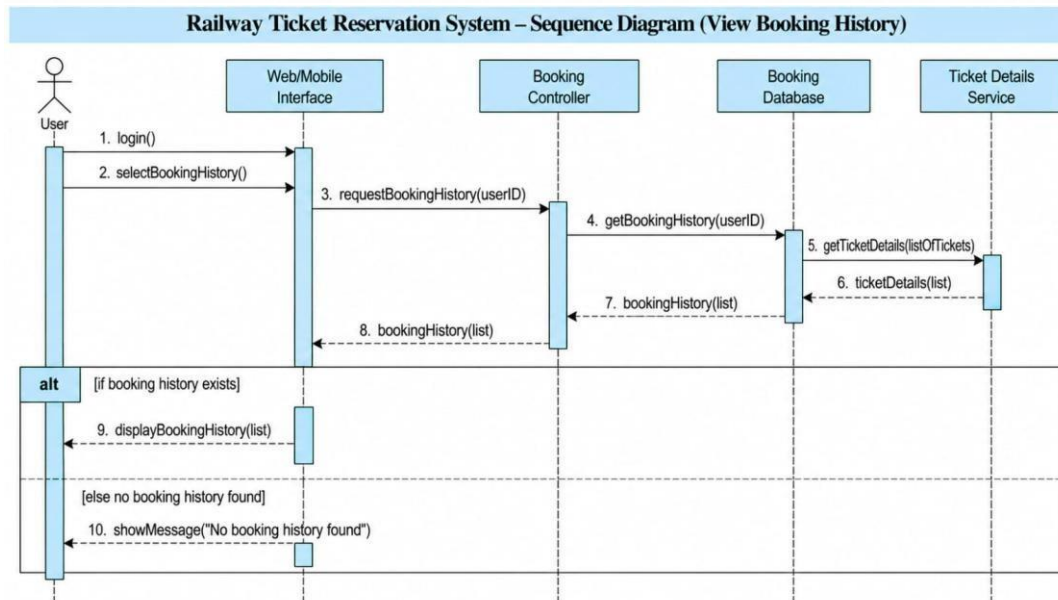
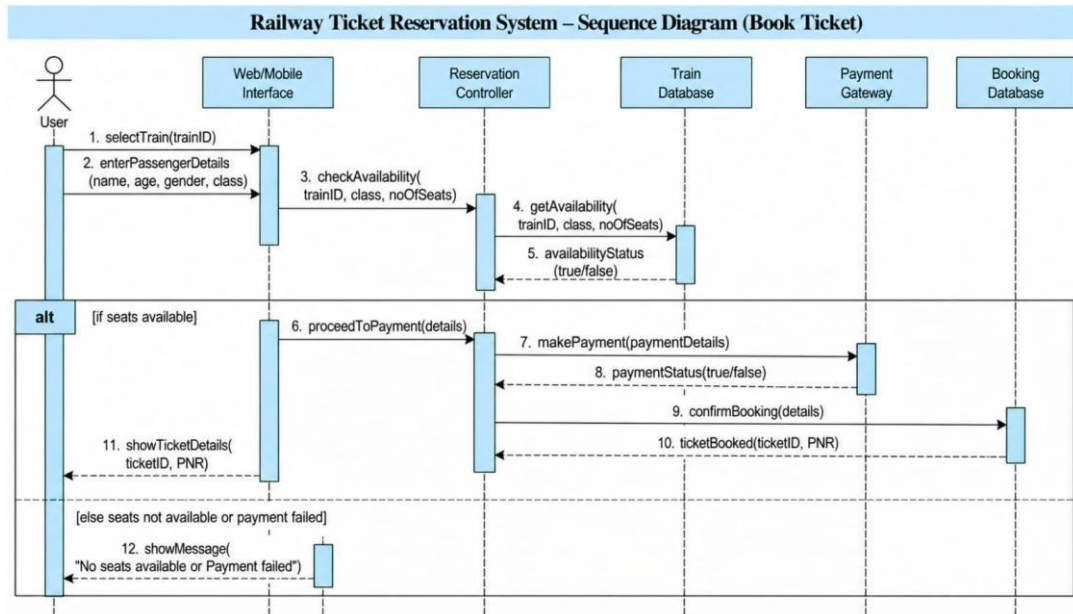
Library Management System – Sequence Diagram (Issue Book)



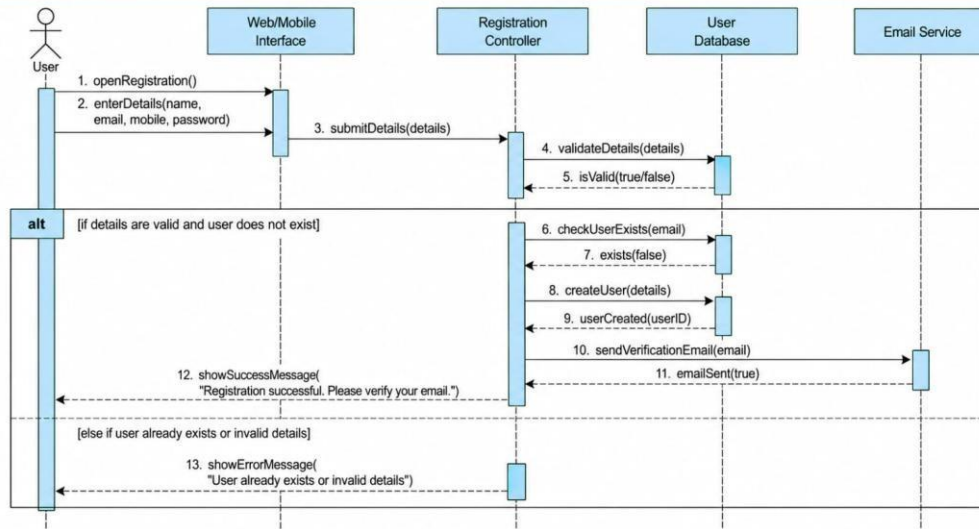
Library Management System – Sequence Diagram (Return Book)



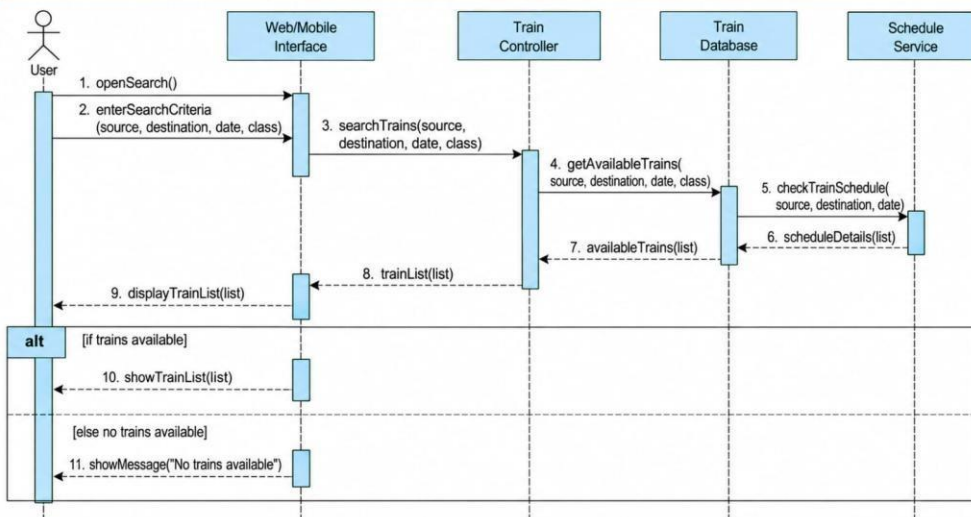
EX.NO.5.4 SEQUENCE DIAGRAM FOR RAILWAY RESERVATION SYSTEM

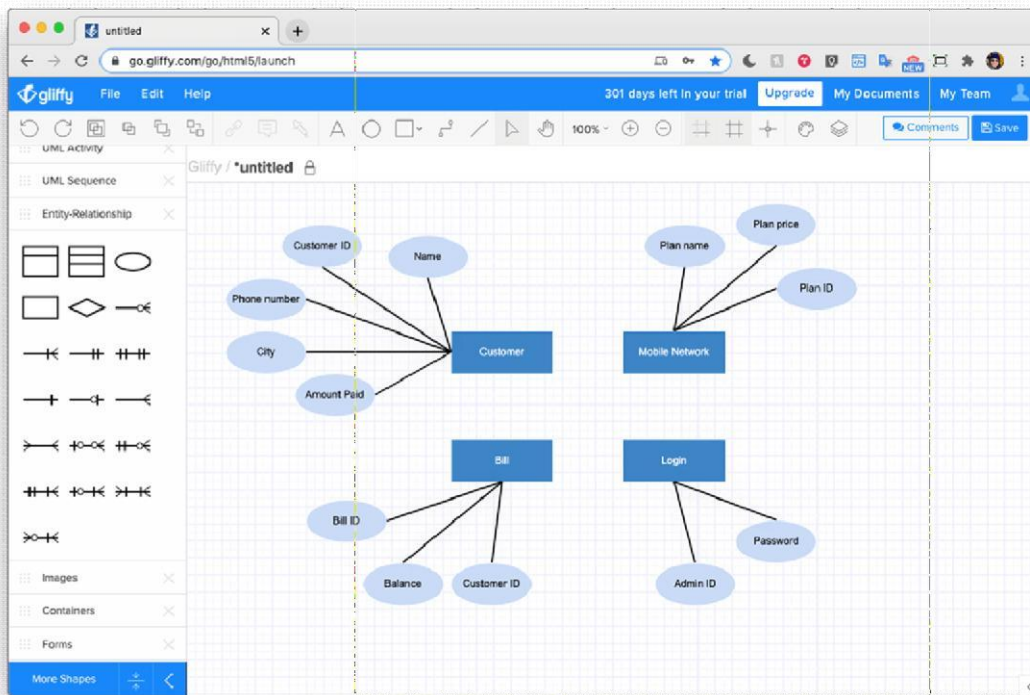


Railway Ticket Reservation System – Sequence Diagram (User Registration)



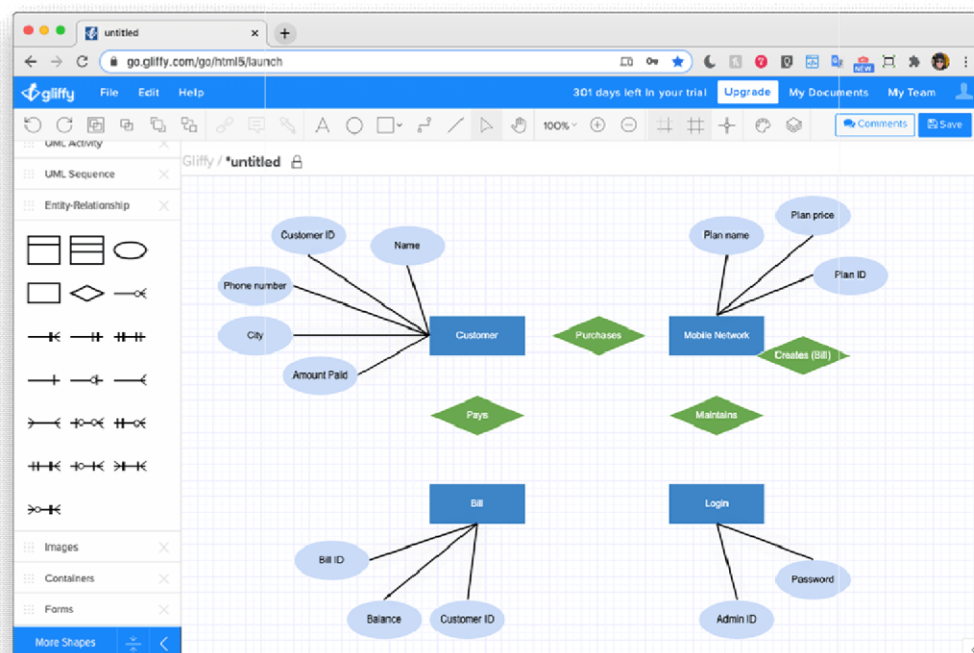
Railway Ticket Reservation System – Sequence Diagram (Search Trains)





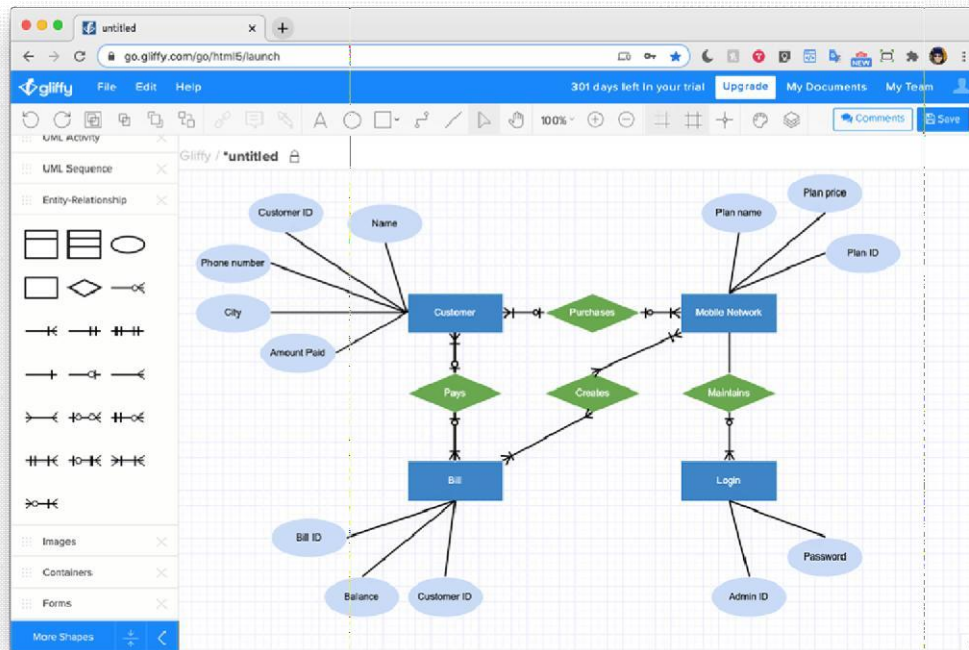
o the relevant entity and position your attributes to the outside of your diagram, which leaves room for relationships.

3. Define the Relationships Between Entities



Now, think through the relationships or verbs taking place within your system. The easiest way to do this is to look at each entity and try to connect it to another by saying, “What does the ___ do with the ___.” The customer purchases the phone. The cell service maintains the phone. The cell service creates a bill. The customer pays the bill.

4. Add Cardinality to Every Relationship in your ER Diagram



The final step for this simple ER diagram is to define the amount of data that will come from each entity. Cardinality is a simple notation that quickly tells your ERD reader whether there are zero, one, many, or some combination of those factors between each entity. Your customer can purchase one or many phones. The cell service maintains many phones. The customer pays one bill.

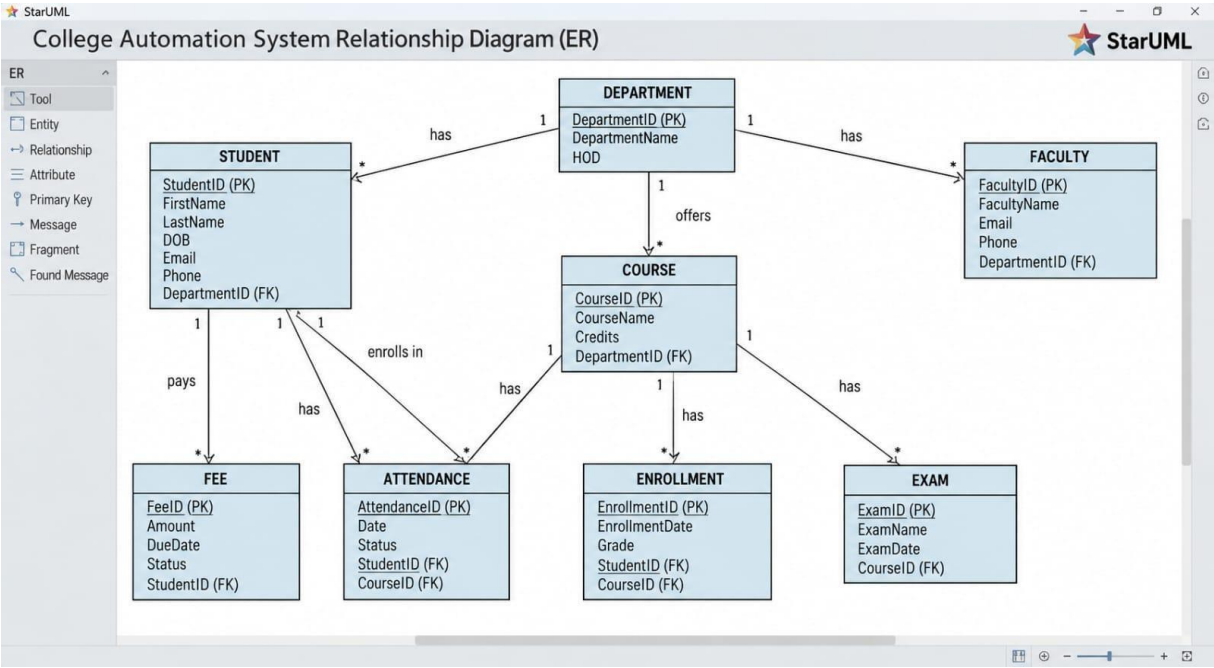
5. Finish and Save Your ER Diagram

This is just a high-level ER model, but it provides enough detail that you should now have a teammate or partner check your work. One of the best ways to do this is to simply have them try to read your diagram out loud. If they end up telling a different story than you intended, you need to do some tweaking.

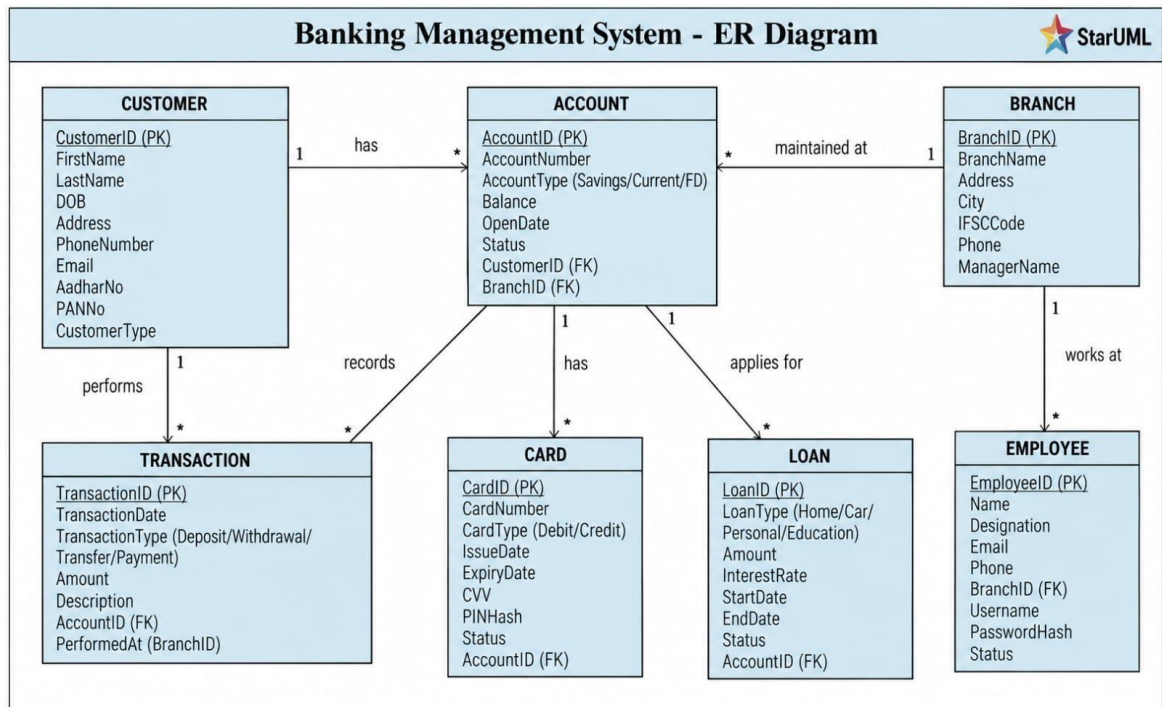
Another good step to take is to clean up or polish your diagram — if you were drawing by hand, you might have some stray eraser marks. Take a moment to finalize your diagram by aligning shapes, adding colour, or redrawing lines to more clearly connect your entities, attributes, and relationships. All these steps are easy if you're using an online diagramming tool like Gliffy

EX.NO.6.1

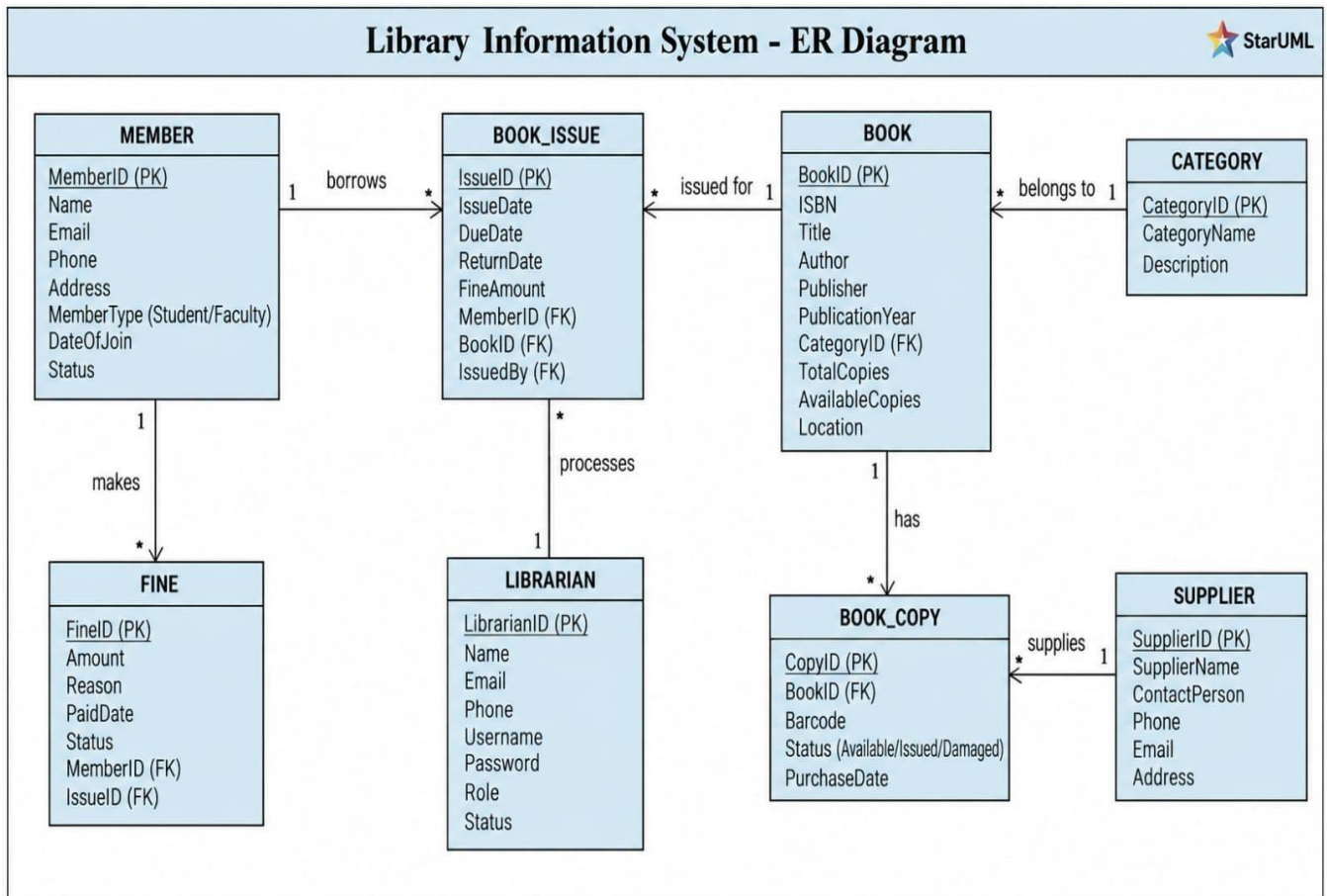
ER DIAGRAM FOR COLLEGE AUTOMATION SYSTEM



EX.NO.6.2 ER DIAGRM FOR BANKING MANAGEMENT SYSTEM



EX.NO.6.3 ER DIAGRAM FOR LIBRARY INFORMATION SYSTEM



EX.NO.6.4 ER DIAGRAM FOR RAILWAY TICKET RESERVATION SYSTEM

